

atomic weight 157.25, melting point 1313 °C (1586 K), boiling point 3273 °C (3546 K) and specific gravity from 7.8 to 7.896. It occurs with other rare earth elements in many minerals, such as monazite, samarskite, gadolinite and some varieties of Norwegian ytterspar. It is the 41st element in order of abundance in Earth's crust.

GAIA HYPOTHESIS (GAIA THEORY; GAIA PRINCIPLE): The theory, based on an idea put forward by the British scientist, James Ephraim Lovelock that the biosphere and the physical components of the Earth (atmosphere, cryosphere, hydrosphere and lithosphere), function as a single self-regulating system and that the response of living organisms to environmental conditions ultimately brings about changes that make the Earth better adapted to support life and that the system would rid itself of species that adversely affect the environment.

GAIN: The ratio of the output signal strength of a circuit or system to the input or the output voltage of an amplifier compared to its input voltage or the mean ratio of the signal output of a system to the signal input of the same system. It may also be defined on a logarithmic scale, in terms of the decimal logarithm of the same ratio ("dB gain"). Voltage Gain = Output Voltage/Input Voltage. It is a measure of amplification.

GAIT: 1. The speed at which something moves or progresses. 2. The pattern of movement of the limbs of animals, including humans, during locomotion over a solid substrate. An example is the four paces of a horse, namely gallop, canter, trot and walk, each having a specific pattern of leg movements.

GALACTAN: A structural polysaccharide with the molecular formula $C_6H_{12}O_6$. It commonly occurs in lactose and in certain pectins, gums and mucilage that is less sweet than glucose. The major monosaccharide subunit is galactose, though other sugars, notably arabinose, are often also present.

GALACTIC BULGE: The spheroidal distribution of stars that is centred on the nucleus of the Milky Way Galaxy and extends to a distance of about 3 kiloparsecs from the centre. It is the gravitational centre of the Milky Way Galaxy. The Sun and other stars of the Galaxy revolve about this centre. Stars that populate the bulge are normally old, Population II objects, dating back to their galaxy's earliest period. Studying bulges can therefore tell astronomers about how galaxies formed and evolved.

GALACTIC CENTRE: The centre of the Milky Way Galaxy or of any other galaxy. It is the point about which the disk of a spiral galaxy rotates and it lies within the galactic nucleus.

GALACTIC COORDINATES: Coordinates based on the plane of the Milky Way Galaxy, which is inclined about 63° to the celestial equator and centred on the Sun, with the zero point of longitude and latitude pointing directly at the galactic centre.

GALACTIC DISK: The flat distribution of stars and interstellar matter in the spiral arms and plane of the Milky Way Galaxy. The biggest and brightest of stars, the O stars and B stars lie in the galactic disk together with many stars such as the Sun and large tracts of interstellar material from which new stars are continually formed.

GALACTIC EQUATOR: The imaginary great circle on the celestial sphere that lies 90° from both galactic poles. It closely follows the middle of the visible Milky Way.

GALACTIC HALO: A large, relatively dust-free, spherical region surrounding a spiral galaxy, such as the Milky Way. The inner, visible part of the halo, which has roughly the same diameter as the galactic disk, is occupied by Population II objects, including globular clusters and old, individual stars. Beyond this is a much larger region, called the **dark halo** or **extended halo**, containing large amounts of dark matter, the presence of which is revealed by its gravitational effect on the galaxy's rotation. In the case of the Milky Way, most of the stellar

halo lies closer to the galactic centre than the Sun does, while most of the dark halo lies farther from the galactic centre than the Sun.

GALACTIC MAGNETIC FIELD: A weak and largely disordered magnetic field, with a strength of about 5×10^{-10} tesla, that pervades the disk of the Milky Way Galaxy and controls the alignment of interstellar dust particles. Similar fields exist in other disk galaxies. The magnetic fields of elliptical galaxies are more difficult to investigate because of the lack of interstellar matter.

GALACTIC NOISE (GALACTIC RADIATION; GALACTIC RADIO WAVES): A diffuse radio signal that comes from the synchrotron radiation of electrons spiralling in the Galaxy's magnetic field.

GALACTIC NUCLEUS: The small, bright, central part of a galaxy. If the nucleus gives off considerably more energy than can be accounted for by its stellar contents, it is said to be an active galactic nucleus (AGN).

GALACTIC PLANE: The plane in which the disk of the Milky Way Galaxy lies or the plane that may be drawn through the galactic equator. One direction perpendicular to this plane is called above or north and the opposite direction, also perpendicular to the galactic plane, is called below or south.

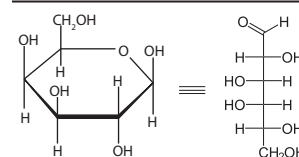
GALACTIC ROTATION: The systematic, differential rotation of stars and interstellar matter in a galaxy about an axis that passes through the galactic centre. Spiral and lenticular galaxies have a well-defined rotational structure whereas the individual stars in elliptical galaxies tend to move in random orbits.

GALACTIC WINDOW: A region of sky near the plane of the Milky Way that is unusually free of dust and gas and thus permits relatively clear views of distant galaxies.

GALACTIC YEAR: The Sun's period of rotation around the centre of the Galaxy. It is equal to about 225 million years.

GALACTOGOGUES: Agents or substances containing properties that increase the secretion of milk from lactating women's breasts. They include herbs such as fennel, hastyberry, alfalfa and hops.

GALACTOSE: A monosaccharide, with the molecular formula $C_6H_{12}O_6$, density 1.723 g/cm³ and melting point, 167 °C (440 K). It is a six-carbon sugar that is a constituent of lactose. It occurs naturally as one of the products of other sugars, for example, in lactose, in polysaccharides and in glycolipids, carbohydrate-containing lipids that occur in the brain and other nervous tissues of most animals.



Chemical structure of galactose

GALACTOSIDASE: An enzyme that catalyses the cleavage of terminal galactose residues from a variety of substrates; several such enzymes exist, each specific for α - or β -linked sugars and further specific for substrates, for example, lactase.

GALAXY: A vast collection of stars, planets, atomic hydrogen gas, molecular hydrogen, complex molecules composed of hydrogen, nitrogen, carbon and silicon, etc., and cosmic rays which make up the universe. Galaxies are held together by gravitational attraction between its components. Galaxies vary in size, composition, structure and activity, but nearly all are arranged in groups or clusters, containing from a few galaxies to as many as 10,000. The Solar System to which the Sun and the Earth belong is in a galaxy called the Milky Way. There are many millions of galaxies in the universe.

GALAXY CLUSTER: A group of galaxies containing from two to several hundred galaxies extending over a radius of up to a few megaparsecs and associated by mutual gravitational attraction galaxies in space. For galaxies, the clusters contain hot intracluster gas, at temperatures between 10^7 and 10^8 K, detectable by its X-ray emission.

GALAXY INTERACTION: An encounter between two or more galaxies that is close enough for the mutual gravitational attraction to alter the appearance and, in some cases, the composition of one or both systems.

GALE: A strong sustained wind, usually between force seven and ten on the Beaufort scale measuring between 45 and 90 km per hour (34 and 47 knots).

GALEA: 1. Shaped like a helmet. 2. In plants, it is the hood-shaped upper lip of double-lipped corollas as seen in some flower.

Galeate is looking like a galea, as shown in the diagram.

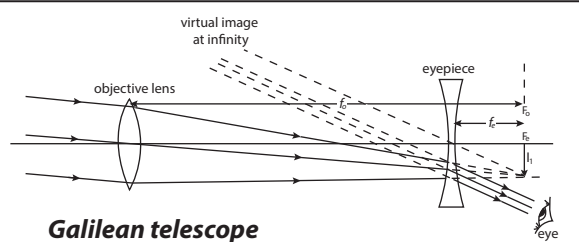


GALENA: The most important ore of lead and silver consisting of lead(II)sulfide (PbS) belonging to the rock salt (NaCl) structure type. It has a specific gravity of 7.5 and hardness of $2\frac{1}{2}$ on Mohs scale. It crystallises in the cubic system and occurs as grey metallic cubes, often with copper, zinc, antimony, arsenic and silver.

GALICIAN BLOND: A breed of cattle from Galicia region in Northern Spain. It is mainly a beef cattle breed, red in colour, with yellow horns.

GALILEAN SATELLITES: The four largest moons of Jupiter, which are easily visible in a small telescope or binoculars. In order of increasing distance from the primary, they are Io, Europa, Ganymede and Callisto. They were discovered independently by Galileo Galilei and Simon Marius in 1610.

GALILEAN TELESCOPE: A refracting telescope having one convex lens and one concave lens. The concave lens serves as the ocular lens or the eyepiece with a short focal length; while the convex lens serves as the objective lens with a large focal length. The lenses are situated on either side of a tube such that the focal point of the ocular lens is the same as the focal point for the objective lens. Galilean telescopes are shorter than equivalent Keplerian telescopes and give bright, upright images, but they have a limited field of view, spherical and chromatic aberration and, except at low magnification (about $30\times$ or less), poor eye relief.



GALILEAN TRANSFORMATION: A set of equations for transforming the position and motion parameters from a frame of reference with origin at 0 and coordinates (x, y, z, t) to a frame with origin at $0'$ and coordinates at (x', y', z', t') . They are: $x' = x - vt$, $y' = y$, $z' = z$, $t' = t$. It is used to transform between the coordinates of two reference frames which differ only by constant relative motion within the constructs of Newtonian physics.

GALILEO: A unit of acceleration used in geological survey. A Galileo is equal to 10^{-2} metres per second.

GALILEO GALILEI (1564- 1642): Italian physicist and astronomer, born in the city of Pisa in Italy, he was taught by monks at Vallombrosa. He entered the University of Pisa in 1581 to study medicine but turned to philosophy and mathematics and left the university in 1585 without a degree. In 1589, he became professor of mathematics at Pisa. Galileo's main contributions were, in astronomy. He invented the astronomical telescope for the observation and the discovery of sunspots, mountains and valleys on the Moon, the four largest satellites of Jupiter and the phases of Venus. He also invented the

thermometer. In physics, he discovered the laws of falling bodies and the motions of projectiles. In the field of inertia, Galileo believed that an object moving uniformly on the Earth's surface without meeting any resistance would continue to move at the same speed without needing any force to keep it going, which is similar to Sir Isaac Newton's first law of motion, which states that a body will continue in its state of rest or uniform motion in a straight line unless interfered with.

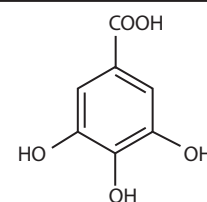
GALL: An abnormal, localised outgrowth on a plant that develops as a result of attack by insects and mites or less commonly by bacteria, fungi, mites or nematodes.

GALL BLADDER: A hollow muscular organ that stores bile made in the liver of vertebrates. It receives dilute bile from the liver, concentrates it and releases it, when food enters the duodenum, through the bile duct.

GALLERY FOREST: A type of forest that grows alongside rivers and streams running through a region that is otherwise treeless.

GALLIC ACID (3,4,5-TRIHYDROXYBENZOIC ACID; GALLATE; 3,4,5-TRIHYDROXYBENZOATE): A

colourless crystalline aromatic compound, with the chemical formula $C_6H_2(OH)_3COOH$, density 1.7 g/cm^3 (anhydrous) and melting point 250°C (523 K). It occurs in gallnuts, sumac, witch hazel, tea leaves, oak bark and other plants and also as part of tannins. It is used to make ink and dyes and on heating yields pyrogallol. It acts as an antioxidant and helps to protect cells against oxidative damage. It is also used to treat albuminuria and diabetes.



Chemical Structure of Gallic Acid

GALLIFORME: An order of medium to heavy-bodied arboreal or terrestrial ground-feeding birds with about 70 genera and more than 250 species found on every continent except Antarctica and almost every habitat on earth including grasslands, deserts and forests. They generally have short wings, and strong legs adapted for running, with four-toed feet. They have various plumage colouration from cryptic, dark to brightly colourful. Some have elaborate head and neck ornamentation including feathers, wattles and casques. Examples of members of this order include turkey, grouse, chicken, quail, partridge, pheasant and jungle fowl. These birds may weigh between 275 g and 9.5 kg, depending on species. They feed mostly on plant matter such as fruits, seeds, leaves, shoots, flowers, tubers and roots as well as insects and worms. They nest mostly on the ground and depending on the species lay from 2 to 35 eggs which hatch as precocial chicks. Some, such as common fowl and turkey are economically important to humans and are kept for meat and eggs. Some others, such as grouse, quail, partridges, pheasants and wild turkeys are hunted as gamebirds. The feathers of some species are collected for ornamentation and for the manufacture of clothing.

GALLIUM: A rare grey metallic element, with the symbol Ga, atomic number 31, atomic weight 69.72, specific gravity 5.907, melting point 29.7°C (303 K) and boiling point 2403°C (2676 K). It is found as a trace element in coal, bauxite and other minerals. It is liquid at room temperature. Gallium has one of the largest liquid ranges for a metal but unlike mercury, has a very low vapour pressure. The melting point of Ga is used as one of the formal temperature reference points in International Temperature Scale of 1990 (ITS-90). It is used in semiconductor technology and as a component of various low-melting alloys.

GALLIUM ARSENIDE: A semiconductor alloy of gallium and arsenic, with the chemical formula GaAs and a melting point of $1,238^\circ\text{C}$ ($1,511 \text{ K}$). It has semi-conducting properties similar to or superior to those of silicon, for example, its chips require less electric power and process data faster than those made from silicon. It is used in

lasers, photocells and microwave generators and in high frequency applications such as cellphones, DVD players and fibre optics.

GALLON: Abbreviated g, it is a measure of volume, which is approximately equal to four litres. The various types of gallons in use are the Imperial gallon (approximately 4.5 litres), which is in unofficial use in the United Kingdom/Ireland. In the United States, there are the liquid (approximately 3.78 litres) and the lesser used dry (approximately 4.4 litres) gallons.

GALLOWAY: A hardy breed of completely black hornless dairy cattle mainly reared for beef originating in Scotland. The coat is distinctive, being formed of long wavy hairs covering a soft undercoat.

GALLSTONE: A deposit of crystallised substances sometimes formed in the gall bladder. Gallstones may lodge in and obstruct the bile duct. The most common type occurs when the liver secretes bile with too much cholesterol to stay in solution.

GALOIS, EVARISTE (1811 - 1832): French mathematician, who made major contributions to the theory of functions, the theory of equations and number theory. He also made crucial contributions to group theory and applied this to the solvability of polynomial equations by radicals.

GALOIS CORRESPONDENCE: Named after French mathematician Evariste Galois, it is a pair of mappings, between completely ordered sets, that are antitone and mutually dominating; thus $f: S \rightarrow S'; f': S' \rightarrow S$ and for each x in S , $x' \in S'$, $x \leq f'(f(x))$; $x' \leq S f f'(x')$. Galois correspondence is sometimes used to mean bijective Galois connection, which is an order isomorphism or dual order isomorphism, dependently on whether we take monotone or antitone Galois connections. **Galois connection** is a particular correspondence (typically) between two partially ordered sets (posets).

GALOIS GROUP: The group of all automorphisms of the splitting field, K , of a given a polynomial equation that leaves all members of the base field, F , fixed. It is denoted $G(F/K)$. It can be regarded as the group of all permutations of the roots of the equation that leave all relations of the roots invariant. The Galois group of the general equation of degree n is the full symmetric group. The Galois group of a polynomial f with respect to the coefficient field F is defined as the group of automorphisms of the splitting field of f that leaves F fixed.

GALOIS THEORY: The algebraic study of the conditions under which a polynomial equation with coefficients in a given mathematical field is solvable by means of rational operations and the extraction of n th roots. This is done by exploiting the correspondence between fields and their respective automorphism groups. In general there is a sequence of sub-fields between the splitting field and the coefficient field and these correspond to a sequence of subgroups.

GALVANI POTENTIAL DIFFERENCE: An electric potential difference between points in the bulk of two phases. It is measurable only when the two phases have identical composition, for example, two copper wires. It is the difference of inner electric potentials in two phases.

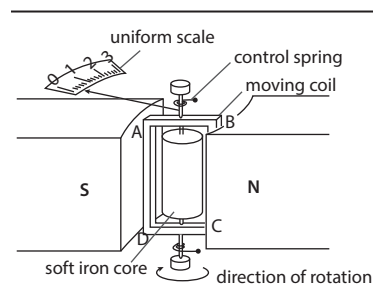
GALVANI, LUIGI (1737-98): Italian physician and physicist. In the late 1770s he observed that the muscles of the dead frog twitched when touched by two different metals. He concluded that the muscle was producing electricity, later disproved by Volta, who showed that electricity is produced from the contact of two different metals in a moist environment. Galvani invented galvanised iron and the galvanometer.

GALVANIC CELL (VOLTAIC CELL): An electrochemical cell that uses an irreversible spontaneous chemical reaction to generate electricity and hence cannot be recharged. It generally consists of two different metals connected by a salt bridge or individual half-cells separated by a porous membrane.

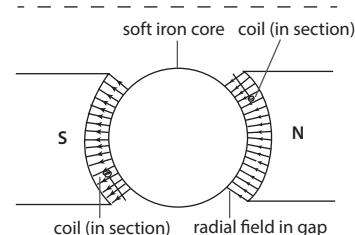
GALVANISATION (GALVANIZATION): An electrochemical process named after the Italian scientist, Luigi Galvani. It is used for coating iron and steel sheeting with a thin layer of another metal, commonly zinc. It protects the iron from rust.

GALVANISED IRON: Iron or steel that has been coated with a layer of zinc to protect it from corrosion.

GALVANOMETER: A very sensitive type of instrument (ammeter) used to detect and measure very small electric currents flowing in a conductor by deflection of a moving coil. It consists of a light coil of wire suspended from a metallic ribbon between the poles of a permanent magnet. As current passes through the coil, the magnetic field it produces reacts with the magnetic field of the permanent magnet, producing a torque, which causes the coil to rotate, moving an attached needle or mirror, which is a measure of the current flowing in the coil.



(a) structure



(b) action (viewed from above)

Moving coil galvanometer

GAMBLER'S RUIN: A term used for a number of related statistical ideas, it is a probability of winning, for example, 1 dollar and a probability of losing 1 dollar. If a gambler bets on repeated trials, in each of which he wins with probability strictly between 0 and 1 until he either doubles his initial capital or loses it all. This gives the transition probabilities of a Markov chain with two absorbing states. The probability of ruin varies both with the initial state and with the probability of winning on each trial.

GAMBUSIA: Any fish of the genus *Gambusia*, native to the eastern and southern United States and northern Mexico that bears living in mangrove creeks and brackish waters of the southern US and northern Mexico. The genus contains over 40 species. Two species, *Gambusia holbrooki* and *Gambusia affinis* feed mosquito larvae and are used to control mosquitoes.

GAME THEORY (THEORY OF GAMES): A mathematical theory that deals with problems by assuming that the people involved invariably try to win and employs optimal strategies for situations in which there is competition or conflict to a desired outcome.

GAMEBOY: A handheld video game that was introduced in 1989, credited as the longest running video game system of all time. It was developed, manufactured and marketed by Nintendo Co., Ltd., a Japanese multinational consumer electronics and video game company. Its screen is tiny, with the graphics limited to shades of gray on a dull green background. Since its introduction, Game Boy has sold over 150 million systems worldwide.

GAMETANGIUM: A plant sex organ in which gametes are produced. The term is usually restricted to the sex organs of algae and fungi.

GAMETE: A name introduced by the Austrian biologist, Gregor Mendel (1822-1884) for reproductive cell that fuses with another gamete during fertilisation in an organism that reproduces sexually. Examples are sperm cells and ova in animals and pollen and ovule in flowers. A female gamete joins with a male gamete during fertilisation to form a zygote. Gametes carry half the genetic information of an individual, 1n of each type.

GAMETOCYTE: 1. An eukaryotic germ cell that divides by mitosis to produce two specialised male and female cell gametes or by meiosis into gametids during gametogenesis. Male gametocytes are called **spermatocytes** and female gametocytes are called **oocytes**. 2. A malaria organism in the stage in its life cycle during which it reproduces in the blood of a mosquito.

GAMETOGENESIS: The production of gametes from gametocytes by cell division or meiosis. Gametogenesis occurs by meiotic division of diploid gametocytes into various gametes or by mitotic division of haploid gametogenous cells depending on the biological life cycle of the organism. For example, plants produce gametes through mitosis in gametophytes.

GAMETOPHYTE: The sexual phase in the life cycle of certain plants such as algae, mosses and fungi when the sex organs and gametes are produced. In the gametophyte phase, male and female organs or gametangia develop and produce eggs and sperm or gametes, which unite in fertilisation or syngamy.

GAMMA: 1. The third letter of the Greek alphabet denoted by "Γ" for the upper case and "γ" for the lower case. They are used as symbols for many scientific functions. 2. In astronomy, it is usually the third brightest star of a constellation. 3. A unit of magnetic intensity, which is equal to one hundred thousandth (10^{-5}) of an oersted. 4. A unit of mass, which is equal to one millionth (10^{-6}) of a gram. 5. In chemistry, it is the third position from a designated carbon atom in an organic molecule at which an atom or a radical may be substituted. Also, it relates one of two or more allotropes or crystal structures of a solid, for example, gamma iron. 6. In physics, involving or relating to photons of very high energy, for example, a gamma detector. 7. In computer science, it refers to the brightness of a monitor or computer display and determines how bright the output of the display is. **Gamma correction** is used to alter the output levels of a monitor. Usually, the settings are 2.2 for PC monitors and 1.8 for Macintosh monitors.

GAMMA CAMERA (SCINTILLATION CAMERA; ANGER CAMERA): A device for imaging gamma radiation emitting radioisotopes. It is used to detect and record the spatial distribution of radioactive compounds in the human body. It consists of a large scintillation crystal of sodium iodide and a collection of photomultiplier tubes above the crystal, connected to it by a transparent material.

GAMMA CORRECTION: An adjustment to the light intensity (brightness) of a scanner, monitor or printer in order to match the output more closely to the original image. Since the standard input signal or voltage level is not high enough to create a bright picture, gamma correction is used to boost the brightness and contrast of the display. PCs typically use a gamma setting of 2.2, while Macs have a default gamma setting of 1.8. Some systems include a display utility that can be used to apply custom gamma settings.

GAMMA DISTRIBUTION: A probability distribution given by $f(x) = \left(\frac{x}{\beta}\right)^{\gamma-1} \frac{\exp(-x/\beta)}{\beta\Gamma(\gamma)}$, $0 \leq x \leq \infty$, $\beta > 0, \gamma > 0$, where γ and β are parameters. The mean and variance of the distribution are: mean = $\gamma\beta$, variance = $\gamma\beta^2$.

GAMMA DIVERSITY (GAMMA RICHNESS): A measure of biodiversity, which indicates the overall species diversity of a range of habitats or communities within a region. It reflects the diversity of species occupying individual habitats called alpha diversity combined with a measure of the heterogeneity between the habitats.

GAMMA FUNCTION: The function defined by $\Gamma(x) = \int_0^\infty t^{x-1} e^{-t} dt$, where $x > 0$ (x need not be an integer). The function is recursive satisfying the relationship $\Gamma(x+1) = x\Gamma(x)$. The integral $\int_0^T t^{x-1} e^{-t} dt$ is known as the incomplete gamma function, satisfying the property $\Gamma(x+1) = x\Gamma(x)$. For the integral n , $\Gamma(n+1) = n!$. Gamma function has three definitions: (i) the Euler and Gauss, defined by $\Gamma(1+z) = \int_0^\infty x^z e^{-x} dx$. The real part of z here is to be greater than -1 . (ii) The second definition is due to Gauss, defined by $\Gamma(1+z) = \lim_{n \rightarrow \infty} \frac{n!n^z}{(z+1)(z+2)\dots(z+n)}$, where z is not a negative integer.

(iii) The definition due to Weierstrass is $\Gamma(1+z) = e^{\gamma z} \prod_{n=1}^\infty \left(1 + \frac{z}{n}\right) e^{\left(-\frac{z}{n}\right)}$.

The gamma function also satisfies the relation $\frac{\pi}{\sin \pi z} = \Gamma(z)\Gamma(1-z)$

for all z whose absolute value satisfies the inequality $0 < z < 1$. Thus $\Gamma(1/2) = \sqrt{\pi}$.

GAMMA IRON: A non-magnetic iron with a face-centred cubic structure. It is the allotropic form of iron existing between the temperature $910^\circ\text{C} - 1400^\circ\text{C}$ ($1183\text{ K} - 1673\text{ K}$). Gamma iron containing carbon or other elements in solution is known as **austenite**. Above 1400°C (1673 K) it transits to delta iron.

GAMMA RADIATION (GAMMA RAYS): An electromagnetic radiation emitted by excited atomic nucleus when it moves to a lower excitation state. Gamma radiation lies above X-Rays in the electromagnetic spectrum. It is emitted by the hottest part of the universe and also emitted by the radioactive decay of certain materials like cobalt-60, zinc-65, caesium-137, and radium-226. Cadmium-109 is also a common source of gamma radiation. Gamma rays can be produced in laboratories by exciting nuclei of a radioactive element by the use of cyclotron and synchrotron. Gamma radiation is one of the three types of radiation resulting from natural radioactivity. It is the most energetic form of electromagnetic radiation, with the shortest wavelength (high frequency). Gamma radiation ranges in energy from about 10^{-15} to 10^{-10} joule (10 keV to 10 MeV) corresponding to a wavelength range of about 10^{-10} to 10^{-14} metre. It is so highly penetrating that 25mm of lead only halves its strength. The speed of gamma rays is equal to the speed of light since it is a type of electromagnetic wave and all electromagnetic waves travel with the speed of light. Gamma rays produced in space are absorbed by the earth's atmosphere. Gamma radiation is used for cancer treatment and for diagnostic purposes. They are also used for killing the harmful bacteria in food (a process called food irradiation) so that it can be preserved for longer duration. It is also used for inspection of castings and welds, etc.

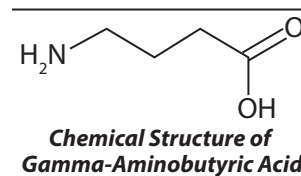
GAMMA-AMINOBUTYRIC ACID (GABA 4-AMINOBUTANOIC ACID): An acid found in the central nervous system, predominantly in the brain, where it acts as an inhibitory neurotransmitter. Its molecular formula is $\text{C}_4\text{H}_9\text{NO}_2$, density 1.11 g/ml , molar mass 103.120 g/mol , melting point 203.7°C (477 K) and boiling point 247.9°C (521 K).

There are two general classes of GABA receptor known as GABA_A in which the receptor is part of a ligand-gated ion channel complex; and GABA_B metabotropic receptors, which are G protein-coupled receptors that open or close ion channels via intermediaries (G proteins). A third one, GABA_C is an ionotropic receptor functions for neuronal inhibition.

GAMMA-RAY ASTRONOMY: The study of astronomical objects and phenomena that emit gamma rays. Many high-energy processes in astrophysics are responsible for the production of gamma rays. Specially designed telescopes are used to observe high-energy astrophysical systems, including stellar coronas, white dwarf stars, neutron stars, black holes, supernova remnants, clusters of galaxies and diffuse gamma-ray background radiation found along the plane of the Milky Way Galaxy. Observations are generally conducted by high-altitude balloons or spacecraft.

GAMMA-RAY BACKGROUND: A diffuse glow of gamma rays seen streaming in all directions. Its exact source is not known but may be due to unresolved active galactic nuclei (AGN), especially blazars.

GAMMA-RAY BURST (GRB): A brief but intense gamma-ray light that have been observed in the universe. It is the most energetic and



brightest source of cosmic gamma-ray photons in the universe and occurs from about a few milliseconds to several minutes. Astronomers think most GRBs get their intense energies enabling them to produce such bright light from the collapse of matter such as stars into a black hole.

GAMMA-RAY LARGE AREA SPACE TELESCOPE (GLAST): The former name for **Fermi Gamma-ray Space Telescope (FGST)**, a NASA spacecraft launched on 11 June 2008 into space to study the most powerful sources of radiation in the universe. It orbits in a low-Earth circular altitude of 550 km, and at an inclination of 28.5 degrees. The most extreme type of radiation is gamma rays.

GAMO: A prefix meaning to unite similar parts.

GAMONE: Any substance released by a gamete that serves to attract another gamete. For example, in certain ferns malic acid is believed to be released from the archegonia to attract the male gametes.

GAMOPETALOUS: Describing a flower such as bougainvillea and morning glory in which the petals are fused to form a corolla tube or funnel.

GAMOSEPALOUS: Having sepals that are fused along their margins to form a tubular calyx.

GANGLION: A cluster of nerve-cell bodies usually enclosed in a connective-tissue sheath outside the central nervous system (CNS). They convey information to the brains. In invertebrates, the majority of neurons are organised as separate ganglia. The spinal ganglion contains the nerve-cell bodies of the nerve fibres that carry impulses toward the CNS (afferent neurons in dorsal root ganglia) or away from it (efferent neurons in ventral root ganglia).

GANGRENE: 1. A condition in which tissues die and decay, as a result of prolonged blood-supply blockage due to conditions such as atherosclerosis, diabetes mellitus or decubitus ulcer and after severe burns or frostbite. 2. Serious rot affecting potato tubers caused by fungi.

GANGRENOUS MASTITIS (MAMMITIS): Inflammation of the mammary gland, a form of the mastitis disease affecting cattle. It may begin as staphylococcal mastitis. The udder becomes blue and cold.

GANGUE: The part of an ore that does not contain the mineral required for extraction. It occurs in a vein or deposit within or alongside a valuable mineral.

GANTRY: 1. A farm machinery consisting of a long steel beam with implement carriers. The engine and cab are at one end of the beam and the drive wheel is at the other end. 2. A frame spanning railroad tracks and used to display signals. 3. A spanning framework used to support machinery, for example, the platform that supports a crane or the structure used to erect and service rockets. 4. **Gantry crane** is a revolving crane, positioned on top of a movable pedestal that travels along tracks. It is designed to reach a more extensive area than other cranes of similar sizes. It is used in ports and harbours for handling cargoes, in drydocks and shipbuilding for lifting heavy equipment, on constructions sites to handle equipment and generally in places where heavy equipment are handled.

GANYMEDE: The largest moon of the planet Jupiter as well as in the solar system. It is slightly bigger than Mercury but with only half its mass. It is one of the four brightest satellites of Jupiter and the seventh in distance from the planet. It has a diameter of 5,262 km, a mass of 0.0247 (Earth = 1), a surface gravity of 0.145 (Earth = 1), a mean distance from Jupiter of 1,070,000 km orbital period of 7.1546 days, rotational period of 7.1546 days and density of 1.94 g/cm³. It was discovered by Galileo and Simon Marius in 1610. Astronomers using the Hubble Space Telescope found evidence of thin oxygen atmosphere on Ganymede in 1996, but the atmosphere is far too thin to support life as known on Earth. Ganymede's surface is a mixture

of two types of terrain, consisting of 40% covered by highly cratered dark regions and 60% covered by a light grooved terrain, which forms intricate patterns across Ganymede.

GAP JUNCTION (NEXUS): A specialised intercellular connection between a multitude of animal cell-types that mediates the transfer of small molecules or ions between interacting cells. Gap junctions abound in epithelial tissues and cardiac muscle.

GAP PHASES: In the cell cycle, the phases that precede cell division gap phase one (G1phase) followed by gap phase two (G2 phase) with the synthesis phase (S phase) in between. In the G1 phase, the cell doubles in size and its enzymes, ribosomes and other cytoplasmic molecules and structures increase in number. In the G2 phase, the replicated chromosomes begin to condense and the structures required for mitosis or meiosis are assembled.

GAP SERIES: A power series with many zero coefficients.

GAPES: A disease affecting the breathing function of poultry, especially young domesticated chickens and turkeys. It is caused by small worms in the windpipe.

GARDEN: A piece of land next to a house where flowers and plants are grown and which often has an area of grass.

GARDEN EGG (AUBERGINE; EGGPLANT; MELONGENE; GUINEA SQUASH; *Solanum melongena*): A plant that produces edible berries with numerous small, soft seeds. It is a delicate tropical perennial in the nightshade family, Solanaceae and is often grown as a tender semi-hardy annual in temperate climates. In tropical and sub-tropical climates, garden eggs can be cultivated directly into the garden. It grows to about 40 - 150 centimetres tall with large, coarsely lobed leaves that are broad and long. It produces white to purple flowers possessing five-lobed petals and yellow stamens. It has a stem that is spiny. The fruit or berry is egg-shaped and glossy with a white, spongy flesh and has a wide range of shapes, sizes and colours (ranging from dark purple to pale mauve and from yellow to white). It is used in many dishes throughout the world and sometimes eaten raw in some areas. It is a source of folic acid, potassium and dietary fibre. Its red and blue colour pigments contain anthozyane that lowers blood cholesterol.

GARLIC: A bulbous perennial plant, *Allium sativum*, of the family Alliaceae, native to central Asia, with a pungent onion-like aroma bulb used as flavouring in cooking. The bulb consists of up to 20 edible wedge-shaped cloves, surrounded by a white papery skin.

GARNET: Any of a group of silicate minerals that conform to the general formula $A_3B_2(SiO_4)_3$ and used as gemstones and abrasives.

GAS: One of three classical states of matter, which has no fixed shape or volume and always fills the whole of the space in which it is contained. It has a density of about 0.1 that of liquids. A gas is distinguished from liquids and solids by the vast separation of the individual gas particles. A substance exists as a solid near absolute zero temperature, but boils into a gas when heated to its boiling point. The average kinetic energy ($KE = nkT$) of a gas is too high for the intermolecular force of attraction to hold the molecules together. That is why molecules of gas are further apart from one another. The shape and size of gas is determined by the containing vessel in which it is introduced.

GAS ANALYSIS INSTALLATION: An assembly for the purpose of determining one or more components in a gaseous mixture. It generally comprises the following elements: sample probe, region of primary treatment of the gases, region of secondary treatment of gases, points for venting to the atmosphere and for calibration sample injection, transfer line and the analytical unit.

GAS BLACK (CARBON BLACK; CHANNEL BLACK; FURNACE BLACK): Finely divided carbon (graphite) produced by incomplete combustion or thermal decomposition of natural gas.

GAS CHROMATOGRAPHY: A technique used for the separation and analysis of complex mixtures of volatile organic and inorganic

compounds. There are two types: **gas-solid chromatography** in which the stationary phase is a solid; and **gas-liquid chromatography** in which the stationary phase is a non-volatile liquid. Most compounds with boiling points less than about 250 °C (523 K) can be readily analysed by this technique.

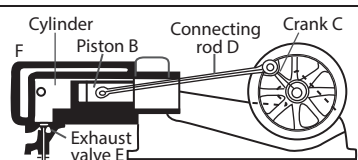
GAS CHROMATOGRAPHY INFRARED (GC-IR): A form of Fourier-transform infrared used to identify small amounts of gas obtained by gas chromatography.

GAS CONSTANT (UNIVERSAL MOLAR GAS CONSTANT):

The constant R , which appears in the general gas equation, $PV = nRT$, which describes the constant of proportionality (R), relating the pressure and volume of a quantity of gas to the absolute temperature, where n is the number of moles and T , the temperature. Some values of R are $8.314 \text{ JK}^{-1} \text{ mol}^{-1}$, $1.985 \text{ Cal K}^{-1} \text{ mol}^{-1}$, $0.082 \text{ Latm K}^{-1} \text{ mol}^{-1}$ and $8.314 \times 10^7 \text{ erg K}^{-1} \text{ mol}^{-1}$.

GAS COOLED REACTOR (GCR): A nuclear reactor in which the cooling medium is carbon dioxide and graphite is the neutron moderator.

GAS ENGINE: An internal combustion engine in which gas such as natural gas (LPG) is used as the fuel. In the diagram, a mixture of natural gas/coal is pumped into the cylinder on the left of the piston B through a valve. The piston is connected by a rod D to the crank C. Ignition is usually done by a spark, with water circulating in the jacket F to prevent the cylinder from becoming too hot.



GAS EXCHANGE: The process by which organisms exchange gases with the environment for the purpose of metabolism. The essential feature of gas exchange is that oxygen is absorbed into the body and carbon dioxide, the waste gas of respiration, is eliminated. An example is the absorption of oxygen into the blood and the removal of carbon dioxide from the blood. It takes place at a respiratory surface, which is a boundary between the external environment and the interior of the organism, which must have a large surface area, a thin permeable surface and a moist exchange surface.

GAS GENERATOR: A combustion chamber used to provide hot gases for a turbine or motor to drive the propellant pumps of a rocket engine or to provide a source of gas at some predetermined pressure. Gas generators are usually operated fuel-rich to maintain the container temperature at reduced values.

GAS GIANT: A large planet composed mainly of hydrogen and helium, together with some methane, ammonia and other light substances, but with a substantial metal-rock core around which the lighter materials that form the bulk of the object have accreted. The four gas giants of the solar system, in order of increasing distance from the Sun are Jupiter, Saturn, Uranus and Neptune.

GAS JAR: A glass jar, typically a plain cylinder that looks like a tube with a broad base and opening, used for collecting and holding gas from experiments.

GAS LASER: A laser in which the active medium is a gas containing a laser active species. It is typically composed of a plasma discharge tube containing a gas that can be excited with an electric discharge.

GAS LAWS (GAS EQUATION): The laws that relate the pressure, volume and temperature of a gas. These relationships have been formulated into what is known as Charles' law, named after J.A.C. Charles (1746-1823). Boyle's law named after Robert Boyle; and Pressure law. They combine to form the General Gas Laws. Charles's law states that, at constant pressure, the volume V of a gas is directly proportional to its absolute (Kelvin) temperature T , or $\frac{V}{T} = k$; and

Boyle's law states that, at constant temperature, the pressure P of a gas varies inversely with its volume V or $PV = k$, where k is a constant. The relationship is used to calculate unknown variables rather than by measurement, i.e. $\frac{PV}{T} = \frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$.

GAS MASK: A device worn over the face to protect the face and to prevent breathing in poisonous gases and fumes. It has a device that filters the air chemically.

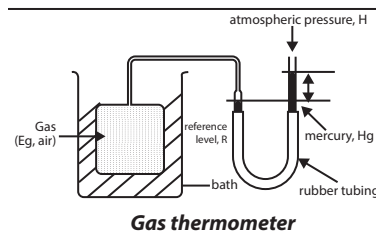
GAS OIL: One of the main fractions of crude oil such as kerosene, diesel fuel, heating oils and light fuel oils boiling within the general range of 232-426 °C (505-1073 K). It is used as fuel oil.

GAS SENSING ELECTRODE: A sensor composed of an indicator and a reference electrode in contact with a thin film of solution which is separated from the bulk of the sample solution by a gas-permeable membrane or an air gap. This intermediate solution interacts with the gaseous species (penetrated through the membrane or an air gap) in such a way as to produce a change in a measured constituent (e.g. the H^+ activity) of the intermediate solution.

GAS SYRINGE (GLASS COLLECTING BOTTLE): A graduated piece of glass apparatus used to measure accurately the volumes of gases and dispense liquids, especially where these liquids need to be kept free from air.

GAS THERMOMETER (CONSTANT-VOLUME GAS THERMOMETER): A

device for measuring temperature in which the working fluid is a gas such as hydrogen or helium. The gas is enclosed in a glass or fused-quartz bulb connected to a mercury manometer. The pressure exerted by a constant volume of a gas is used for the measurement of temperature of the bath in which the bulb containing the gas is immersed.



Temperature t° on this scale is defined as $\frac{t}{100^\circ \text{C}} = \frac{P_t - P_{100}}{P_{100} - P_0}$, where

P_0 , P_{100} and P_t are the pressure exerted by the gas when at ice point, steam point and at temperature t respectively.

GAS TURBINE (COMBUSTION TURBINE): A rotary engine, in which the burning of fuel supplies hot gas to spin a turbine. It has an upstream compressor coupled to a downstream turbine and a combustion chamber in-between. Gas turbines are used in jet engines and some locomotives. Ships also use gas turbines as a power source. It is also used in industry for generating and pumping purposes.

GAS VACUOLE: A structure made up of numerous gas-filled vesicles, many of which may be found in the cells of certain aquatic bacteria and blue-green algae. The density of the cell depends on how much gas is contained within the vesicles. The position of the organism in a column of water can thus be regulated by these structures. The vesicles are unusual in being bound by a protein rather than phosphoglyceride membrane.

GAS-LIQUID CHROMATOGRAPHY: A technique for separating or analysing mixtures of gases, gas compounds or substances that can be vapourised by heat. It comprises all gas-chromatographic methods in which the stationary phase is a liquid dispersed on a solid support. Separation is achieved by partition of the components of a sample between the phases.

GAS-PHASE-GROWN CARBON FIBRES: Carbon fibres grown in an atmosphere of hydrocarbons with the aid of fine particulate solid catalysts such as iron or other transition metals and consisting of graphitisable carbon. Gas-phase-grown carbon fibres transform during graphitisation heat treatment into graphite fibres. These show a very high degree of preferred orientation and are particularly suitable for intercalation treatments.

GAS-SOLID CHROMATOGRAPHY: All gas chromatographic methods in which the stationary phase is an active solid, for example, charcoal and molecular sieves. Separation is achieved by adsorption of the components of a sample. In gas chromatography, the distinction between gas-liquid and gas-solid may be obscure because liquids are used to modify solid stationary phases and because the solid supports for liquid stationary phases affect the chromatographic process.

GASEOUS DIFFUSION SEPARATOR: An instrument used to separate a gas mixture into its components by means of differences in the diffusion rates of the component molecules. This method has been used in separation of various isotopes of uranium (by means of UF_6) and hydrogen (H_2O , HDO , D_2O , HTO , etc.).

GASEOUS EXCHANGE: The process by which oxygen is extracted from inhaled air into the bloodstream and, at the same time, carbon dioxide is eliminated from the blood and exhaled. For exchange to occur, the gases usually have to pass through a liquid boundary layer. The rate of diffusion of gases dissolved in water is much slower than diffusion through air, an important consideration in calculations of gaseous exchange. During photosynthesis, carbon dioxide is taken in and oxygen given off; the reverse occurs in respiration. The passage of water vapour is usually in one direction only (out of the plant).

GASEOUS PROPELLANT: A working substance used in a gaseous-propellant rocket engine. Nitrogen, argon, krypton, dry air and Freon 14 have all been employed in spacecraft.

GASIFICATION: The conversion of a solid or liquid hydrocarbons such as coal, petroleum, biofuel or biomass, into carbon monoxide and hydrogen by reacting the raw material, such as house waste, plastic waste or compost at high temperatures with a controlled amount of oxygen and steam. The resulting gas mixture is a fuel called **synthesis gas** or **syngas**.

GASOHOL: A liquid fuel that is a mixture of 90% petrol and 10% alcohol or 97% petrol and 3% wood alcohol, used in motor vehicles. The alcohol is either ethanol or methanol. The ethanol is produced from high sugar agricultural products such as corn and methanol from coal and organic wastes such as waste-wood products. Gasohol has higher octane or antiknock properties than petrol and burns more slowly and completely, resulting in reduced emissions of some pollutants.

GASOLINE (PETROL): A mixture of hydrocarbons derived from petroleum with a boiling point below $200\text{ }^{\circ}\text{C}$ (573 K). Its main use is as a fuel for internal combustion engines. It is obtained by any of the following means: the fractional distillation of petroleum; condensation or adsorption from natural gas; thermal or catalytic decomposition of petroleum or its fractions; by the hydrogenation of producer gas or coal; or by the polymerisation of hydrocarbons of lower molecular weight. It is composed primarily of the alkanes (paraffins) hexane, heptane and octane, plus smaller amounts of higher-boiling alkanes. Gasoline is a liquid at ambient temperature, but it volatilises readily in air to form a flammable mixture. Gasoline is more volatile than diesel oil, jet fuel or kerosene, not only because of the base constituents, but also because of additives.

GASTRIC JUICE: A strong acidic mixture of inorganic salts, hydrochloric acid and digestive enzymes with pH 1 to 3 in humans produced by glands in the stomach lining, when peptides are detected in the stomach and the hormone gastrin is released into the bloodstream. **Gastric** describes parts and functions of the body related to the stomach.

GASTRIC MILL: A type of digestive tract or gizzard occurring in many crustaceans. It is a muscular tube situated in the anterior region of the stomach and consists of a set of bones that grind food particles.

GASTRIN: A peptide hormone produced by the parietal cells of the stomach, which stimulates the release of gastric acid (HCl) after a meal. It is a polypeptide, occurring in several different molecular forms, the most important of which are molecules of 17 and 34 amino acid residues. It is released by G cells in the antrum of the stomach, duodenum and the pancreas. It binds to cholecystokinin B receptors

to stimulate the release of histamines in enterochromaffin-like cells and it induces the insertion of K^+/H^+ adenosine triphosphate (ATP) ase pumps into the apical membrane of parietal cells.

GASTRODERMIS: The layer of cells lining the digestive tract of certain invertebrates, including the nematode worms and cnidarians.

GASTROENTERITIS: Inflammation of the stomach and the intestines, characterised by abdominal pain, nausea, vomiting and diarrhoea. It may be caused by food poisoning, allergy or infections by viruses, bacteria and parasites or through contact with an infected person. It may be complicated by severe dehydration, especially in infants and people with weakened immune systems as in those with some chronic disorders such as diabetes.

GASTROESOPHAGEAL SPHINCTER (CARDIAC SPHINCTER): A ring of smooth muscle at the junction of the oesophagus and the stomach that remains closed except during swallowing to prevent the stomach contents from entering the oesophagus.

GASTROLITH (GIZZARD STONES): A stone swallowed by animals, some birds, reptiles such as crocodile and certain lizards and found in the gastrointestinal tract. This stone is used to grind food because the animals lack suitable grinding teeth. Gastroliths, which are typically very smooth and polished are also found in dinosaur fossils.

GASTROPODA (GASTROPODS): A taxonomic class of molluscs that includes the snail, whelks, limpets, land/sea slugs and conches. They occupy marine, freshwater and terrestrial habitats.

GASTROVASCULAR CAVITY (ENTERON; COELENTERON): The internal body cavity of the coelenterates such as sea anemones and jellyfish, which has one opening that serves both as mouth and anus.

GASTRULA: The stage in the development of an animal embryo following the blastula, when double-walled sac or cup is formed. The outer wall is called the **ectoderm** and the inner wall is the **endoderm** containing a cavity known as the **primitive gut**.

GATE: 1. An electronic circuit that has one or more inputs with a single output, which is energised only when a certain combination of pulses is present at the inputs. In the transmission gate, the output waveform is a replica of a selected input during a specified interval. In the switching gate, a constant output is obtained for a specified combination of inputs. These gates are the basic components of digital computers. 2. A circuit in which one signal, generally a square wave, serves to switch another signal on and off. 3. One of the electrodes in a field-effect transistor. 4. An output element of a cryotron. 5. An electric waveform in radar which is applied to the control point of a circuit to alter the mode of operation of the circuit at the time when the waveform is applied.

GATE ARRAY (UNCOMMITTED LOGIC ARRAY, ULA): In computer science, it is a type of partially finished integrated circuit in which the logic gates are laid down to a general-purpose design but are not connected to each other. The interconnections can then be set in place according to the requirements of individual manufacturers. Producing ULAs may be cheaper than using a large number of TTL (transistor-transistor logic) chips or commissioning a fully customized chip.

GATEPOST (TURNSTILE): In logic, the symbol, $\vdash$ that represents a binary relation and is used in different contexts, for example, to represent logical consequence when inserted between expressions to form a sequent or when prefixed to a single expression indicating that it is a theorem. For example, the expression $P \vdash Q$ means that Q is derivable from P in the system. Different variants of the symbol are often used to distinguish syntactic and semantic notions.

GATEAUX DERIVATIVE: Named after René Gateaux, a French mathematician, it is the mapping determined by the Gateaux differential when this exists in all directions. It is defined for functions between locally convex topological vector spaces such as Banach spaces.

GATEAUX DIFFERENTIAL: The directional derivative at x with increment h , of a given function f defined on an open domain. If

this limit exists for all h , then the function is said to be Gateaux differentiable at x and, if it is linear in h , the mapping $T = \delta f(x; \cdot)$ is said to be the (linear) Gateaux derivative of f at x , which is often referred to as the gradient of f , written $\Delta f(x)$. It is given by

$$\delta f(x; h) = \lim_{t \rightarrow 0} \left[\frac{f(x + th) - f(x)}{t} \right].$$

GATED ION CHANNEL: A specific ion channel that opens and closes only under defined circumstances to allow the cell to alter its membrane potential. Channels may be either voltage gated, such as the sodium channel of neurons or ligand gated such as the acetylcholine receptor of cholinergic synapses. Channels tend to be relatively ion specific and allow fluxes of typically 1000 ions to pass in around 1ms, they are thus much faster at moving ions across a membrane than transport adenosine triphosphate ATPases.

GATED PHOTO DETECTOR: Where photo detectors are switched on or off usually in a repetitive manner employing electronic switches, they are described as gated.

GATED PHOTOCHROMISM: Special form of photochromism in which one or both forms of a photochromic system are transformed reversibly, either chemically or electrochemically, into a non-photochromic form.

GATEWAY: In computer science, it refers to either hardware or software that acts as a bridge between two networks so that data can be transferred between a number of computers. For example, there is a gateway that allows a connection to take place, when you send an e-mail to a friend or when you log in to a Web site. Often, the connection to a Web site will involve many smaller connections to other servers along the way. In these cases, a number of gateways are used.

GATTERMANN REACTION: A synthesis of an aldehyde from an aromatic or heterocyclic compound, hydrogen cyanide, hydrogen chloride and a catalyst of aluminium chloride or zinc chloride. It is a variation of the Sandmeyer reaction for preparing chloro- or bromoarenes by reaction of the diazonium compound. In the Gattermann reaction, the aromatic amine is added to sodium nitrite and the halogen acid (10 °C), then fresh copper powder (e.g. from $\text{Zn} + \text{CuSO}_4$) is added and the solution warmed. The diazonium salt then forms the haloarene, for example, $\text{C}_6\text{H}_5\text{N}_2 + \text{Cl}^- \rightarrow \text{C}_6\text{H}_5\text{Cl} + \text{N}_2$. The copper acts as a catalyst.

GAUGE: 1. A standard or scale of measurement. 2. The width of a railway or tramway track. 3. A standard dimension, quantity or capacity. 4. An instrument for measuring or testing. Also known as **valuation**, it is a mapping g from an integral domain $E \setminus \{0\}$ into the non-negative integers, such that $g(ab) \geq g(a)$ for all a and b in $E \setminus \{0\}$ and such that for b in E and a in $E \setminus \{0\}$ there exist q and r in E for which $b = qa + r$ and either $r = 0$ or $g(r) < g(a)$. For example, the degree of polynomials over a field is a gauge.

GAUGE BOSON (FIELD PARTICLE; GAUGE PARTICLE): Any of the particles such as the photon and gluons that carry the four fundamental forces of nature.

GAUGE FUNCTION (MINKOWSKI FUNCTION): A non-negative, positively homogeneous, convex function with value zero at zero. Given a convex set C an associated gauge is constructed by setting $g_C(x) = \inf\{t > 0 : x \in tC\}$. An everywhere finite (symmetric) gauge is a norm and there is a duality between gauge and support functions.

GAUGE PRESSURE: Pressure measured relative to ambient atmospheric pressure. At sea level, it is 1 atm less than the pressure in the atmosphere.

GAUGE THEORY: Any of a number of quantum field theories put forward and which are important as the successful field theories explaining the dynamics of elementary particles. A gauge theory involves a symmetry group for the fields and potentials. In the case of electrodynamics, the group is Abelian, whereas the gauge theories for strong and weak interactions use non-Abelian groups.

GAUGE TRANSFORMATION: A small coordinate change that transforms one nearly Cartesian coordinate system into another of the same kind.

GAUSE'S PRINCIPLE (COMPETITIVE EXCLUSION PRINCIPLE; GAUSE'S LAW OF COMPETITIVE EXCLUSION): In biology, the law that states that two species that compete for the exact same resources cannot stably coexist. When the populations of two species compete for the same limited resources, one population will use the resources more efficiently and have a reproductive advantage that will eventually lead to the elimination of the other population. It was derived by G. F. Gause in 1934.

GAUSS: The centimetre-gram-second (cgs) unit of magnetic flux density, which is equal to 10^{-4} weber per square metre. Its symbol is G. The International System of Units (SI) is the currently used system of units, and the tesla (T) has replaced the gauss.

GAUSS METER: A magnetometer or an instrument for measuring direction and intensity of a magnetic field, whose scale is graduated in gauss or kilogauss and usually measures only the intensity and not the direction, of the magnetic field.

GAUSS-MARKOV LEAST SQUARES THEOREM: A mathematical theorem which states that the minimum variance approach to estimation is due to Markov and the least squares method is due to the Carl Gauss. The theorem also holds for multivariate distributions. It is stated as the least squares estimate of β for the model $Y_i = \beta X_i + \epsilon_i$; $E[\epsilon_i] = 0$; $\text{var}(\epsilon_i) = \sigma^2$; $\text{cov}(\epsilon_i, \epsilon_j) = 0$, has uniform minimum variance among all unbiased linear estimates of β .

GAUSS-SEIDEL ITERATION (LIEBMANN METHOD; METHOD OF SUCCESSIVE DISPLACEMENTS): An iterative method similar to the Jacobi iteration, which is used to solve a linear matrix system of equations, $Ax = b$ by repeatedly solving for x_{NEW} from $x_{\text{NEW}(i)} = \left(\frac{1}{a_{ii}} \right) \left[b_i - \sum_{j < i} a_{ij} x_{\text{OLD}(j)} - \sum_{j > i} a_{ij} x_{\text{NEW}(j)} \right]$. This assumes that $a_{ii} \neq 0$ and will not always converge even then.

GAUSS, KARL FRIEDRICH (1777-1855): A German mathematician, astronomer and physicist, who became director of Göttingen Observatory in 1806. One of the greatest mathematicians of all time, Gauss made important contributions to physics and astronomy and pioneered the application of mathematics to gravitation, electricity and magnetism. His collaboration with Wilhelm Weber on electromagnetism led to the invention of an electric telegraph. He worked out the relationship between electric flux and electric field.

GAUSS' HYPERGEOMETRIC EQUATION (HYPERGEOMETRIC DIFFERENTIAL EQUATION): The differential equation, arising in many physical contexts, defined by $x(1-x)y'' + [c - (a+b+1)x]y' - aby = 0$.

GAUSS' LEMMA: The result that if a polynomial with integer coefficients has a factorisation over the rationals, then it has a factorisation over the integers as well.

GAUSS' LAW (GAUSS'S FLUX THEOREM): A law formulated by Carl Friedrich Gauss in 1835 relating the distribution of electric charge to the resulting electric field. It states that the total electric flux normal to a closed surface in an electric field is proportional to the algebraic sum of the electric charges within the surface. A similar law applies to the surfaces, drawn in a magnetic field and the law can be generalised for any vector field through a closed surface. It is one of four of Maxwell's equations which form the basis of classical electrodynamics; the other three being Gauss's law for magnetism, Faraday's law of induction and Ampère's law with Maxwell's correction. Gauss's law can be used to derive Coulomb's law and vice versa.

GAUSS' TEST: The test for the convergence of a power series that if $\left| \frac{u_n}{u_{n+1}} \right| = 1 - \frac{L}{n} + O\left(\frac{L}{n^{1+\epsilon}}\right)$, then u_n converges absolutely if $L > 1$ and diverges or converges conditionally, if $L \leq 1$.

GAUSSIAN CURVATURE: A measure of the curvature at a point of a space surface, which is equal to the product of the principal curvatures, $k_1 k_2$. It has the dimension of $1/\text{length}^2$ and is positive for spheres, negative for one-sheet hyperboloids and zero for planes. It determines whether a surface is locally convex (when it is positive) or locally saddle (when it is negative).

GAUSSIAN DISTRIBUTION (NORMAL DISTRIBUTION; BELL CURVE): A probability distribution, $f(x)$, of a random variable, x that is assumed by many statistical methods. This is given by

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{1}{2} \frac{(x-\mu)^2}{\sigma^2}\right], \text{ where } \mu \text{ is the mean and } \sigma^2 \text{ is the variance.}$$

All normal distributions are symmetric and have bell-shaped density curves with a single peak.

G

GAUSSIAN DOMAIN (UNIQUE FACTORISATION DOMAIN):

An integral domain in which every non-zero non-unit is uniquely representable as a finite product of irreducible elements, up to permutation. Since the Gaussian integers are a Gaussian domain and $5 = (1 + 2i)(1 - 2i)$, 5 is reducible in the Gaussian integers; 3 is irreducible therein and hence prime, because it is in a Euclidean domain.

GAUSSIAN ELIMINATION (GAUSSIAN PIVOTING): A method for solving a set of linear equations in many unknown variables. It is usually carried out by the use of elementary row operations to the augmented matrix.

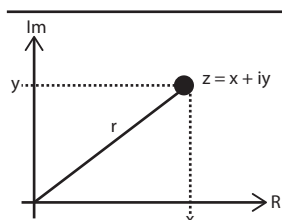
GAUSSIAN FIELD: The field that comprises complex numbers $u + iv$, with u and v being rational.

GAUSSIAN FUNCTION: The function $y = \exp(-x^2)$, whose integral from $-\infty$ to ∞ converges to $\sqrt{\pi}$. In one dimension, the Gaussian function is the probability density function of the Gaussian distribution (normal distribution).

GAUSSIAN INTEGER: A complex number of which the real and imaginary parts are integers, a number of the form $a + ib$, where a and b are integers. They constitute the algebraic integers in the Gaussian field.

GAUSSIAN PLANE (ARGAND DIAGRAM):

A two-dimensional method for representing complex numbers by Cartesian coordinates x, y . Along the x -axis or horizontal axis are plotted the real numbers and along the y -axis or the vertical axis the non-real or imaginary numbers. The number $x + iy$ is represented by the point (x, y) or by the corresponding vector. If the polar coordinates of (x, y) are (r, θ) , r is the modulus and θ the argument of $x + iy$.



An Argand diagram indicating the imaginary axis (y-axis) and real axis (x-axis).

GAUSSIAN RECIPROCITY

(QUADRATIC RECIPROCITY): The number-theoretic result that if p and q are distinct odd prime numbers, p is a quadratic residue of q if and only if q is a quadratic residue of p , unless both are congruent to 3 modulo 4, in which case the opposite holds, in that one case, p is a quadratic residue of q if and only if q is not a quadratic.

GAUSSIAN UNIT: A system of units for the Gaussian system (electric and magnetic quantities based upon electrostatic and electromagnetic units). In this unit, the electric and magnetic constants are both equal to unity. For example, one Gaussian unit of capacitance (statfarad) = $1.112650 \sim \text{pF}$; electric potential (statvolt) = $299.7925 \sim \text{V}$; and electric charge (statcoulomb) = $333.5641 \sim \text{pC}$.

GAY-LUSSAC, JOSEPH LOUIS (1778-1850): French scientist whose discovery of the laws of chemical combination in gases helped

to establish the atomic theory. It also led to Avogadro's law. He showed that all gases expand by the same fraction of their volume for a given temperature increase. This led to the devising of a new temperature scale whose profound thermodynamic significance was later established by Lord Kelvin (1824 - 1907), British physicist.

GAY-LUSSAC'S LAW: The principle or law named after the French chemist, Joseph Louis Gay-Lussac that, when gases combine in a chemical reaction, both reactants and products do so in simple ratios of their volumes and that any gaseous product is also produced in a simple ratio or that when gases react, they do so in volumes which bear simple whole number ratios to one another and to the volume of the product, if temperature and pressure remain constant.

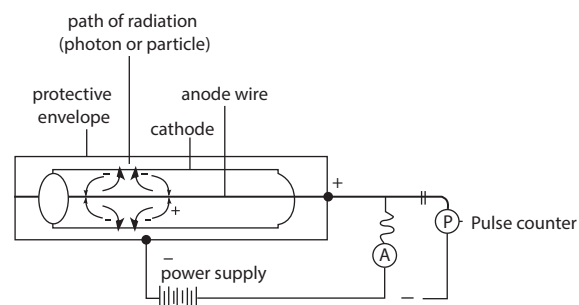
GAYLUSSITE: A translucent, yellowish-white hydrous carbonate mineral, with the chemical formula $\text{Na}_2\text{Ca}(\text{CO}_3)_2 \cdot 5\text{H}_2\text{O}$. It crystallises in the monoclinic system. It is an unstable mineral which dehydrates in dry air and decomposes in water. It is formed as an evaporite from alkali lake waters.

GEAR: A toothed wheel whose function is to transfer turning motion from one shaft to another when the centre distance of the shafts is not too large. They provide a positive drive, maintaining exact velocity ratios between driving and driven shafts, a factor that may be lacking in the case of friction gearing which is subject to slippage.

GECKO: A small lizard, belonging to the family Gekkonidae of the reptilian order Squamata, found especially in the tropics. It has wide feet that can stick to vertical surfaces such as wall. They make chirping sounds in social interactions with other geckos.

GEGENSCHNEIN: A faint elliptical patch of light visible on a moonless night on the ecliptic at a point 180° from the position of the Sun, caused by the reflection of sunlight by meteoric particles. It was discovered by the Danish astronomer, Theodor Brerssen in 1854.

GEIGER COUNTER (GEIGER-MÜLLER COUNTER): A device invented by the German physicist Hans Geiger (1882 - 1945) and used to detect and measure nuclear radiation intensity by counting the number of ionising particles produced. It consists of a sealed metal tube (the cathode) filled with an inert gas such as argon or neon with methane. In the centre of the tube runs a fine-wire anode. A potential difference of about 1000 volts is maintained between the electrodes. Although the tube carries such a high voltage, no electric current passes between the electrode and the walls because the inert gas does not normally conduct electricity. But when a radioactive particle passes through the tube, it makes the gas discharge and suddenly the gas begins to conduct. Due to the discharge of the gas in the tube, a momentary electric current is created between the electrode and the tube which can subsequently be amplified to either make it audible or visible. This current can be measured on a sensitive measuring ammeter or as a pulse by a pulse counter from the collection of each primary particle.



Geiger Counter (Geiger- Muller Counter)

GEIGER-MARSDEN EXPERIMENT (GOLD FOIL EXPERIMENT):

An experiment carried out by Ernest Rutherford's assistants Hans Geiger and Ernest Marsden under his supervision at the Physical Laboratories of the University of Manchester between 1908 and 1913. This led to the discovery of the nuclear (proper) structure of an atom. In the experiment, a beam of high energy alpha particles from a radioactive source was directed at a thin gold foil of approximately 8.6×10^{-6} cm having a circular fluorescent zinc sulfide (ZnS) screen around it. Whenever a particle hit the screen, a tiny flash of light was produced at that point. It was observed that alpha particles elastically scattered from the nuclei of gold atoms. Although the majority of the alpha particles passed through undeflected, some were strongly deflected into hyperbolic paths while others were scattered through almost 180° . This led to the development of the Rutherford model (planetary model) of the atom and eventually to the Bohr model.

GEIGER-NUTTALL LAW: It states that for an alpha-emitting radioactive substance, the logarithm of the decay constant and the logarithm of the range in air of the emitted alpha rays are in linear relation to each other. It is expressed as $\ln \lambda = -a_1 ZE^{-1/2} + a_2$, where λ is the decay constant, Z is the atomic number, E is the binding energy and a_1 and a_2 are constants.

GEIGER, HANS WILHELM (1882-1945): German physicist, who carried out research with British physicist Ernest Rutherford (1871-1937) at Manchester University before returning to Germany in 1912. In 1908, he and Ernest Rutherford produced the Geiger-Muller counter. In 1909, his scattering experiments with alpha particles led to Rutherford's nuclear theory of the atom.

GEISSLER TUBE: A high-voltage discharge tube invented by the German physicist and glass blower Heinrich Geissler in 1857. It was an evacuated glass cylinder with an electrode at each end containing gasses such as neon, argon or air; mercury or other conductive liquids; or ionisable minerals or metals, such as sodium. When a high voltage is applied to the terminals, electrical current flows through the tube. The current will disassociate electrons from the gas molecules, creating ions and when electrons recombine with the ions, different lighting effects are created.

GEITONOGAMY: Transfer of pollen grains from the anther of one flower, to the stigma of another flower on the same plant.

GEL: 1. A colloidal solution, of solid particles in a liquid, with the particles forming less than 10% volume of the gel that gives the suspension some degree of elastic firmness. Examples are sweet fruit jellies and hair styling gel. 2. Also called **gelatine**, it is a coloured transparent sheet attached to the front of a lighting instrument to produce coloured light.

GEL FILTRATION CHROMATOGRAPHY: A type of size exclusion chromatography in which molecules in solution are separated based on their size or hydrodynamic volume, using an aqueous solution to transport the sample through the column. Gel filtration is used mainly to separate polar or water soluble polymers and macromolecules such as proteins, nucleic acids and polysaccharides.

GELATINE: A colourless or pale yellow soluble animal protein substance obtained by boiling collagen with water and evaporating the solution. It is used in various pharmaceutical products, foods, medicine, glue and photography.

GELATINOUS PRECIPITATE: A precipitate that is viscous and jelly-like when formed.

GELDING: The castrated male of the horse family. Castration eliminates hormonally driven behaviour, making an animal calmer and better-behaved.

GELIGNITE (BLASTING GLYCERIN; GELATIN; NITROGELATIN):

A high explosive made from a mixture of jellylike mass of nitroglycerine, gun-cotton, sodium nitrate and wood pulp. It is used in quarries and mining.

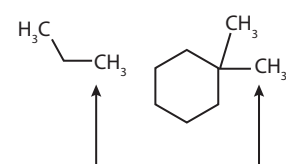
GELL-MAN, MURRAY (1929 -): US theoretical physicist, who entered Yale University at 15 and earned his Ph.D. from the Massachusetts Institute of Technology in 1951. He held a professorship at the California Institute of Technology, becoming Millikan Professor of Theoretical Physics in 1967. In 1955, he proposed the property of strangeness for certain fundamental particles, a quantum property that accounted for decay patterns of certain mesons. In 1964, he and Yuval Ne'eman (1925-2006) proposed the eightfold way to define the structure of particles. This led to Gell-Mann's postulate of the quark. In 1969, he was awarded the Nobel Prize in Physics.

GELFAND TRANSFORM: Named after Russian mathematician, Israil Moiseyevich Gelfand, it is the mapping that associates a continuous function on the compact space, Δ , of all maximal ideals in the induced weak topology, to an element of a commutative Banach algebra, A . Its formula is $\hat{x}(h) = h(x)$.

GELFOND-SCHNEIDER THEOREM: Named after Russian mathematician, Alexander Gelfond (1906-68) and Theodor Schneider (1911-1988), German mathematician. It is the theorem that for complex algebraic numbers α and β with α not 0 or 1 and β irrational, α^β is transcendental. Thus $\sqrt{2}^{i\sqrt{2}}$ and $e^\pi = (-1)^i$ are transcendental.

GEM: 1. A mineral or other material that is valuable by virtue of its durability, rarity and beauty, cut and polished for ornamental use. Examples are organic substances such as pearl, amber, coral and jet and inorganic substances of variable composition, such as natural glass. 2. Molecules in which two functional groups are attached to the same atom.

GEM-DIMETHYL GROUP: Two methyl groups on the same carbon atom.



GEMINATE ION PAIR: Ion pair formed from a precursor that constitutes a single kinetic entity, by electron transfer or ion transfer in an encounter complex.

GEMINATE PAIR: A pair of molecular entities in close proximity in solution within a solvent cage and resulting from reaction, for example, bond scission, electron transfer and group transfer of a precursor that constitutes a single kinetic entity.

GEMINATE RECOMBINATION: A term used to describe the reaction of two transient species produced from a common precursor in solution, with each other. If reaction occurs before any separation by diffusion has occurred, this is termed **primary geminate recombination**. If the mutually reactive entities have been separated and come together by diffusion, this is termed **secondary geminate recombination**.

GEMINI: A prominent zodiacal constellation in the Northern Hemisphere lying between Cancer and Taurus represented as the twins Castor and Pollux.

GEMMA: An asexual unicellular or multicellular outgrowth of a plant or animal which develops into a new organism as in certain mosses and liverworts.

GEMMATION (GEMMA): A type of vegetative propagation occurring in certain multicellular animals, such as the cnidarians, sponges, mosses, liverworts and tunicates, in which small clumps of undifferentiated cells develop on the surface of a plant, which later separate to develop into a new organism identical with the parent.

GENBANK: The genetic sequence database of the US National Institutes of Health. It was started in 1982 and now contains nearly 70 million annotated sequence entries and over 70 billion base pairs.

GENDER DIFFERENCES: The typical differences between men and women that are not anatomical or biological but are based on social factors that vary from culture to culture.

GENE: A unit of heredity in a living organism, which are chromosomes consisting of lengths of DNA which controls the hereditary characteristics of organisms.

GENE AMPLIFICATION: A technique by which selected region of DNA from a single cell can be replicated naturally or technologically until there is a sufficient amount to be used by the cell or for analysis. For example, the genes that code for proteins involved in ribosomes are amplified early in the process of cell development so that there are sufficient numbers of them to assemble the cell.

GENE BANK: A facility, where a collection of seeds or other forms of genetic material, such as tubers, spores, bacteria or yeast cultures, live animals and plants, frozen sperm and eggs or frozen embryos are available, which researchers can use to develop new plant varieties with desired characteristics such as higher yield and better resistance to disease and weather.

GENE CENTRE (CENTRE OF DIVERSITY): An area that has a high degree of genetic variation for a particular species or genus of plants that can also be the centre of origin for that species. Gene centres are typically found in mountainous regions. Some gene centres, the primary centres, are believed to correspond to the region where a particular crop originated. For example, numerous forms of wheat exist in the Middle East and, since many of the wild relatives of wheat are also found in this area, it is believed the crop was first domesticated there. Other centres, the secondary centres, do not contain wild relatives and are believed simply to be areas where the crops have been cultivated for a considerable period. The recognition of gene centres is important in conservation and plant breeding work, as such areas are important reservoirs of natural genetic variability.

GENE CLONING (DNA CLONING): The process of isolating and obtaining the exact copies of a particular gene or DNA sequence using genetic engineering techniques. Cloning is frequently employed to amplify DNA fragments containing genes, but it can be used to amplify any DNA sequence such as promoters, non-coding sequences, chemically synthesised oligonucleotides and randomly fragmented DNA.

GENE CONVERSION: The irreversible change from a normal (wild-type) allele to a mutant allele or vice versa, as a result of the transfer of genetic material from one homologous chromosome to another, thereby converting a normal (wild-type) allele to a mutant allele or vice versa. Loss of wild-type alleles due to gene conversion is implicated in the initiation of certain hereditary diseases and cancers in humans.

GENE EXPRESSION: The process by which information from a gene is used in the synthesis of functional structures operating in a cell. These structures are often proteins, but in non-protein coding genes such as rRNA genes or tRNA genes, the product is a functional RNA. Gene expression is the most fundamental level at which genotype gives rise to the phenotype.

GENE FAMILY (MULTIGENE FAMILY): A group of related similar sequence and function genes that have arisen by duplication of an ancestral gene. Such genes show similarities of nucleotide sequence that reveal their common origin and if they have evolved recently they occur close together on the same chromosome.

GENE FLOW: The exchange of genetic material by interbreeding between populations of the same species or between individuals within a population. Gene flow increases variability within the population and makes possible new combinations of traits. In humans, gene flow usually comes about through human migration.

GENE IMPRINTING: The differential expression of a single gene according to its parental origin. An example is the symptoms of

Huntington's disease. Imprinted genes are either expressed only from the allele inherited from the mother such as H19 or CDKN1C or in other instances from the allele inherited from the father, such as IGF-2.

GENE LIBRARY (GENOMIC LIBRARY): A collection of cloned DNA fragments created by recombinant DNA representing the entire genetic materials of an organism, which enables screening and isolation of any particular gene.

GENE MANUPULATION (GENETIC ENGINEERING; BIOTECHNOLOGY; RECOMBINANT DNA TECHNOLOGY): The use of *in vitro* techniques to produce DNA molecules containing novel combinations of genes or altered sequences and the insertion of these into vectors that can be used for their incorporation into host organisms or cells in which they are capable of continued propagation of the modified genes. For example, gene manipulation can modify bacterial cells to synthesise completely new substances such as hormones, vaccines, monoclonal and antibodies or introduce novel traits into plants or animals.

GENE MUTATION: A change in the DNA sequence within a gene or chromosome of an organism resulting in the creation of a new character or trait not found in the parental type that involves a change in a single base or base pair of the nucleotides in a gene. A gene in which such a change has occurred is known as a mutant gene or allele. Gene mutations can be inherited from a parent or acquired during a person's lifetime. Mutations that are passed from parent to child are called **hereditary mutations** or **germline mutations** because they are present in the egg and sperm cells, which are also called germ cells. This type of mutation is present throughout a person's life in virtually every cell in the body. The DNA sequence can be altered in several ways; for example by insertion, substitution, deletion and inversion. **Point mutation** is a single nucleotide base change in DNA or RNA molecule. Point mutations result in a misreading of the genetic code during the translation phase of protein synthesis and usually change the order of amino acids point mutation making up a protein, which may or may not affect the function of that protein. For example, a point mutation is the cause of sickle cell disease.

GENE POOL: The complete set of unique alleles that is present in a population of a species, which may be used to help determine gene frequencies or the ratio between different types of genes in a population. Gene pool gives an idea of the number of genes, the variety of genes and the type of genes existing in a population. A large gene pool indicates high genetic diversity, increased chances of biological fitness and survival. A small gene pool indicates low genetic diversity, reduced chances of acquiring biological fitness and increased possibility of extinction. Gene pool increases when mutation occurs and survives. Gene pool decreases when the population size is significantly reduced, for example, famine and genetic disease. Some of the consequences of small gene pool are low fertility and increased probability of acquiring genetic diseases and deformities.

GENE PROBE (DNA PROBE): A single-stranded DNA or RNA fragment marked by a chemical or radioactive substance that binds to a given gene. It is used in genetic engineering to search for a particular gene or other DNA sequence.

GENE REPLACEMENT THERAPY (GRT): A hypothetical treatment for hereditary diseases in which affected cells from a sufferer could be removed from the body, the DNA repaired in a laboratory and the functioning cells reintroduced.

GENE SHEARS: In genetic engineering, they are pieces of messenger RNA that can catalyse chemical reactions by binding to other pieces of messenger RNA, recognising specific sequences and cutting them at that point to lose their ability to convey genetic information. If a piece of DNA which codes for the shears can be inserted in the chromosomes of a plant or animal cell, that cell will then destroy all messenger RNA of a particular type. Genetic shears may be used to protect plants against viruses, which infect them and cause disease. They might also be useful against AIDS.

GENE SPLICING (GENE-SPLICING): The sequences of cutting and rejoining of DNA of an organism or the process of inserting a foreign gene into laboratory cultures of bacteria to generate commercial biological products. Gene splicing is often used in industry to allow single-celled organisms to produce useful products, such as human insulin. It is also used in the production of genetically modified organisms.

GENE THERAPY: The use of genetic engineering techniques to alter or replace defective genes. There are many techniques of gene therapy, all of them still in experimental stages. The two basic methods are called *in vivo* and *ex vivo* gene therapy. The **in vivo method** inserts genetically altered genes directly into the patient. The *ex vivo* method removes tissue from the patient, extracts the cells in question and genetically alters them before returning them to the patient.

GENE TRACKING: A method for determining the inheritance of a particular gene in a family. It is used in the diagnoses of genetic diseases such as cystic fibrosis or Huntington's Chorea.

GENERAL: Of a statement, theorem, etc., not specifying an individual subject but quantifying over a domain. Existential as well as universal statements are general in this sense. Usually, the domain is non-trivial, but in formal contexts this may not be so.

GENERAL CIRCULATION (ATMOSPHERIC CIRCULATION): In atmospheric science, it is the large-scale movement of air together with ocean circulation by which heat is distributed on the surface of the Earth. It is caused by temperature differences in the atmosphere and the rotation of the Earth, which transfers heat from warm zones (the tropics) to cooler zones (toward the poles). **GENERAL FORMULA:** A formula which shows the relative numbers of different atoms in terms of the variable n for all the members of a particular family of compounds.

GENERAL INDUCTION (COMPLETE INDUCTION; SECOND-KIND INDUCTION): A mathematical induction in which the step is from all integers less than n to all less than $n + 1$, rather than from the single integer n to $n + 1$.

GENERAL LINEAR GROUP (FULL LINEAR GROUP): The group, often written $GL(V)$, of all linear transformations of a finite-dimensional vector space, V , that are invertible. Where V is over a field F , it is defined as the group of all non-singular $n \times n$ matrices over F , written $GL(n, F)$.

GENERAL PROTECTION FAULT, GPF (GENERAL PROTECTION ERROR): An error caused when an application program such as Microsoft Word or the Netscape Web browser tries to access storage memory assigned to another application. The only way for a user to deal with this error is to exit the application or reboot the system.

GENERAL RECOMBINATION (HOMOLOGOUS RECOMBINATION): Recombination that occurs between two similar or identical molecules of DNA. It is most widely used by cells to accurately repair harmful breaks that occur on both strands of DNA, known as double-strand breaks. Homologous recombination also produces new combinations of DNA sequences during meiosis, the process by which eukaryotes make gamete cells, like sperm and egg cells in animals.

GENERAL SOLUTION: A solution of partial differential equation of n^{th} -order and of n parameters such that any numerical values assigned to the parameters yields a solution to the equation.

GENERAL THEORY OF RELATIVITY (GENERAL RELATIVITY): The geometric theory of gravitation published by Albert Einstein in 1916. It is the current description of gravitation in modern physics. General relativity generalises special relativity and Newton's law of universal gravitation, providing a unified description of gravity as a geometric property of space and time or spacetime. In this theory the interactions of bodies, which had been ascribed to gravitational forces, are explained as the influence of bodies on the geometry of space-time. In particular, the curvature of spacetime is directly related to the four-momentum (mass-energy and linear momentum)

of whatever matter and radiation are present. The relation is specified by the Einstein field equations, a system of partial differential equations. Many predictions of general relativity differ significantly from those of classical physics, especially concerning the passage of time, the geometry of space, the motion of bodies in free fall and the propagation of light. Examples of such differences include gravitational time dilation, the gravitational redshift of light and the gravitational time delay. General relativity's predictions have been confirmed in all observations and experiments to date. Although general relativity is not the only relativistic theory of gravity, it is the simplest theory that is consistent with experimental data. However, unanswered questions remain, the most fundamental being how general relativity can be reconciled with the laws of quantum physics to produce a complete and self-consistent theory of quantum gravity.

GENERALISATION (GENERALIZATION): 1. Making a theory or hypothesis that what one observes in a special single case applies more widely to form an essential basis of all valid deductive inferences. For example, after studying how one acid reacts with a metal, one draws the conclusion that all acids react with the metal that way. Generalisations postulate the existence of a domain or set of elements, as well as one or more common characteristics shared by those elements. 2. The extension of a number of results to form a rule. For example, the computations $3 + 5 = 5 + 3$ and $1.5 + 2.7 = 2.7 + 1.5$ could be generalised to $a + b = b + a$. 3. In logic, **universal generalisation** is any statement ascribing a property to every member of a class. **Existential generalisation** is any statement ascribing a property to one or more members of a class. 4. The act or process of perceiving similarity or relation between different stimuli, as between words, colours, sounds, lights, concepts or feelings. 5. **Stimulus generalisation** is the act or process of responding to a new stimulus by an organism similar to but distinct from the conditioned stimulus. 6. **Response generalisation** is the act or process of an organism making a different but similar response to the same stimulus. 7. **Mediated generalisation** is the act or process of responding to a stimulus not physically similar to the conditioned stimulus and not previously encountered in conditioning.

GENERALISED CONTINUUM HYPOTHESIS: In mathematics, the assertion that for any infinite cardinal the next greater cardinal is that of its power set.

GENERALISED COORDINATE (LAGRANGIAN COORDINATES): A set of variables used to specify the position and orientation of a system, in principle defined in terms of Cartesian coordinates of the system's particles and of the time in some convenient manner. The number of such coordinates equals the number of degrees of freedom of the system.

GENERALISED EIGENVALUE PROBLEM: The problem of finding scalars λ and vectors x solving $Ax = \lambda Bx$ where A and B are given matrices or linear operators and B is positive definite. In the classical eigenvalue problem, B is the identity matrix or operator.

GENERALISED FORCE: The generalised force corresponding to a generalised coordinate is the ratio of the virtual work done in an infinitesimal virtual displacement, which alters that coordinate and no other, to the change in the coordinate.

GENERALISED INVERSE (PSEUDO-INVERSE): In mathematics, any of several generalisations of the inverse of a matrix or a bounded linear operator with closed range on Hilbert space, often denoted A^+ for a given matrix or operator A . It is a linear operator that coincides with the inverse for an invertible operator. The most commonly encountered pseudoinverse is the Moore-Penrose matrix inverse, which is a special case of a general type of pseudoinverse known as a matrix 1-inverse.

GENERALISED LINEAR MODELS: They are models that arise from generalisation of ordinary linear models. Some functions of the expected value of the response variable is modelled as a linear combination of the explanatory variables, $x_1, x_2, \dots, x_n$ thus $f(E(y)) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n$, where f is the link function.

GENERALISED MAXIMUM LIKELIHOOD RATIO TEST

STATISTIC: The ratio of the maximum likelihood of drawing a given sample under a given hypothesis to the maximum likelihood of drawing that sample under the hypothesis that the given hypothesis is false. The term is also sometimes used for the inverse of this ratio.

GENERALISED NILPOTENT: An element x in a Banach algebra such that $\lim_{n \rightarrow \infty} \|x^n\|^{\frac{1}{n}} = 0$. The set of such elements is called the radical of the algebra.

GENERALISED POLYNOMIAL: A function of the form $\sum_{i=1}^n c_i p_i$ for fixed continuous functions P_i and arbitrary scalars.

GENERALISED MEAN (HÖLDER MEAN; POWER MEAN): In mathematics, a family of functions for aggregating sets of numbers that include as special cases the Pythagorean means (arithmetic, geometric and harmonic means).

GENERALISED RATIO TEST (D'ALEMBERT'S RATIO TEST): The test for the convergence of an infinite series by testing whether there is a $k < 1$ such that the ratio of the absolute values of a term to its predecessor is less than k for all terms after some term. If this ratio is always greater than unity, then the series diverges, allowing the limit to be placed in the preceding limit by an appropriate limit inferior or limit superior. The ratio test is given by $\frac{u_{k+1}}{u_k} = p$, if $p < 1$ the series converges, the series diverges when $p > 1$ and inconclusive if $p = 1$.

GENERALISED T-DISTRIBUTION (GT-DISTRIBUTION): A probability distribution, $f(x)$ which is related to Student's t-distribution

$$\text{and is given by; } f(x) = \frac{p}{2\sigma q^p \beta(p^{-1}, q)} \left[1 + \frac{1 \times 1}{(q\sigma p)} \right]^{q+p-1},$$

where β is the beta function. In $f(x)$, σ is a scale parameter while p and q control the shape of the distribution. Large values of p and q are associated with lighter tails for the distribution. When $p = 2$ and $\sigma = \sqrt{2a}$, this becomes Student's t-distribution with 2 degrees of freedom.

GENERATE: 1. In mathematics, to provide a precise criterion for membership in a set, in the form of an algorithm whose application recursively yields all and only the members of the set. For example, the formation rules of a language generate all and only its well-formed expressions; and the basis elements in a vector space generate the space. 2. Of a subset of a structure such as a ring, group or module, to enable all elements of the group to be constructed by recursive application of the operations defined on the structure to the members of the subset; thus the structure is contained in the closure of the set of its generators under these operations. For example, the set of all transpositions generates the symmetric group and the set of all cycles generates the alternating group. A finitely generated structure is the closure of a finite set of generators. Any basis of a vector space generates the space.

GENERATE CONIC: A conic section obtained when intersecting plane includes the vertex of the cone. A degenerate conic can be two lines that may or may not be parallel, a single line (actually two coinciding lines), a single point or the null set (no points). Equivalently, the degenerate conics have equations that, when regarded as quadratic forms in x and y equated to zero, have singular matrices.

GENETIC CODE: The information on DNA, which determines how sequences of triple nucleotides, called codons, specify the amino acid sequence of proteins, i.e., which protein to synthesise in living cells. The genetic code is very similar in all organisms.

GENERATING FUNCTION: In mathematics, a generating function $f(x)$ is a formal power series for a sequence, $f(x) = \sum_{n=0}^{\infty} a_n x^n$, whose coefficients are the given sequence $\{a_0, a_1, \dots\}$. This is often used in

the study of the sequence by analytic techniques in both combinatorics and analysis. For example, the generating function for the Fibonacci numbers is $\frac{x + x^2}{1 - (x + x^2)}$.

GENERATING SET: In mathematics, a set whose elements generate a given algebraic structure. In other words, a smaller set of objects can be used to generate larger objects (the **generated set**) by doing a set of operations.

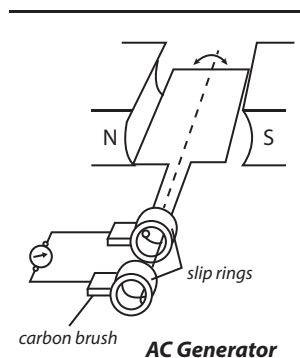
GENERATION: 1. The members of a family, a group of animals that have similar ages, all the people of about the same age within a society or within a particular family; or a stage or degree in a succession of natural descent such as a grandfather, a father and the father's son comprising three generations. 2. In technology, it is the stages of successive improvement in the development of a technology such as the internal combustion engine or successive iterations of products with planned obsolescence, such as video game consoles or mobile phones. 3. The act of producing offspring, also known as **procreation**. 4. In physics, it is one stage of a chain reaction involving the splitting of one atom. It takes 10^{21} generations in 10^{-9} seconds to have an atomic bomb.

GENERATION TIME: 1. The interval between the beginnings of consecutive cell divisions, which may be as short as 20 minutes in bacteria or the time interval required for a population to double. 2. The mean time required for a neutron arising from fission to produce a new fission.

GENERATIVE CELL: One of the cells found in the pollen tube of seed plants. In gymnosperms, it gives rise to the body cell and stalk cell whereas in angiosperms it gives rise directly to the two male gametes or generative nuclei.

GENERATIVE NUCLEUS: One of the two male gametes in the pollen tube of angiosperms producing two sperm nuclei by mitosis. On entering the nucellus and embryo sac they participate in the fertilisation process. One fuses with the egg nucleus to form the zygote and the other usually fuses with the polar nuclei or definitive nucleus to form the primary endosperm nucleus.

GENERATOR: 1. Any machine that converts mechanical energy into electrical energy. Generators usually work on Faraday's law of electromagnetic induction that by rotating a conductor in a magnetic field, an emf is induced, which creates a flow of current. There are two main ways to induce emf: moving a wire in a magnetic field; or moving a magnet in a coil of wire. A simple generator consists of a magnetic field produced by North and South poles of a strong horseshoe type magnet, a coil of wire made of a large number of turns of insulated copper wire called the armature and two copper slip rings of a commutator to which the two ends of the coil are connected. Two carbon brushes press lightly against the two half rings. When the coil is rotated, the two slip rings touch the two carbon brushes one by one and the current produced in the rotating coil is collected through the commutator slip rings and into the carbon brushes. From the carbon brushes current can be supplied to various electrical appliances such as radio, television and electric bulb. The mechanical power for an electric generator is usually obtained from a rotating shaft and is equal to the shaft torque multiplied by the rotational or angular, velocity (speed). The mechanical power may come from various sources such as turbines powered by water, wind, steam or gas; or diesel and petrol engines. Most of the generators that supply power for domestic, commercial or industrial uses generate alternating current (AC), which reverses polarity at a fixed frequency, usually 50 or 60 cycles per second (50



or 60 Hz). These generators are known as synchronous generators or alternators. Generation of AC power enables the use of transformers to convert electrical power at any voltage and current it is generated to high voltage and low current for long-distance transmission and then stepping it down to a low voltage suitable for domestic, commercial or industrial uses. 2. In mathematics, also called **generatrix**, it is a point, line or plane whose motion along a certain path called the directrix, sweeps out a given geometric figure. For example, a line moving in a circle traces out a cylinder. Also, it is one of a set of elements that together generate a group. 3. An element of a cyclic group whose powers exhaust the group.

GENERATOR POTENTIAL: The local charge that develops in the sensitive part of a receptor cell, known as the generator region. It is caused by depolarisation of the membrane surrounding this part of cells.

GENERIC: 1. In biology, relating to a genus, as all the members of a genus, class, group or kind. 2. Of a product, such as drugs or consumer products like milk, having no brand name or registered trademark, as in generic drugs. 3. In mathematics, applicable or referring to an entire class of objects; general. In logic, for example, a generic set is a general set of conditions, which is consistent and is also such that every sentence from a specified set of sentences is decided. Also, properties that hold for "typical" examples are called generic properties. For example, a generic property of a class of functions is one that is true of "almost all" of those functions, as in the statement; "A generic polynomial does not have a root at zero." 4. In mathematics, of a set in a complete metric space, expressible as a dense G-delta set and so (in the sense of Baire category) a residual set.

GENETIC ALGORITHM: A search heuristic that mimics the process of natural selection based on the mechanisms of Darwinian evolution. This heuristic helps in the generation of solutions to optimisation and search problems. It uses random mutation, crossover and selection procedures to breed better models or solutions from an originally random starting population or sample.

GENETIC CODE (GENE CODE): A sequence of nucleotides in DNA and RNA that determines the amino acid sequence of proteins forming the basis of heredity through their role in protein synthesis.

GENETIC CONTROL: 1. The control of gene expression. Genetic control may be on the transcriptional or translational level. Transcriptional control works by controlling the number of RNA transcripts of a region of DNA, indirectly controlling protein synthesis. Translational control of protein synthesis works by regulating the step of translating RNA into protein. 2. Pest control using resistant varieties of crops.

GENETIC COUNSELLING: An educational counselling process for helping people who have a genetic disease or may be at risk for such a disease to facilitate informed decision-making. It includes the following: (i) interpretation of family and medical histories of any genetic disorders for individuals who are at risk of developing them or concerned that they may pass them on to their children; (ii) education about inheritance, testing, management, prevention, resources and research; and (iii) counselling of individuals at risk of developing a genetic disorder in adult life to promote informed choices and adaptation to the risk or condition, for example, by the removal of the ovaries in women at high risk of developing ovarian cancer.

GENETIC DISEASE (GENETIC DISORDER): Any disorder or illness caused at least partly by defective genes or chromosomes. In humans, there are some 3,000 genetic diseases. Genetic diseases can be classified as: single-gene defects also called a Mendelian disorder, chromosomal disorders and multifactorial.

GENETIC DRIFT (SEWALL WRIGHT EFFECT; ALLELIC DRIFT): The random change in the frequency of alleles in a population over successive generations due to random sampling and chance. Each new generation differs from its parent generation in alleles due to

random variation in the distribution of gametes. The alleles in offspring are a random sample of those in the parents and chance has a role in determining whether a given individual survives and reproduces.

GENETIC EFFECTS: In respect to radiation, they are effects, mainly mutations present in the offspring of those exposed to or produced by the absorption of ionising radiation's. Currently, these effects are additive and irreversible.

GENETIC ENGINEERING (RECOMBINANT DNA TECHNOLOGY; GENE MANUPULATION; BIOTECHNOLOGY): The development of techniques for the artificial manipulation, modification and recombination of DNA or other nucleic-acid molecules in order to modify the biological processes of an organism or population of organisms. For example, genetic engineering can modify bacterial cells to synthesise completely new substances such as hormones, vaccines, monoclonal and antibodies or introduce novel traits into plants or animals.

GENETIC EROSION: The process in which an already limited gene pool of an endangered species of plant or animal diminishes even more when individuals from the surviving population die off without getting a chance to meet and breed with others in their endangered low population. The situation is particularly serious in gene centres, where valuable genes for such characters as disease resistance may be lost.

GENETIC FINGER PRINTING (DNA FINGER PRINTING; DNA TYPING; DNA PROFILING; DNA TESTING; DNA TYPING): A technique employed by forensic scientists to assist in the identification of individuals on the basis of their respective DNA profiles. The individual's DNA is analysed, usually using small samples of body tissues such as hair, blood, semen or vaginal secretions to reveal the pattern of repetition of particular short nucleotide sequences called variable number tandem repeats, throughout a particular genome. This pattern is usually very unique to the individual concerned. It has, therefore, found widespread application for identification purposes in forensic science and paternity disputes, as well as in veterinary science. Although 99.9% of human DNA sequences are the same in every person, the small percentage, which is different, can distinguish one individual from another.

GENETIC INHERITANCE (HEREDITARY): The passing of genetic information in the form of DNA from parents to their offspring.

GENETIC ISOLATION: The absence of genetic exchange between populations or species. This may be caused by geographic separation or premating or postmating mechanisms such behavioural, anatomical or physiological that prevent reproduction. This may lead to speciation or formation of new species and endangering species to extinction due to a small gene pool. Examples of genetic isolation are the ecological community that are secluded away from others (such as Tristan da Cunha or Pitcairn Islands), crossing of plants which produce sterile offspring and human influence on genetic isolates through restricted breeding of dogs.

GENETIC LOAD: The average number of unfavourable recessive gene mutations per individual in a population, which is the reduction in the mean fitness for a population compared to the maximum fitness.

GENETIC MAP (GENE MAP; LINKAGE MAP): The arrangement of genes or other identifiable sequences on a DNA molecule or chromosome arrayed in linear order with distances between loci expressed as percent recombination (map units, centimorgans). Genes on the same chromosome are described as **linked** or **syntenic**. The genes on a chromosome can be represented as a single linear structure that goes from one end of the chromosome to the other. Humans have 24 linkage groups, corresponding to the 22 autosomes, plus X and Y chromosomes. Groups of genes that are widely separated on a chromosome may show independent assortment; however, all such groups can eventually be tied together by mapping additional loci between them.

GENETIC MATERIAL: The nuclear (chromosomal) and cytoplasmic (mitochondrial and chloroplast) material that plays a fundamental role

in determining the nature of all cell substances, cell structures and cell effects. The genes have properties of self-propagation and variation. Two different kinds of genetic material exist: deoxyribonucleic acid (DNA) and ribonucleic acid (RNA). Most organisms are made of DNA, but a few viruses have RNA as their genetic material.

GENETIC MODIFICATION: The alteration and recombination of genetic material under laboratory conditions, resulting in trans-genetic organisms or change in the genes in a living organism, as occurs in nature and which has been used for many years in selective breeding.

GENETIC PAN: In soils, a natural subsurface soil layer of low or very low permeability having a high concentration of small particles and differing in certain physical and chemical properties from the soil immediately above or below the pan.

GENETIC RECOMBINATION: Formation of new combinations of alleles in offspring (viruses, cells or organisms) as a result of exchange of DNA sequences between molecules. It occurs naturally, as in sexual reproduction or experimentally, as a result of genetic engineering. This process involves pairing between complementary strands of two parental duplex or double-stranded DNAs and results from a physical exchange of chromosome material.

GENETIC RESOURCES: The diversity and availability of alleles in both natural and artificially maintained stocks of organisms. A broad genetic base increases the capacity for variation and successful adaptation, such as resistance to pathogens or hostile environments. Both national and international programmes exist to maintain genetic resources of agriculturally important organisms. This has led to the establishment of a number of gene banks or seed banks, where potentially valuable seeds are evaluated, multiplied and stored under ideal conditions.

GENETIC SCREENING: The process by which the genome of human or other organisms is analysed for genetic markers that indicate the presence of particular genes, especially one that causes certain diseases. Prospective parents or an embryo or foetus is tested when a specific genetic disorder is suspected. Genetic screening begins with a complete medical history of both parents.

GENETIC SPIRAL (DEVELOPMENTAL SPIRAL): A hypothetical spiral formed by drawing a line through and joining the centres of successive leaf primordia at a shoot apex. The form of the spiral depends on the angle of divergence and the distance between successive primordia. The first is often 137.5° and the second depends on the rate of vertical growth of the apex. It is often found that this distance increases geometrically.

GENETIC VARIATION: Deviation from the genotype in structure, form, physiology or behaviour as a result of genetic mutation. Gene variation is important to the process of natural selection. Natural selection is the result of the interactions between genetic variations in a population and the environment. The environment determines which variations are more favourable. More favourable traits are thereby passed on to the population as a whole. The genetic variations that arise in a population happen by chance, but the process of natural selection does not.

GENETICALLY MODIFIED ORGANISMS, GMOs (GENETICALLY ENGINEERED ORGANISM, GEO): Organisms whose genetic material incorporate and express gene from another species. This is achieved by the insertion of a modified gene or a gene from another organism using the techniques of genetic engineering. Genomes of various organisms, from single celled microbes to multicellular plants and animals, can be purposely transformed using recombinant DNA technology. GMOs are used mainly in the production of pharmaceuticals, gene therapy and agriculture. However, there are ethical issues concerning their use and production. The limitation of the knowledge of modern science as well as the possibility of unforeseen negative effects in the health and the environment are some of the issues being raised up against their use.

GENETICIST: A scientist who studies and practises the science of heredity and variation of organisms. A geneticist can work in

academics, in industry, in medicine or in research. Geneticists are involved in identifying genes responsible for basic biological traits, including genes that cause disease or mediate a response to a medication. Geneticists also determine how those genes function in organisms, including plants, animals and humans. They may treat individuals with a particular genetic disorder or counsel families regarding the genetics of human disease. Geneticists are also involved in identifying genetic mechanisms to improve agricultural processes, such as breeding pest-resistant fruits and vegetables. They are also interested in the distribution of genetic variations in populations and how those variations arise.

GENETICS: The branch of biology concerned with heredity and variation in living organisms, which is the transmission of characteristics from parents to offspring via the genes in the chromosomes. This is based on the work of Austrian monk and biologist Gregor Mendel (1822-1884) whose experiments with cross-breeding (hybridisation) of peas to conclude that the heritable units (now called genes) were not blends of parental traits but rather were separate physical entities passed individually from one generation to the next.

GENICULATE: Having a joint or abrupt knee-like bends.

GENITAL HERPES (HERPES GENITALIS): A highly contagious sexually transmitted disease characterised by painful blisters in the genital area and caused by the herpes simplex virus type 2 (HSV type 2). The virus can also be transmitted from mother to newborn through the birth canal. The other type of the herpes virus is simplex virus type 2 (HSV type 2), also known as **human herpesvirus 1 (HHV-1)** causes common inflammation, the cold sore and fever blisters in the mouth and around it.

GENITALS (GENITALIA): The external parts of an animal's reproductive organs used to facilitate internal fertilisation. For example, the human male genitals are the penis and the scrotum and the female genitals are the clitoris and vulva.

GENITOURINARY SYSTEM (URINOGENITAL SYSTEM): The organ system of the reproductive organs and the urinary system. In mammals, the urinary system structures include kidneys, ureters, bladder, and urethra. The major structures of the reproductive system in males are the testes, sperm ducts, urethra, and penis; in females, they are the ovaries, fallopian tubes, uterus, and vagina.

GENOME: The genetic composition of an organism, consisting of all of the genes carried on chromosomes. Humans have 22 pairs of autosomes and 2 sex chromosomes and require 46 separate chromosome sequences in order to represent the completed genome.

GENOME PROJECT (HUMAN GENOME PROJECT): An international scientific undertaking that seeks to understand the entire genetic blueprint of a human being. The bacterium *Haemophilus influenzae* was the first organism to be sequenced. The first eukaryotic organism to be sequenced was that of the budding yeast, *Saccharomyces cerevisiae*.

GENOMIC IMPRINTING: The parental effect on gene expression. The offspring produced depends on the source of the chromosome containing the imprinted gene, (whether it is from the female or from the male parent). Identical alleles may have different effects on offspring, depending on whether they arrive in the zygote via the ovum or via the sperm. An imprinted gene is one in which the DNA is methylated, causing gene expression to be suppressed.

GENOMIC LIBRARY: A collection of DNA segments from a genome, derived from restriction fragments that have been cloned in vectors and includes all or part of the genetic material of an organism.

GENOMICS: A branch of genetics concerned with the study of genomes of organisms. It is primarily concerned with determining the entire sequence and structure of an organism's DNA and then to determine how that DNA is arranged into genes.

GENOTOXICITY: The extent to which a chemical or other agent damages cellular DNA, resulting in mutations or cancer.

GENOTYPE: The genetic composition of an organism, a cell or an individual that code for a trait, as opposed to its physical

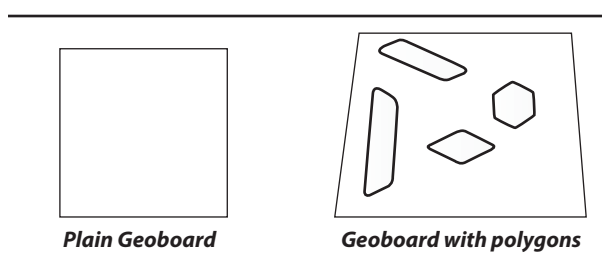
characteristics phenotype. The term was coined in 1909 by German scientist Wilhelm Johannsen.

GENOTYPE FREQUENCY: The percentage or proportion of individuals in a population that possess a specific genotype.

GENUS: 1. A category or low-level taxonomic rank below family or subfamily and above species used in the classification of living and fossil organisms that consists of a number of similar or closely related species. A genus includes fewer members than a family and more members than a species. For example, lions, leopards and tigers belong to the genus *Panthera*, while domestic cats and relatives belong to the genus *Felis*. 2. In topology, also called **geometric genus**, it is the number of holes in a geometric shape. It is the largest number of nonintersecting simple closed curves that can be drawn on the surface without separating it. Topological shapes are classified according to how many holes they have. Shapes with no holes at all such as spheres, eggs, and bowls are regarded as genus 0; torus, a wedding band, a pipe, or anything with a looped handle (a teacup) with one hole is a genus 1 shape; the figure 8 with two holes is a genus 2 shape. A torus and a teacup are regarded as topologically equivalent because both are genus 1 (they have one hole each). 3. A class of non-equivalent primitive binary quadratic forms with given discriminant, each form representing the same integers. 4. The least natural number m such that an entire function has a Weierstrass product expansion.

GEO-TARGETING (GEOTARGETING): An Internet-based advertising method that markets advertisements for a products or services directly to a website visitor based on his/her geographical location such as country, city, state and others derived from the website visitor's IP address and other personal information that is provided by the visitor or others as well as the particular service or product needs for that region. Its main aim is to improve the cost-effectiveness of marketing programs. For example, a website visitor's geographical location can be derived from IP address, as the first three digits of an IP address corresponds to a country code, while the succeeding digits often refer to specific areas within that domain.

GEOBOARD: A mathematical manipulative often used to explore basic concepts in plane geometry such as perimeter, area or the characteristics of triangles and other polygons. It consists of a flat board into which a certain number of nails have been half driven in a regular rectangular pattern. These nails represent the lattice points in the plane on which rubber bands are placed to illustrate shapes and their properties.



GEOCENTRIC UNIVERSE (GEOCENTRISM; GEOCENTRICISM; GEOCENTRIC MODEL): A theological, rather than scientific, view of the universe in which the Earth is regarded as being at its centre and other objects go around it.

GEOCHEMICAL ANALYSIS: A chemical technique that investigates the chemical compounds that make up the Earth, its atmosphere and its seas. It involves taking soil and rock samples at regular intervals from the surface of a site and its surroundings for chemical analysis. The process is useful in predicting the location of petroleum, metals, water and commercially valuable minerals.

GEOCHEMISTRY: The science of chemistry as it applies to geology. It deals with the relative and absolute abundances of the chemical elements in the Earth and their isotopes, examination of the distribution and movements of elements in different parts of the Earth, distribution of elements and their isotopes in the cosmos, study

of the role of processes and compounds that are derived from living or once-living organisms, etc.

GEOCHRONOLOGY: The branch of geology that deals with the dating of the Earth by studying its rocks and the fossils it contained.

GEODE: In geology, a subspherical cavity which occurs in certain sedimentary and volcanic rocks into which crystals have grown from the outer wall into the centre. It is formed when mineral-rich water entering a cavity in a rock undergoes a sudden change in pressure or temperature, causing crystals such as quartz, calcite and other minerals to form from the solution and line the cavity's walls.

GEODESIC: Also called geodetic, it is relating to or involving the geometry of curved surfaces. **Geodesic curve** is the shortest curve between two points on a curved surface lying wholly on the surface. On a sphere, a geodesic is an arc of a great circle through the two given points. This arc is unique unless the two points are antipodal. **Geodesic line** is the shortest distance between two points on a curved surface or curves whose tangent vectors remain parallel if they are transported along it.

GEODESY: The science concerned with surveying and mapping the Earth's surface to determine, for example, its exact size, shape and gravitational field. The information obtained is used in civil engineering, boundary demarcations, navigation, resource management and exploration and geophysical studies of the dynamics of the Earth.

GEODETIC EFFECT (GEODETIC PRECESSION; DE SITTER PRECESSION; DE SITTER EFFECT): The effect, predicted by the general theory of relativity in which a mass distorts spacetime. The geodetic effect represents the effect of the curvature of spacetime, predicted by general relativity, on a vector carried along with an orbiting body. For example, the vector could be the angular momentum of a gyroscope orbiting the Earth, as carried out by the Gravity Probe B experiment. The geodetic effect was first predicted by Willem de Sitter, Dutch astronomer and mathematician (1872-1934) in 1916, who provided relativistic corrections to the Earth-Moon system's motion. De Sitter's work was extended in 1918 by Jan Schouten, Jan Arnoldus Schouten (1883-1971) Dutch mathematicians and in 1920 by Adriaan Fokker (1887-1972), Dutch physicist.

GEODYNAMICS: The study of the processes leading to motions of the Earth resulting in deformation of planetary mantle and crust and the related earthquakes and volcanism that shapes the structure of the Earth and other planets.

GEOENGINEERING: Projects designed to stabilise the climate system by using scientific processes to improve the environment, in order to counteract the effects of global warming.

GEOGRAPHIC INFORMATION SYSTEM (GIS): A computer application that enables a user to create, display, store, interpret or manipulate geographic information in many ways. GIS finds applications in many fields and can be used for scientific investigations, resource management and development planning.

GEOGRAPHICAL ISOLATION (ALLOPATRY): The separation of a population of the same species into two geographical regions or by a physical barrier such as a mountain or a body of water. It is a key factor in speciation, the formation of new species.

GEOGRAPHICAL RANGE: The geographic area in which a population lives. Among organisms, abiotic factors such as climate can create physiological barriers to dispersal. Also, biotic factors such as predation and competition can cause limitations to a species geographic range.

GEOGRAPHY: The science that studies the Earth's surface, its topography, climate and physical conditions and how these factors affect people and society. Its two fundamental branches are systematic and regional geography. Systematic geography deals with individual physical and cultural elements of the Earth. Regional geography is concerned with various areas of the Earth, particularly the unique combinations of physical and cultural features that characterise each

region and distinguish one region from another. **Geographical** describes anything relating to geography or to the geography of a specific region.

GEOID: An imaginary shape of the Earth defined by mean sea level and its imagined continuation under the continents at the same level of gravitational potential.

GEOLOGIC COLUMN: A graphic representation of the layers of rock that make up the Earth's crust based on the relative ages of the rocks and in which the oldest rocks are at the bottom.

GEOLOGICAL TIME (GEOLOGICAL TIME SCALE): The history of the Earth since it formed thousands of millions of years ago that reflects a consistent sequence of historical periods, grouped into four eras: Precambrian, Paleozoic, Mesozoic and Cenozoic. It is estimated to be about 4.6 billion years ago to today.

GEOLOGIST: A person who studies the structure of the Earth or another planet, especially its rocks, soil and minerals and its history and origins.

GEOLOGY: The scientific study of the origin, structure, physical properties, history and composition of the Earth. Geology is commonly divided into sub disciplines concerned with the chemical makeup of the Earth. It includes **mineralogy**, the study of minerals; **petrology** is concerned with rocks; **structural geology** is concerned with the structure of the Earth; **volcanology** is concerned with volcanic phenomena; **geomorphology** is concerned with landforms and the processes that produce them; and **glaciology** is concerned with the scientific study of glaciers and their effects on the landscape; **palaeontology** is the scientific study of geologic history, including the study of fossils; **stratigraphy** is the development of sedimentary strata; **astrogeology** is concerned with the evolution of planetary bodies and their satellites; and **economic geology** and its various branches, such as mining geology and petroleum geology. Some major fields closely allied to geology are geodesy, geophysics, geochemistry and environmental geology.

GEOMAGNETIC FIELD: The magnetic field observed in and around the Earth. It has two poles, the northern and southern magnetic poles on the Earth's surface. Away from the surface, the field becomes distorted. The intensity of the magnetic field at Earth's surface is approximately 32 microteslas (0.32 gauss) at the equator and 62 microteslas (0.62 gauss) at the North Pole.

GEOMAGNETISM: The science concerned with the magnetic field observed near the Earth's surface, within the Earth and extending upward to the magnetospheric boundary. For example, if a bar magnet is suspended at any point on the Earth's surface so that it can move freely in all planes, the north seeking end of the magnet will everlastingly come to rest in a broadly northerly direction.

GEOMETER (GEOMETRICIAN): A specialist or one who studies geometry.

GEOMETRIC (GEOMETRICAL): 1. Of, relating to, using the methods of or according to the principles of geometry. 2. Representing, consisting of, formed by or characterised by abstract mathematical points, lines, surfaces, etc., rather than actual physical shapes. A geometric figure is usually taken to be a diagram, but mathematics is concerned only with its abstract and not with its physical properties, so that diagrams are not themselves geometric figures properly so called, but merely representations of them.

GEOMETRIC ATTENUATION: The reduction of a radiation quantity due to the effect only of the distance between the point of interest and the source, excluding the effect of any matter present.

GEOMETRIC DISTRIBUTION: The distribution of the number of independent Bernoulli trials before a successful result is obtained. In probability theory and statistics, the geometric distribution is either of two discrete probability distributions: the probability distribution of the number X of Bernoulli trials needed to get one success, supported on the set $\{0, 1, 2, 3, \dots\}$; and the probability distribution of the

number $Y = X - 1$ of failures before the first success, supported on the set $\{0, 1, 2, 3, \dots\}$.

GEOMETRIC FORM OF THE HAHN-BANACH THEOREM (SEPARATION THEOREM OF MAZUR): The theorem that in n -dimensional Euclidean geometry, disjoint convex sets lie entirely on opposite sides of some closed hyperplane. This requires one of the sets to have a non-empty topological interior.

GEOMETRIC ISOMERISM (CIS/TRANS ISOMERISM): Isomerism in which the atoms in molecules are connected in the same order but in different spatial arrangement that cannot be inter-converted without breaking a chemical bond. Groups attached to each atom may be on the same side of the bond or on opposite sides. When two particular atoms are adjacent to each other on the same side of the structure formula, the term *cis* is used. When the atoms are on the opposite side of the structure formula, the term *trans* is used.

GEOMETRIC MEAN: The average of a set of n values described mathematically as the n^{th} root of the product of the positive numbers. For example, the geometric mean of two numbers, say 2 and 8, is the square root of their product expressed as $\sqrt{2 \times 8} = 4$.

GEOMETRIC OPTICS: The branch of physics that deals with light and vision, especially the generation, propagation and detection of electromagnetic radiation having wavelengths greater than x-rays and shorter than microwaves.

GEOMETRIC PROGRESSION (GEOMETRIC SEQUENCES): A sequence of numbers called terms in which each term is obtained by multiplying the previous term by a constant called a common ratio denoted by r . If a is the first term, the next or second term is ar , the third term is ar^2 , etc. The terms (number of terms, n) of a geometric sequence or progression are given generally as: $a, ar, ar^2, ar^3, \dots, ar^{n-1}$. For example, 1, 2, 4, 8 has a common ratio of 2. In general, a geometric series can be written as $a + ar + ar^2 + \dots + ar^{n-1}$ and the sum of n terms is given by $S_n = \frac{a(1-r^n)}{1-r}$ for $r < 1$ and $S_n = \frac{a(r^n-1)}{r-1}$, $r > 1$. It is also called an **exponential sequence**, with the following examples: $1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \dots$ or exponential series, e.g., $1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{16} + \dots$

GEOMETRIC SEQUENCE: A sequence of numbers such that each term after the first term can be obtained from the previous term by multiplying by a constant term.

GEOMETRIC SERIES: A series whose terms form a geometric progression, for example, $1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots$. This converges only if the absolute ratio of successive terms is less than 1. The sum of finite initial segments of the series is $a + ar + ar^2 + \dots + ar^{n-1} = \frac{a(r^n-1)}{r-1}$, so that if the infinite series converges, its sum is $\frac{a}{(1-r)}$.

GEOMETRIC SOLID: 1. The bounding surface of a three-dimensional portion of space, which includes width, depth and height. An example is the kind of space we live in. 2. **Solid Geometry** is the branch of mathematics that deals with three-dimensional figures and surfaces.

GEOMETRIC STANDARD DEVIATION: It measures the deviation of observations around geometric mean. Given the set $x_1, x_2, \dots, x_n$ for n observations, the geometric standard deviation is given by $\log \sigma_g = \frac{1}{n} \sum_{i=1}^n (\log x_i - \log G)^2$, where $G = \sqrt[n]{x_1 \cdot x_2 \cdot \dots \cdot x_n}$ is the geometric mean observations and σ_g is the geometric standard deviation.

GEOMETRIZE: To use or apply geometric methods to something; or to represent in geometric form.

GEOMETRIZED UNITS (GEOMETRIZED UNIT SYSTEM; GEOMETRIC UNIT SYSTEM): A system of natural units, used

mainly in general relativity, in which all quantities that have dimensions involving length, mass and time are given dimensions of a power of length only.

GEOMETRY: The branch of mathematics concerned with the properties of space. It deals with size, shape, relative position of figures of plane (two dimensional) and solid (three dimensional) figures. The term also applies to some specific mathematical system or axiomatisation of these properties and relations, such as Euclidean and Riemannian geometries.

GEOMETRY FACTOR: The average solid angle in steradians at source subtended by the aperture or sensitive volume of the detector, divided by 4π .

GEOMETRY TEMPLATE: A piece of clear plastic with cut-out shapes that include a millimetre ruler, half-circle and full-circle protractors, a percent circle, pattern-block shapes, and other geometric figures that is used in mathematics.

GEOMORPHOLOGY: The branch of geology that deals with topographic features or the nature and origin of surface landforms such as mountains, valleys, plains and plateaux. A specialist in geomorphology is called geomorphologist.

GEO-NEUTRINOS: Neutrinos that are emitted as a result of radioactivity inside the Earth. Small numbers of geo-neutrinos have been detected in experiments.

GEOPHYSICS: The geological science, which is a major discipline of the Earth sciences and a sub discipline of physics in which the principle of mathematics and physics are applied to the study of the Earth's crust and interior.

GEOPHYTE: A plant life form in Raunkiaer's system of classification. They are perennial herbaceous plants in which the perennating buds are below ground, giving rise to bulbs, corms or rhizomes.

GEOPOTENTIAL: The potential energy of a unit mass relative to sea level. It is numerically equal to the work that would be done in lifting the unit mass from sea level to the height at which the mass is located, against the force of gravity.

GEOSPACE (SOLAR-TERRESTRIAL ENVIRONMENT): The domain of Sun-Earth interactions, consisting of the particles, fields and radiation environment from the Sun to Earth's space plasma environment and upper atmosphere. Geospace is considered to be the fourth physical geosphere, after solid earth, oceans and atmosphere.

GEOSTATIONARY ORBIT (GEOSTATIONARY EARTH ORBIT- GEO): A geosynchronous orbit directly above the Earth's equator (0° latitude), from west to east at an altitude of approximately 35,900 kilometres and at a speed matching that of Earth's rotation, thus remaining stationary in relation to Earth. Communication satellites and weather satellites are often given geostationary orbits, so that the satellite antennas that communicate with them do not have to move to track them, but can be pointed permanently at the position in the sky where they stay. Due to the constant 0° latitude and circularity of geostationary orbits, satellites in GEO differ in location by longitude only.

GEOSTATISTICS: A branch of applied statistics that focuses on mathematical description and analysis of geological observations. It describes spatial relationships among sample data and to apply this analysis to the prediction of spatial and temporal phenomena. It is used in application of problems in many fields such as geology, hydrology, petroleum geology, hydrogeology, meteorology, oceanography, geochemistry, geomettallurgy, geography, forestry, environmental control, landscape ecology, soil science and agriculture.

GEOSPHERE: The parts of Earth that are solid. It includes the rocks, minerals and ground as well as the mountains on the surface, the liquid rock in the mantle below and the minerals and metals of the outer and inner cores.

GEOSYNCHRONOUS ORBIT (GSO): An orbit around the Earth with an orbital period that matches the Earth's sidereal rotation

period (23 hours, 56 minutes, 4 seconds). The synchronisation of rotation and orbital period means that for an observer on the surface of the Earth, the satellite appears to constantly hover over the same meridian (north-south line) on the surface, moving in a slow oscillation alternately north and south with a period of one day, so it returns to exactly the same place in the sky at exactly the same time each day.

GEOSYNCLINE: A large basic or syncline (concave fold in the rock strata) which has become filled with sediment and whose floor has subsided, so that it contains vast thickness of sedimentary rock.

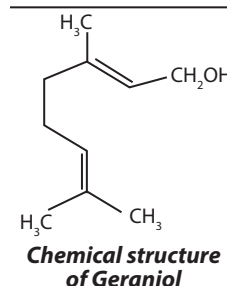
GEOTAXIS: The directional stimulus of a cell or a microorganism in response to gravity. The direction of movement can be either positive or negative. Geotaxis is observed on surfaces (especially inclines), in water, air, sand or mud. The most pronounced examples of geotaxis are found in invertebrates that live in sand or mud. For example, certain cnidarian larvae that swim towards the seabed exhibit positive geotaxis.

GEOHERMAL ENERGY (GEOHERMAL POWER): The heat energy stored and extracted from the core of the Earth or sometimes escaping from the core of the Earth in a volcano. Sending water down boreholes can release steam energy to run a power station and provide electricity. This geothermal energy originates from the original formation of the planet, from radioactive decay of minerals and from solar energy absorbed at the surface.

GEOTROPISM (GRAVITROPISM): The growth of plant organs in response to gravity. An example is the downward growth of plant roots. Plant shoots grow away from gravity (negative geotropism), but most roots are positively geotropic.

GERADE: Describing a molecular orbital of a homonuclear diatomic molecule with even parity. This means that during the process of inversion, the sign of the orbital is unchanged upon going from any point in the molecule through the centre of inversion to the other side.

GERANIOL (3,7-DIMETHYLOCTA-2,6-DIEN-1-OL): An alcohol, with the chemical formula $C_{15}H_{26}O$, density 0.889 g/cm^3 , melting point 15°C (288 K) and boiling point (229°C , 502 K). It is present in a number of essential oils such as rose oil, palmarosa oil and citronella oil and also in small quantities in geranium and lemon. It is a clear to pale-yellow oil, with a rose-like odour and insoluble in water, but soluble in most common organic solvents. It is used in perfumes; and also in flavours such as peach, raspberry, grapefruit, red apple, plum, lime orange, lemon, watermelon, pineapple and blueberry.



GERBER TEST: A volumetric test to determine the quantity of fat in milk. Milk fat is separated from proteins by adding sulfuric acid. The separation is facilitated by using amyl alcohol and centrifugation. The fat content is read directly via a special calibrated butyrometer. It is the European counterpart of the Babcock test used primarily in the United States, which is a fast and easy way to determine the richness of milk by measuring butterfat content.

GERGONNE POINT: Named after the French projective geometer, Joseph Gergonne (1771 - 1859), it is the point at which the lines from the vertices of a triangle to the points of contact of the opposite sides meet with the inscribed circle.

GERM: 1. A microorganism that causes disease. Examples are certain bacteria and viruses. 2. Anything, such as fertilised egg, a spore or the embryo of a seed which is capable of developing into a complete organism or growing into a complete adult or part. 3. Embryo wheat plant inside the kernel.

GERM CELL: Any cell in the series of cells that eventually produce gametes of organisms that reproduce sexually originating near the gut and migrating to the developing gonads in many animals. They

undergo cell division of two types, mitosis and meiosis, followed by cellular differentiation into mature gametes, either eggs or sperm.

GERM LAYER: The layers of cells in an animal embryo at the gastrula stage, which appear early in the development of animals and from which are derived the various organs of the animal's body. The three kinds of germ layers are the ectoderm or outer skin, the endoderm or inner skin and the mesoderm or middle skin.

GERM LINE: Inherited material that comes from the ova (eggs) or sperm and used by sexually reproducing organisms to pass on genes to offspring.

GERM-LINE GENE THERAPY: A medical procedure in which genes are introduced into reproductive cells or embryos to correct inherited genetic defects that can cause disease.

GERMPLASM: The cytoplasm of a germ cell, especially that part containing the chromosomes that is transmitted from one generation to another. In other words, it is the collection of diverse genetic resources such as seeds, a leaf, a piece of stem, pollen or tissues that are available to be used in research, for example, for the development of improved breeding lines and varieties.

GERM PORE: A thin-walled area on a spore or pollen-grain wall through which the germ tube may emerge on germination. In the pollen grain, the germ pore is an area where the exine is either thinner or absent. Pollen grains may be identified to some extent by the shape and number of the germ pores.

GERM TUBE: The filament that emerges on germination of a spore. The structure that emerges from a pollen grain is termed a pollen tube.

GERMAN MEASLES (RUBELLA): A mild viral disease, usually contracted in childhood causing skin rash, lymph node enlargement (swollen glands) and, especially in adults it causes joint pain. It is caused by the rubella virus (Rubivirus). It is spread through contact with fluid droplets expelled from the nose or throat of an infected person.

GERMAN SILVER (NICKEL SILVER; ARGENTANN; ALPACCA): An alloy of copper, zinc and nickel, containing 60% copper, 20% nickel and 20% zinc. It is used in making wire and cutlery.

GERMANIUM: A grey-white semi-metallic element in Group 14 (IVA) of the Periodic Table, with the symbol Ge, atomic number 32, atomic weight 72.59, melting point 937.4 °C (1210 K) and boiling point 2830 °C (3103 K). It occurs in some silver ores. It is an important semiconductor, used for making solid-state diodes and transistors. Its oxide is used in optical instruments and infra-red cameras.

GERMICIDE (DISINFECTANT): A chemical substance capable of killing many different types of microorganisms or a substance which is used to reduce or prevent the growth of microorganisms. Germicides are used both in the home to deal with minor cuts and grazes and in hospitals during the treatment of wounds and in surgery.

GERMINAL EPITHELIUM: A layer of epithelial cells on the surface of ovary/testis that continuously divides by mitosis and meiosis to form secondary oocytes and spermatogonia respectively.

GERMINATION: The beginning of the growth of a seed, spore or other reproductive body, usually after a period of dormancy resulting from the imbibition of moisture and a high enough temperature. If carefully controlled, mass germination of cereal seeds can supply enzymes for the making of alcoholic beverages and for other industries.

GERSCHGOREN CIRCLE THEOREM: The result that all the eigenvalues of a matrix $\{a_{ij}\}$ lie within the circles centred at a_{ii} and with radii $R_i = \sum_{j=1}^n a_{ij}$. In addition, the union of any k of these circles,

if disjoint from the remaining circles, contains exactly k characteristic values counting multiplicity.

GESTALT: A concept of a unified whole that is greater than or different from the sum of its parts; that is, a complete structure whose nature is not explained simply by analysing the constituent

elements. In other words, the parts are so integrated together that they are perceived as one object. A chair, for example, will generally be recognised as a chair despite great variations between individual chairs in such attributes as size, shape and colour.

GESTATION: The length of time between the fertilisation of an ovum and the birth of the offspring. The carrying of an embryo or foetus inside a female viviparous animal, called pregnancy. The gestation period of humans is 9 months. The elephant's gestation period is 22 months, the longest of any land animal. Depending on the species of whale the average gestation period is between 10 and 18 months.

GETTER: A reactive material with a strong affinity for other specified substances. It may be used to remove the traces of gas from vacuum systems.

GEV: A unit of energy, used primarily in high-energy physics, which is equal to 10^9 electronvolts or approximately 11.602×10^{-10} joule.

GEYSER: A natural spring that intermittently discharges an explosive column of stream of water accompanied by a vapour phase into the air due to the build-up of steam in underground chambers.

GHERKIN: A vine, *Cucumis anguria* or its prickly mature fruits similar in form and nutritional value to a cucumber. The immature fruits are widely used for pickling.

GHOST: 1. A faint permanent image that can be caused by displaying the same picture on a monitor for too long a time. 2. Acronym for **General Hardware-Oriented System Transfer**, it is a discontinued disk cloning and backup tool.

GHRELIN: A peptide hormone that is mainly secreted by cells in the stomach, lining the fundus and epsilon cells of the pancreas. It stimulates hunger by stimulating the release of neuropeptide Y, a potent stimulant of appetite, from the arcuate nucleus in the hypothalamus. Ghrelin levels increase before meals and decrease after meals. It also stimulates the release of growth hormone by binding to receptors called growth hormone secretagogue in the anterior pituitary. Its role in regulating the body's energy reserve has prompted interest as a potential target for anti-obesity treatments.

GIANT COVALENT STRUCTURE (GIANT MOLECULAR STRUCTURE; MACROMOLECULAR STRUCTURE): A solid structure made up of many similar atoms joined together by covalent bonds. For example, graphite consists of flat layers of carbon atoms joined by covalent bonds, where each layer is a giant molecule; and diamond consists of carbon atoms joined by covalent bonds which are all the same length, making it the hardest naturally occurring substance. Although graphite has a similar structure as that of diamond, it has longer and weaker bonds between layers of carbon atoms allowing the layers to slide and making graphite not as hard as diamond. Another substance having a giant covalent structure similar to that of diamond is silicon dioxide (sand), but has oxygen atoms sandwiched between pairs of silicon atoms. Substances having giant covalent structures do not conduct electricity as they have no free electrons as solids and no free ions when molten. They are some of the hardest structures known because of the strong bonds between their atoms. These bonds also give them high melting and boiling points.

GIANT IONIC STRUCTURE: A structure of a substance in which a large number of oppositely-charged ions attract each other to form an ionic lattice. Because the force of attraction is spread throughout the lattice, it makes these substances have high melting and boiling points. Examples of substances having giant ionic structure are sodium chloride and magnesium oxide. These substances dissolve in water because the water molecules carry small electrical charges which attract the charged ions.

GIANT MAGNETORESISTANCE: A very large decrease in resistance, about 10 to 20% that occurs in thin films consisting of alternating layers of ferromagnetic and nonmagnetic metals when subjected to a magnetic field. Giant magneto resistance compares with a maximum sensitivity of a few percent for other types of magnetic sensors. It was discovered by Albert Fert and Peter Grünberg, who were

awarded the 2007 Nobel Prize in Physics. The effect is observed as a significant change in the electrical resistance depending on whether the magnetisation of adjacent ferromagnetic layers is in a parallel or an antiparallel alignment. The overall resistance is relatively low for parallel alignment and relatively high for antiparallel alignment. GMR is used commercially by hard disk drive manufacturers for devices such as MP3 Player.

GIANT MOLECULAR CLOUDS (GMC): A large and relatively dense cloud of cold gas and dust in interstellar space from which new stars are born, several hundreds of light-years across, consisting of molecular hydrogen (H_2) gas. There are as many as 5000 in the Milky Way and two of them occur in the constellation Orion.

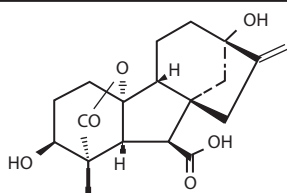
GIANT MOLECULAR STRUCTURE (GIANT COVALENT STRUCTURE; MACROMOLECULAR STRUCTURE): A solid structure of covalent substances such as diamond, graphite and silicon dioxide (silicon(IV) oxide), made up of a huge number of similar atoms joined by covalent bonds in one dimension (long chains, such as polymers); two dimensions found in flat sheets as in graphite, where each layer is a giant molecule; or in three dimensions as found in diamond and silica. Giant molecules do not conduct electricity, as they have no electrons as solids and no free ions when molten. They are some of the hardest structures known because of the strong bonds between their atoms. These bonds also give them high melting and boiling points. Graphite, an allotrope of carbon, is an anomaly here as free electrons between its layers enable it to conduct electricity.

GIANT PLANET: A planet that is much larger and more massive than Earth, characterised by large mass, low density and an extensive atmosphere and not primarily composed of rock or other solid matter. They are Jupiter, Saturn, Uranus and Neptune,

GIANT STAR: A very large star having radii between 10 and 100 solar radii and luminosities between 10 and 1,000 times that of the Sun. Examples are Alcyone (η Tauri), a blue-white (B-type) giant; Thuban (α Draconis), a white (A-type) giant; σ Octantis, a yellow-white (F-type) giant and Arcturus (α Bootes), a red K-type giant.

GIANT STRUCTURE: Atoms or ions having large numbers of particles present in a crystal lattice or a substance that contains particles arranged in a regular repeating pattern. For example, diamond is a giant structure of carbon atoms, in which each carbon atom is joined to four other carbon atoms, forming a giant covalent structure, hence making diamond very hard with a high melting point; sugar crystal is a giant structure composed of sucrose molecules arranged in a repeating pattern that extends in all three dimensions; and metals are giant structures composed of closely packed atoms held together by metallic bonds rather than covalent bonds, hence are malleable and conduct electricity because the electrons are not held tightly by the atoms, but flow throughout the piece of metal. Another example of a giant structure is **giant ionic structure** in which a large number of oppositely-charged ions attract each other to form an ionic lattice. Because the force of attraction is spread throughout the lattice, it makes these substances have high melting and boiling points. Examples of substances having giant ionic structure are sodium chloride and magnesium oxide. These substances dissolve in water because the water molecules carry small electrical charges which attract the charged ions.

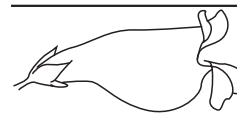
GIBBERELLINS: A class of plant hormones that stimulates growth and seed germination or any of a group of plant growth substance chemically related to terpenes and occurring naturally in plants and fungi. Gibberellins are named in order of discovery, for example GA_1 , GA_2 , GA_3 ... GA_n . Gibberellic acid, which was the first gibberellin to be structurally characterised is GA_3 . There are currently 136 GAs identified from plants, fungi and bacteria. They promote the elongation of stems and assist in the



Chemical structure of Gibberellins

mobilisation of food reserves in germinating seeds and are influential in inducing flowering and fruit development. Gibberellins are classified on the basis of structure as well as function. All gibberellins are derived from the ent-gibberellane skeleton.

GIBBOUS: 1. A term used to describe a structure, which is swollen or bulging on one side. The calyx of the narrow-leaved vetch (*Vicia angustifolia*) is an example of a gibbous structure. 2. A term used to describe the Moon or a planet, denoting that it is more than half but less than fully illuminated.



Gibbous calyx

GIBBS ENERGY DIAGRAM: A diagram showing the relative standard Gibbs energies of reactants, transition states, reaction intermediates and products, in the same sequence as they occur in a chemical reaction.

GIBBS FREE ENERGY (GIBBS FUNCTION): Denoted by G , it is a measure of the energy that would be liberated or absorbed during a reversible process or a measure of the capacity of a system to do work, such as the likelihood of a particular chemical reaction to form products. It is expressed as $G=H-TS$, where H is heat content, T is thermodynamic temperature and S is entropy.

GIBBS PARADOX: The paradox in which there is an increase in entropy when two separate volumes of gases of the same kind, at the same temperature and pressure, are mixed.

GIBBS PHENOMENON: Named after the American theoretical physicist and chemist Josiah Willard Gibbs (1839-1903), it is the necessary behaviour of partial sums of a Fourier series near a jump discontinuity of a function of bounded variation; since convergence is not uniform the curves of the partial sums tend to approximate vertical segments longer than the jump, by a proportion of precisely

$$\frac{2}{\pi} \int_0^{\pi} \frac{\sin x}{x} dx = 1.17898...$$

GIBBS SURFACE (GIBBS DIVIDING SURFACE): A geometrical surface chosen parallel to the interface and used to define the volumes of the bulk phases in the calculation of the extent of adsorption and of other surface excess properties.

GIBBS-DUHEM EQUATION: An equation describing the relationship between the chemical potentials of species in a mixture named after Josiah Gibbs and Pierre Duhem.

GIBBS, JOSIAH WILLARD (1839-1903): US mathematician and physicist, who became the first person to earn an engineering doctorate from Yale University, where he taught from 1871 until his death. During the 1870s, he developed the theory of chemical thermodynamics, devising functions such as Gibbs free energy. He also derived the phase rule. In mathematics, he introduced vector notation.

GIBBS' PHASE RULE: A rule stating that the number of degrees of freedom in a material system at equilibrium is equal to the number of components minus the number of phases plus the constant 2. It is expressed by the equation: $F = C - P + 2$, where C is the number of chemically distinguishable components present, P is the number of phases in equilibrium and F is the number of degrees of freedom of the system, i.e. the number of independently variable intensive properties that must be specified in order to describe completely the state of the system. For example, the system of water vapour, liquid water and solid ice has zero degrees of freedom because the three phases of vapour, liquid and solid coexist in the one component, water.

GIBBSITE: One of the mineral forms of hydrated aluminium hydroxide ($Al(OH)_3$), with the chemical formula $Al(OH)_3 \cdot 3H_2O$. It was named after US mineralogist George Gibbs (1776 - 1833). Bauxite and laterite are sources of gibbsite, used as an ore for aluminium.

GIBIBYTE: A unit of data storage that refers to true binary numbers and equals 2 to the 30th power. It can be estimated as 10^9 but is defined as exactly 1,073,741,824 bytes. It is one of the set of units with binary prefixes defined by the International Electrotechnical Commission (IEC) to accurately reflect the difference between decimal and binary numbers. Its symbol is GiB. When referring to true binary

numbers, the traditional kilo, mega, giga, etc. designations were changed to kibi, mebi, gibi, etc.

GIBLETS: The edible offal of a poultry consisting of heart, liver and gizzard.

GID (STURDY): A brain disease of herbivores, especially young sheep, also occurring in cattle. It is caused by ingestion of the tapeworm's (*Multiceps multiceps*) eggs voided by dogs and foxes. It results in the presence of the larvae of the tapeworm in the brain.

GIF (GRAPHICS INTERCHANGE FORMAT): A compressed image file format based on indexed colours, which is a palette of at most 256 colours. Together with JPEG (Joint Photographic Experts Group), they are the two most commonly used file formats for pictures on the World Wide Web because pictures saved in these formats take up relatively small amount of space and help greatly reduce their file size. These compressed image files can be quickly transmitted over a network or the Internet, often seen on Web pages. GIF files are useful for small icons and animated images, but they lack the colour range to be used for high-quality photos.

GIGA: A prefix used in the metric system to denote one thousand million times ($1\,000\,000\,000$, 10^9). Its symbol is G.

GIGABYTE: In computing, denoted GB, it is a measure of memory capacity, which is equal to 1,024 megabytes (2^{30} bytes). It is also used sometimes for 1,000 billion bytes.

GIGAFLOPS: A unit of the processing power of a processor's floating point unit (FPU). One gigaflops (gigaFLOPS) is equal to one billion FLOPS or floating point operations per second.

GIGAHERTZ: A unit of measurement of frequency, which is equal to 1,000 megahertz (MHz) or 10^9 Hz. It is commonly used to measure computer processing speeds.

GIGANTISM (GIANTISM): 1. Excessive growth resulting from heredity, diet or growth due to overproduction of growth hormone by the pituitary gland in humans before the end of adolescence. 2. The quality or condition of being very large, tall or bulky.

GIGO (GARBAGE IN GARBAGE OUT): A phrase coined by George Fuechsel, an IBM 305 RAMAC technician/instructor in New York in the field of computer science or information and communication technology (ICT), which denotes that a computer program or process is only as good as the ideas or data put into it.

GILBERT: The centimetre–gram–second (cgs) unit of magnetomotive force, which is equal to 0.7958 ampere-turns in the SI system. It was named after William Gilbert. Its symbol is Gb. The International System of Units (SI) is the currently used system of units, and the ampere (A) has replaced the gilbert.

GILBERT, WILLIAM (1544-1603): English physician and physicist, who studied medicine at Cambridge, where he was elected a Fellow of St. John's College and set up practice in London, becoming president of the College of Physicians (1599) and physician to Queen Elisabeth I. In 1600, he published his famous book, *De Magnete*, about magnetism, in which he likened the Earth to a huge bar magnet. He was the first to use the term 'magnetic pole and electricity'. He found that an iron bar that is left in alignment with the Earth's magnetic field will slowly become magnetised and that sufficient heating will cause a magnet to lose its magnetism.

GILGAI: A small, ephemeral lake formed from a depression in the soil surface, found in expanding soils that contain large amounts of clay. The term is also used to refer to the overall microrelief (local and small variations in the elevation on the surface of an area of land) in such areas, consisting of mounds and depressions produced by expansion and contraction caused by changes in moisture. Gilgai soils experience a high level of nutrient exchange and cycling between the soil surface and subsurface layer due to the cracking nature of the soil.

GILL: 1. A respiratory organ used by some aquatic animals to obtain oxygen from the surrounding water and also to excrete carbon dioxide. 2. One of the thin, radially arranged lamellae that hang from the underside of the cap in certain fungi of the Basidiomycotina. The gills, which are covered by the hymenium, are usually wedge shaped in longitudinal section and attached at the wider end of the wedge. 3. A unit of volume

for liquid measure, which is equal to a pint or four ounces (118 millilitres U.S.) and one quarter of a British Imperial pint (142 millilitres).

GILL ARCHES (BRANCHIAL ARCHES): Cartilaginous structure located on each side of the pharynx to provide support to the gills of aquatic as well amphibian animals.

GILL BAR: A cartilaginous support for the tissue between the gill slits found in some stages of lower animals such as amphibians.

GILL NET: A net used by commercial and artisanal fishermen of all the oceans and in some freshwater and estuary areas that catches fishes by the gill. It is set vertically in the water so that fish swimming into it are entangled by the gills in its mesh. Its use is closely monitored and regulated by fisheries management and enforcement agencies.

GILL SLITS: 1. Also called **branchial cleft** or **gill cleft**, it is an opening in aquatic vertebrates and some amphibians leading from the pharynx to the exterior, characteristic of sharks and related fishes, through which water passes to the exterior, thereby bathing the gills. 2. Also called **branchial groove**, it is one of several rudimentary invaginations in the surface of the embryo, present during development of all air-breathing vertebrates and corresponding to the functional gill slits of aquatic species.

GILT: 1. Female of the swine family that has not given birth. 2. A thin layer of gold or of a substance that looks like gold applied to a surface.

GIMBAL (GIMBAL RING): A type of mount for an instrument such as a gyroscope or compass consisting of two rings mounted on axes at right angles to each other, in which the instrument is free to rotate about two perpendicular axes.

GINGER: 1. The spicy rhizome of a tropical plant, *Zingiber officinale*; family Zingiberaceae or its aromatic, pungent rhizome, which is used in pieces or in powder form in cooking or eaten preserved in sugar. 2. A brownish yellow colour.

GINGIVA: The gum or the part of the epithelial tissue consisting of the mucosal tissue that lies over the alveolar bone.

GINGIVITIS: An infection on the gums, which makes them red, swollen and painful and can cause bleeding.

GINI INDEX (GINI COEFFICIENT; GINI RATIO): A measurement of inequality in the distribution of incomes for a given population named after its developer, Corrado Gini, an Italian statistician. Its symbol is I_G . Graphically, the Gini index is represented by the surface area between a line at 45° and the Lorenz curve. The Gini coefficient is zero if there is no inequality in income distribution and a value of 1 if the income distribution is completely unequal, so that $0 \leq I_G \leq 1$.

GINNING: The process of removing the seeds from cotton, cleaning and baling. The first step in the ginning process is vacuuming the cotton into tubes that carry it to a dryer to reduce moisture to improve the fibre quality. It is then cleaned to remove leaf trash, sticks and other foreign matter. There are two methods of ginning: (i) by **saw gins**, used for shorter staple or fibre length cotton, which involves the use of circular saws that grip the fibres and pull them through narrow slots. (ii) By **roller gins**, used for long fibre cotton because saw gins can damage the delicate fibres. The raw fibre, now called **lint**, is compressed into bales for sale to processors into threads and other users.

GIORGI UNITS (MKS SYSTEM): A metric system of units devised by A. Giorgi in 1901, because electric measurements could not be explained in terms of the three fundamental units of length, mass and time. Giorgi units are based on the metre, kilogram and the second derived from the earlier centimetre–gram–second (cgs) units. The electrical unit chosen to augment these three basic units was the ampere and the permeability of space (magnetic constant) was taken as 10^{-7} Hm^{-1} . To simplify electromagnetic calculations the magnetic constant was later changed to $4\pi \times 10^{-7} \text{ Hm}^{-1}$ to give the rationalised MKSA system. This system, with some modifications, formed the basis of SI units, now used in all scientific work.

GIRAFFE: An ungulate mammal (*Giraffa camelopardalis*) belonging to the order Artiodactyla and native to Africa. It has an extremely long neck and legs and is the tallest living terrestrial mammal.

GIRDER: 1. In engineering, the term refers to a main load-supporting beam. Steel or iron girders may be made of a single piece or of laminated strips built up of plates, latticework or bars. When the cross-section is formed in the shape of an I by riveting and welding plates or by rolling the girder, greater stiffness is achieved and larger spans are possible. Girders may also be made of reinforced or pre-stressed concrete. 2. In botany, it is a structure composed of tissue, which provides mechanical support for a stem or leaf.

GIRDLE: 1. A bony structure to which the limbs of vertebrates are attached.

GIRTH: 1. The measurement around a cylindrical and thick object such as a tree trunk or the waist. 2. A broad band fastened around the belly of a horse to keep a saddle in place. 3. The length of the shortest cycle contained in a graph.

GIS (GEOGRAPHIC INFORMATION SYSTEM): Mapping software that relates the location information with other pertinent information to give it meaning and value. This software is responsible for capturing, storing, checking and displaying data related to positions on earth's surface. It can be used for finding what is nearby, routing information alerts, finding patterns, mapping densities and mapping quantities, among others.

GIVEN: In mathematics: 1. Known or determined independently. 2. Stipulated or assumed for the purposes of a specific construction or proof; for example, an epsilon-delta proof of continuity typically begins 'Given $\epsilon > 0$ '. 3. An axiom or assumption of a proof.

GIZZARD (VENTRICULUS; GASTRIC MILL; GIGERIUM): The muscular compartment of the alimentary canal of many animals including birds, reptiles, earthworms and some fish that is specialised for breaking up food.

GIZZARD STONES (GASTROLITH): A stone swallowed by animals, some birds, reptiles such as crocodile and certain lizards and found in the gastrointestinal tract. This stone is used to grind food because the animals lack suitable grinding teeth. Gastroliths, which are typically very smooth and polished are also found in dinosaur fossils.

GLABROUS: In biology, it refers to a surface that is smooth, glossy and devoid of hairs or other projections.

GLACIAL ACETIC ACID: A crystalline form of anhydrous acetic acid below its freezing point that is at least 99.8 percent pure.

GLACIAL DEPOSITION: The laying down of sedimentary material consisting of particles such as debris, sand, soil and rocks carried by glacier and dropped when the ice melts. These particles are known collectively as **glacial till**, which often forms into piles known as **moraines**. Glacial deposition can occur beneath the ice called **subglacial deposition**, inside the ice called **englacial deposition**, on top of the ice called **supraglacial deposition**, at the side of the ice called **marginal deposition** and in front of the ice called **proglacial deposition**.

GLACIAL DRIFT: Sediment and rocks transported by glaciers and deposited directly on the land or indirectly in streams, lakes and oceans. It consists of a heterogeneous mixture of clay, silt, sand, gravel and boulders ranging in size and shape. There are two distinct types of glacial drift which are: (i) **till**, which is unstratified sediment of many particle sizes; and (ii) **stratified drift**, which is relatively well-sorted sediment according to the size and the weight of the fragments.

GLACIAL EROSION: The wearing down and removal of rocks and soil by a glacier. Glaciers erode by three main processes: abrasion/scouring, plucking and ice thrusting. In an **abrasion erosion**, debris in the basal ice scrapes along the bed, polishing and gouging the underlying rocks, similar to sandpaper on wood. In **plucking erosion**, glaciers cause pieces of bedrock to crack off. In **ice thrusting erosion**, the glacier freezes to its bed and then as it surges forward, it moves large sheets of frozen sediment at the base along with the glacier.

GLACIAL POLISH: The abrasion of bedrock surfaces by materials carried on the bottom of a glacier, leaving the surfaces smooth and shiny.

GLACIAL TROUGH (U-SHAPED VALLEY): A steep-sided, flat-bottomed valley formed by a glacier.

GLACIATE: 1. To cover something with a glacier or become covered with a glacier. 2. To affect something by the action of a glacier, especially by erosion.

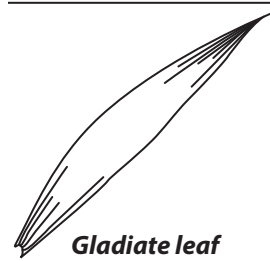
GLACIER: A perennial mass of ice originating in snowfields above the snow line, which moves slowly downhill and is constantly replenished from its source, sometimes causing the formation of features such as valleys on the land.

GLACIER BUDGET (ICE BUDGET): The balance between accumulation and ablation in a glacier. If accumulation exceeds ablation the glacier will grow, called a **positive regime**; if ablation exceeds accumulation the glacier will retreat, called a **negative regime**. Accumulation is the addition of ice to the glacier due to snow being compacted into ice; and ablation is the loss of ice by melting and evaporation.

GLACIOFLUVIAL DEPOSITS: Sediments moved by glaciers and subsequently sorted and deposited by streams formed by the melting of ice and glaciers. The coarse materials are deposited near the end of the glacier, and the finer materials are carried further away. Glaciofluvial deposits are stratified and may occur in the form of outwash plains, deltas, eskers, kames and kame terraces. **Esker** is a long, winding ridge of gravel and other sediment, deposited by glacial meltwater. **Kames** are small rounded hills with heights of between 10 and 12 metres, composed of stratified sand, gravel, pebbles and fine clays. **Kame terraces** are features of stratified sand and gravel deposited by streams between a glacier and an adjacent valley wall.

GLACIOLOGY: The study of the physical and chemical properties of snow and ice.

GLADIATE: In botany, it refers to a structure or organ, which is shaped like a sword, for example, gladiate leaves, as shown in the diagram.



GLAND: A specialised organ of the body of animals, plants or insects that manufactures and secretes enzymes, hormones or other chemical substances into the body either through a duct or direct into the blood stream. There are two principal types of glands: the **endocrines**, which are responsible for internal secretion and the **exocrine**, responsible for external secretion. Some glands such as the pancreas produce both internal and external secretions.

GLANDERS (MELIOIDOSIS): A serious contagious disease of horses, donkeys and mules. It is caused by a Gram-negative bacterium, *Burkholderia pseudomallei*, found in soil and water. Symptoms may include pain in chest, bones or joints, cough, skin infections, lung nodules and pneumonia.

GLANDULAR FEVER (INFECTIOUS MONONUCLEOSIS; MONONUCLEOSIS): An acute viral disease that mostly affects young adults and caused by the Epstein-Barr virus that can spread through close, personal contact and is transmitted by saliva. It is characterised by fever, sore throat, fatigue, swollen lymph nodes and glands.

GLASER, DONALD ARTHUR (1926): US physicist, who contributed greatly to the understanding of atomic function and provided the technology for the discovery of new atomic particles. He received his B.Sc. degree in Mathematics and Physics from Cleveland's Case Institute of Technology (now Case Western Reserve University) in 1946 and his Ph.D. degree from the California Institute of Technology in 1950. In 1949, Glaser accepted a position as instructor at the University of Michigan, where he began his bubble chamber experiments. In 1959, he became a Professor of Physics at the University of California, Berkeley. In 1952, at the University of Michigan, he devised the bubble chamber for detecting ionising radiation, for which he was awarded the 1960 Nobel Prize.

GLASS: An amorphous usually transparent solid made by heating together lime, sodium carbonates and sand. Glass in molten form is plastic and can be shaped by means of several techniques. At low temperatures, glass is brittle and breaks with a shell-like fracture on the broken face. Glass can also be translucent or opaque. It is used in making materials such as bottles, windows and doors, lenses, etc.

GLASS CERAMIC: A glass material formed from a mixture of lithium oxide, sand and alumina that crystallises, when heated to high temperatures. Glass ceramics have no pores between crystals and have high flexural strength and shock resistance. They are mechanically strong and can sustain repeated and quick temperature changes, up to 800-1000 °C (1073-1273 K). They have low heat conduction coefficient and can be made nearly transparent. Glass ceramics are used in making fire-resistant materials for coatings, moulded mechanical and electrical parts, heat-exchanger tubes, missile cones and cookware and dinnerware.

GLASS ELECTRODE: A glass membrane electrode or half-cell used to measure hydrogen ion concentration or pH. It consists typically of an electrode membrane that responds to pH, a highly isolating base material to support the unit, solution inside the glass electrode, an internal electrode, a lead wire and a glass electrode terminal. Generally, silver chloride is used as the material for the internal electrode. Potassium chloride solution maintained at pH 7 is usually used as the internal solution.

GLASS LASER: A solid state laser in which the active medium is a glass host that supports the active species. An example is theneodymium-doped phosphate glass laser.

GLASS ROD (STIRRING ROD; STIR ROD): A device that is used in science laboratories for agitating or stirring a solution in order to cause a quicker reaction. It is also used to spread liquid substances on a solid surface and to pour liquids slowly, especially dangerous chemicals. Its length is between 10 and 40 centimetres with a thickness of about half a centimetre and usually made of borosilicate. It is mostly used in chemistry and biology experiments.

GLASS TRANSITION: A second-order transition in which a supercooled melt yields, on cooling, a glassy structure. Below the glass-transition temperature the physical properties vary in a manner similar to those of the crystalline phase. For example, lithium disilicate crystals melt at 1031.85 °C (1305 K). The melt can be supercooled to the glass transition temperature at approximately 499.85 °C (773 K) below which the viscous liquid freezes to a rigid amorphous glass.

Glass-transition temperature is the temperature at which the viscosity of the glass is 10^{13} dPas.

GLASS WOOL: A material that consists of fine glass fibres, arranged into a texture similar to wool, used in filters as a thermal insulator and for making fibreglass.

GLASS-LIKE CARBON: A granular non-graphitisable carbon with a very high isotropy of its structural and physical properties and with a very low permeability for liquids and gases. The original surfaces and the fracture surfaces have a pseudo-glassy appearance.

GLASS-REINFORCED PLASTIC, GRP (FIBRE-REINFORCED PLASTIC): A solid composite material that is commonly called fibreglass, made by mixing two resins with reinforcing fibres of glass. It has a good weight to strength ratio, rust resistance and ability to be moulded in a wide variety of ways. It is used in the manufacture of car body panels, boat hulls, furniture, tennis rackets, etc.

GLASSHOUSE: A large climate-controlled structure made of glass inside which plants are grown, for commercial or for scientific purposes.

GLAUBER MODEL: A model describing critical dynamics in which the dynamics is dissipative and the order parameter is not conserved.

GLAUBER'S SALT (SODIUM SULFATE DECAHYDRATE): White or colourless monoclinic crystals, with the chemical formula $\text{Na}_2\text{SO}_4 \cdot 10\text{H}_2\text{O}$. Upon exposure to fairly dry air it effloresces, forming powdery anhydrous sodium sulfate. It is water soluble, has a salty, bitter taste and is sometimes used in medicine as a mild laxative. It is also used in dyeing.

GLAUBERITE: A mineral consisting of a mixed sulfate of sodium and calcium, with the chemical formula $\text{Na}_2\text{SO}_4\text{CaSO}_4$. Because of its solubility, glauconite is often dissolved away from the crystal matrix leaving a mould into which other minerals are deposited, giving them an anomalous crystal form.

GLAUCOMA: A group of eye diseases marked by unusually high pressure within the eyeball that leads to damage of the optic disk resulting in progressive, permanent vision loss.

GLAUCONITE: A greenish clay mineral of the illite group, found chiefly in marine sands.

GLAUCOUS: A term used to describe surfaces having a waxy greyish-blue bloom, such as the leaves of rape or swede (*Brassica napus*). Glaucous means sparingly or slightly glaucous or becoming glaucous.

GLEBA: The mass of spore-producing tissue that forms in the fruiting bodies of fungi such as truffles, puffballs, earth-stars and stinkhorns. It is enclosed by the peridium.

GLEIZATION: A soil formation process that occurs in poorly drained environments. It results in the development of extensive soil organic layer over a layer of chemically reduced clay that takes on a blue colour.

GLENOID CAVITY: The socket shaped cavity in the scapula that holds the head of the humerus in a ball and socket joint, bringing about movement in all directions.

GLEYS: A thick sticky rich soil found in waterlogged ground with reduced amounts of iron and other elements. It has a lot of clay and is usually a light grey colour. Land with this type of soil is often used for grazing animals such as sheep and goats. Although gley soil can be drained for agriculture purposes, it is expensive to do this and lime must be added to reduce the acidity.

GLEYSOL: A type of soil that is formed under waterlogged conditions which may be produced by either a high groundwater table or temporary accumulation of water above a relatively impermeable layer, or both. In the tropics and subtropics, rice may be planted in gleysols.

GLIA CELLS: The supportive tissue and fibres that nourish nerve cells, including the network of branched cells located in the central nervous system. Glial cells not only provide physical support, but also respond to injury, regulate the chemical environment of neurons, form the myelin insulation of nervous pathways, help guide neuronal migration during development and play an important role in the blood-brain barrier. Glial cells come in four main types: astrocytes, oligodendrocytes, ependymal cells and microglia. They are the most common cellular component of the brain. There are five to 10 times more glial cells than neurons. Unlike neurons, glial cells have the ability to divide and multiply.

GLIADIN: A storage protein found in the caryopses of wheat.

GLIDE: 1. To move smoothly and continuously along, as if without effort or resistance, as sometimes done by birds. 2. To move in the air, especially at an easy angle downward, with less engine power than for level flight, solely by the action of air currents and gravity or by momentum already acquired. 3. A symmetry element in a crystal lattice that consists of a combination of a translation with a reflection about a plane.

GLIDE ANGLE: The angle or slope, of glide during descent.

GLIDE REFLECTION: A combination of a reflection about a line and a translation parallel to the line.

GLIDER (SAILPLANE): An unpowered airplane which, once launched by air or ground towing or by using a winch, is kept aloft by its light, aerodynamic design and the skill of the pilot in exploiting "thermals" and other rising air currents.

GLIOMA: A tumour of glial cells, the supporting cells of the brain. Unlike neurons, glial cells have the ability to divide and multiply. If this process occurs too rapidly and without control, a glioma forms. They never metastasise, but produce signs of focal damage to the brain, such as weakness, visual disturbance, personal change or epilepsy and often a characteristic type of headache. Surgery and radiation therapy may be helpful, but glial cell destruction cannot be reversed.

GLOBAL (IN THE LARGE, *IM GROSSEN*): Of a mathematical relation, property, etc., holding for all values of the variables without restriction.

GLOBAL ADDRESS LIST (GAL): A single address list created by Microsoft that contains all addresses of all users and their respective email addresses in an organisation that users can use to select names, e-mail addresses and distribution lists. It can be accessed over the computer network using LDAP protocol, CardDAV or other electronic means.

GLOBAL OCEAN CONVEYOR (THERMOHALINE CIRCULATION): The flow of ocean water caused by changes in temperature and salinity in the deep ocean and wind-driven currents on the surface.

GLOBAL NAVIGATION SATELLITE SYSTEM (GLONASS): The Russian GPS (global positioning system) and the equivalent of the American NAVSTAR. Both systems use the same techniques for data transmission and positioning. GLONASS is based on a constellation of active satellites that continuously transmit coded signals in two frequency bands. These signals can be received by users anywhere on Earth to identify their position and velocity in real time based on ranging measurements. The GLONASS orbits are roughly circular, with an inclination of about 64.8° and a semi-major axis of 25,440 km.

GLOBAL OSCILLATION NETWORK GROUP (GONG): A network of six observatories around the world that provides almost continuous, highly sensitive monitoring of the Sun's pulsations. The data collected is used for detailed studies of solar internal structure and dynamics. The sites are located in the Canary Islands (Teide Observatory), Western Australia (Learmonth Solar Observatory), California (Big Bear Solar Observatory), Hawaii (Mauna Loa Solar Observatory), India (Udaipur Solar Observatory) and Chile (Cerro Tololo Inter-American Observatory).

GLOBAL POSITIONING SYSTEM (GPS): A satellite based navigational system originally developed for U.S. military use. With the use of GPS receiver, it can determine location and time information in all weather at all times and anywhere on or near the Earth with a high degree of accuracy, when and where there is an unobstructed line of sight to four or more GPS satellites. The system uses a network of 24 communications satellites designed and controlled by the U.S. Department of Defence that transmit signals globally around the clock. The Global Positioning System (GPS) is freely accessible by anyone with a GPS receiver.

GLOBAL REGULAR EXPRESSION AND PRINT (GREP): A Unix command line text search utility that allows full-text searching within files for lines that match a given regular expression and prints the matching lines to the program's standard output.

GLOBAL SURFACE TEMPERATURE: An estimate of the Earth's average surface air temperature. This is based on the average ocean temperature close to the surface and the temperature found between the land surface and 1.5 metres above.

GLOBAL SYSTEM FOR MOBILE COMMUNICATION (GSM): In telecommunications, it is a second generation (2G) standard for mobile networks. It is a circuit-switched technology based on Time Division Multiple Access (TDMA). It operates on three distinct frequencies: 900 MHz, 1800 MHz and 1900 MHz. The 900 and 1800 MHz frequencies are used in Europe and 1900 MHz frequency is used in North America. Data transfer rate of GSM is 9.6 Kbps.

GLOBAL VARIABLE: In computing, it is a variable that is visible and accessible throughout the program, unless shadowed. Global variables retain their values throughout the execution of a program. It can be used to avoid having to pass frequently-used variables continuously throughout several functions.

GLOBAL WARMING: Gradual increase in the global average surface temperature resulting from enhancement of the greenhouse effect, primarily by air pollution. During the last 100 years, the Earth's temperature has risen by about 0.18°C due to increase in greenhouse gases arising from anthropogenic (caused by humans) processes. Global warming is expected to cause sea level to rise, changes in rainfall pattern, destroy forests, increased desertification, etc.

GLOBAL WARMING POTENTIAL (GWP): An index, describing the radiative characteristics of well-mixed greenhouse gases, that represents the combined effect of the differing times these gases remain in the atmosphere and their relative effectiveness in absorbing outgoing infrared radiation. This index approximates the time-integrated warming effect of a unit mass of a given greenhouse gas in today's atmosphere, relative to that of carbon dioxide.

GLOBAL WINDS: Winds comprising three prevailing winds: Tradewinds, Westerlies, and Polar Easterlies that appear all over the world and are caused by uneven heating of the Earth's surface, pressure differences, and the Coriolis effect.

GLOBALISATION (GLOBALIZATION): A process of interaction and integration between the people, companies/organisations and governments of different nations beyond domestic and national level. It is often driven by international trade and investment and aided by information technology. It has effects on the environment, on culture, on political systems, on economic development and prosperity and on human physical well-being in societies around the world.

GLOBALLY UNIQUE IDENTIFIER (GUID): A 128-bit (16 byte) number used by software programs to uniquely identify the location of a data object. Some examples of data that include GUIDs are streaming media files, Windows registry entries, database keys and various file types. GUIDs are typically written in hexadecimal notation, containing 32 digits.

GLOBALSTAR: A consortium of international telecommunications companies formed in 1991 to provide a worldwide satellite-based telephony service. Together with Iridium, ICO and others, Globalstar is one of the new comsat companies supporting global mobile voice and data communications with a constellation of satellites in low Earth orbit. The Globalstar constellation, launched in 1998-99, consists of 48 satellites (with four in-orbit spares) in 1,400-km-high orbits inclined at 52°, allowing a concentration of coverage over the temperate regions of Earth from 70° S to 70° N. The satellites operate in the L-band.

GLOBE: A spherical model of the Earth, which is used to represent the world or the heavens. It is the most accurate map of the Earth, duplicating its spherical shape and relative size. There are two types of globes: terrestrial and celestial. **Terrestrial globes** represent the Earth's surface. Some of the terrestrial globes represent the political features or relief (physical) features of the Earth by colours. Globes are often made to rotate around a rod, which serves as the axis of the globe and passes through the north and south poles and attached to a stand. The axis is usually inclined at an angle of about 23.5°, which represents the degrees by which the axis of the Earth is tilted off the axis of the Sun. **Celestial globe** has the relative positions of the stars mapped without distortion. It is a model of the heavens as seen from the Earth. Each star is plotted on the globe at the point corresponding to the spot on the Earth where it appears to be directly overhead.

GLOBIGERINA: A genus of single-celled organisms of the order Foraminifera of the class Rhizopoda, members of which are found floating on the surface of the oceans. They have a perforated shell, through which they can extend pseudopodia. After death the shells sink to the ocean floor to form much of the organic oozes. One species is an indicator of sea temperature and therefore of ocean paleoclimatology. Shells coiling to the left indicate that the animals lived in cold water; shells coiling to the right indicate animals that lived in warm water.

GLOBIN: A protein, belonging to the histone family. Combined with haeme it forms the respiratory pigment haemoglobin.

GLOBOSE: Spherical; shaped like a globe or ball.

GLOBULAR CLUSTER: Approximately spherical clusters of stars, usually containing 104 to 106 stars. Because they are so distant from the solar system, most are not visible to the unaided eye, but Omega Centauri and a few others can be seen without a telescope as hazy patches of light.

GLOBULAR PROTEIN (SPHEROPROTEIN): One of the two main protein classes, which is a folded protein that adopts an approximately globular shape. It is highly soluble and plays an important role in body metabolism. Examples of globular proteins are albumin, globulin, casein, haemoglobin, all of the enzymes and protein hormones.

GLOBULAR-CHAIN CRYSTAL: A type of crystal comprised of macromolecules having globular conformations. They usually occur with globular proteins.

GLOBULIN: Any of a group of globular proteins that are generally insoluble in water but soluble in dilute saline (salt) solutions and present in blood, milk, eggs and as a reserved protein in seeds. Globulins in animal fluids include enzymes, antibodies (the gamma globulins), lipoproteins, complement components, transport proteins and various types of fibrous and contractile proteins.

GLOBUS PALLIDUS: A structure in the brain involved in the regulation of voluntary movements at a subconscious level. It is one of three nuclei that make up the basal ganglia, the others being the putamen and the caudate. It relays information from the other nuclei to the thalamus.

GLOCHID: Sharp, specialised, short hair-like spines that have their tips barbed, as seen in *Opuntia*, Cactus and prickly pear. Glochids can be broken off easily and can attach to skin or clothing.

GLOMERULAR FILTRATE: The fluid in the lumen composed of substances such as water, salts and other substances collected from the Bowman's capsule of the nephron that has been filtered from the capillaries of the glomerulus. The glomerular filtrate has the same composition as the plasma except that it does not contain any of the larger components, such as plasma proteins or cells.

GLOMERULAR FILTRATION RATE (GFR): The volume of fluid that is filtered from the capillaries of the glomeruli into the kidney tubules per unit time. It is influenced chiefly by blood pressure and by changes in the diameter of arterioles in the glomeruli. In humans, GFR is about 125ml/min and about 99% of this is reabsorbed as it passes along the kidney tubules to form urine.

GLOMERULE: A dense cluster or a dense, headlike cluster of sessile or subsessile flowers. **Glomerulate** means arranged in very small, dense cluster.

GLOMERULUS: A tangled mass of blood capillaries enclosed by the cup-shaped structure called Bowman's capsule. Waste products and water pass from the glomerulus into the Bowman's capsule of the mammalian kidney.

GLOSSO-PHARYNGEAL NERVE: The ninth of twelve pairs of cranial nerves. It arises from the medulla oblongata and travels with the internal jugular vein, through the jugular foramen, from the cranium to the neck. It contains sensory, motor and autonomic fibres and supplies the tongue, soft palate, pharynx and parotid gland.

GLOTTIS: The opening of the larynx into the pharynx of vertebrates consisting of the true vocal cords (vocal folds) and the opening between them.

GLOW: 1. To emit a soft steady light. 2. To shine brightly and steadily, especially without a flame such as coal or charcoal in fire. 3. To have a bright, warm, usually reddish colour. 4. A light produced by a body heated to luminosity; incandescence.

GLOW DISCHARGE (COLD-CATHODE DISCHARGE): An electrical discharge that passes through a gas at low pressure and causes the gas to become luminous. It causes a voltage drop in the vicinity of the cathode that is much higher than the ionisation voltage of the gas.

GLOW-WORM (FIRE-FLIES): Any of various different groups of insect larvae and adult larviform females, which glow due to bioluminescence.

GLOWING SPLINT TEST: A test for a gas sample to determine whether it is an oxidising gas, such as oxygen or any other oxidising gas such as nitrous oxide. This is done by lighting a splint which burns for a few seconds. Whilst the ember at the tip is still glowing hot, the splint is introduced to the gas sample that has been trapped in a vessel. When exposed to any concentrated oxidising gas, the

glowing ember flares, and re-ignites to produce a sustained flame.

GLUCAGON: A hormone made by endocrine glands in the pancreas. Glucagon is responsible for glycogenolysis, the breakdown of liver glycogen to sugar thereby increasing the concentration of sugar in the bloodstream. Glucagon may also be involved in the regulation of protein and fat metabolism, gastric acid secretion and gut motility, excretion of electrolytes such as sodium, potassium and chloride by the kidney, contractility of heart muscles and the release of insulin from the pancreas.

GLUCAN: Any polysaccharide of D-glucose monomers linked by glycosidic bonds. Examples are glycogen and starch.

GLUCOCORTICOID: Any of a family of steroid hormones characterised by ability to bind with the cortisol receptor and trigger similar effects. Glucocorticoids are distinguished from mineralocorticoids and sex steroids by their specific receptors, target cells and effects. Both glucocorticoids and mineralocorticoids are classed as corticosteroids and originate in the cortex of the adrenal glands.

GLUCONEOGENESIS: The metabolic pathway that results in the synthesis of glucose from non-carbohydrate sources, such as fat and protein. It occurs when glycogen is exhausted in the liver.

GLUCONIC ACID (DEXTRONIC ACID; D-GLUCONIC ACID): An optically active organic compound, with the molecular formula $C_6H_{12}O_7$ and melting point 131 °C (404 K). It is made by the reaction of certain moulds or a soluble crystalline organic acid, which is an oxidation product of glucose. Gluconic acid occurs widely in nature because such species arise from the oxidation of glucose. It is used in paint strippers.

GLUCOSAN: 1. Any of a class of polysaccharide compounds yielding glucose upon hydrolysis. They include dextrin, starch and cellulose. 2. Any of several anhydrides of glucose.

GLUCOSE (DEXTROSE; GRAPE SUGAR): 1. Also known as **D-glucose**, **D-glucopyranose**, **grape sugar**, **corn sugar**, **dextrose** and **cerelose**, it is an organic compound, which is a white crystalline sugar with the chemical formula $C_6H_{12}O_6$. It occurs widely in nature, present in all plants and animals, fruits and honey. It is made in plants by photosynthesis. It is the source of energy in cell function and crucially important in the metabolism of living organisms. Glucose is the result of digestion by animals of carbohydrates. Glucose units in long chains make up polysaccharides such as cellulose, glycogen and starch. 2. A colourless to yellowish syrupy mixture of dextrose, maltose and dextrans containing about 20 percent water. It is used in confectionery, alcoholic fermentation, tanning and treating tobacco.

GLUCOSE 1-PHOSPHATE: A phosphorylated derivative of glucose, formed by isomerisation from glucose 6-phosphate. It is an intermediate in the synthesis of UDP-glucose, from which starch and cellulose are formed. It is also the predominant breakdown product of starch and cellulose.

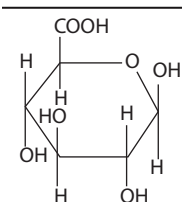
GLUCOSE 6-PHOSPHATE: A phosphorylated derivative of glucose. It can be formed by the action of hexokinase on glucose or by isomerisation from glucose 1-phosphate. It is an important activated form of glucose. In glycolysis and in the pentose phosphate pathway the first step is activation of glucose to glucose 6-phosphate. It is also an intermediate in sucrose formation.

GLUCOSE TOLERANCE TEST (GTT): In medicine, a diagnostic method of assessing the efficiency of the body to metabolise glucose, which is used primarily in the diagnosis of type II diabetes and gestational diabetes (high blood sugar levels developed only during pregnancy). It is done after the individual has fasted overnight, but before breakfast, given a specific amount of glucose orally and the levels of sugar in the blood and the urine are measured at intervals of, often between 30-60 minutes over several hours. The result depends on a number of factors, including the ability of the intestines to absorb glucose, the power of the liver to take up and store glucose, the capacity of the pancreas to produce insulin and the amount of 'active' insulin.

GLUCOSIDE: A carbohydrate-containing compound that yields a glucose and a non-sugar component (either an alcohol or phenol), when decomposed by the process of hydrolysis. The natural glucosides are important in plant metabolism and many polysaccharides, such as cellulose, starch and glycogen, are regarded chemically as glucosides. They may be used as drugs, colouring agents and aromatics.

GLUCURONIC ACID (β -D-GLUCOPYRANURONIC ACID; GLYCURONIC ACID): A compound, with the molecular formula

$C_6H_{10}O_7$, which is derived from the oxidation of glucose. It is formed in the liver of all animals, including humans and other primates. The main function of glucuronic acid is to combine with other substances such as drugs, toxins and hormones and either carry them to other parts of the body or eliminate them. It is used to determine urinary steroids and steroid conjugates in blood. It is a precursor of ascorbic acid in all plants and mammals, except guinea pigs and primates. Consuming alcohol and other substances that cause harm to the liver lowers the body's ability to produce glucuronic acid.



Chemical structure of Glucuronic Acid

GLUE: A type of adhesive, which is a mixture in a liquid or semi-liquid state that adheres or bonds items together. It may come from either natural or synthetic sources. It hardens by either evaporating a solvent or by chemical reactions that occur between two or more constituents.

GLUEBALL: A hypothetical bound state consisting of two or more gluons particles, without valence quarks. They are thought to be unstable and decay rapidly into hadrons. Glueballs are extremely difficult to identify in particle accelerators, because they mix with ordinary meson states.

GLUINO: The hypothetical supersymmetric partner of a gluon. Gluinos are expected by supersymmetry theorists to be a pair produced in particle accelerators such as the Large Hadron Collider, if they exist. Gluinos have a lepton number 0, baryon number 0 and spin 1/2. They decay via strong interaction to a squark and a quark provided that an appropriate mass relation is satisfied. The squark subsequently decays to another quark and the lightest supersymmetric particle, LSP. This means that a typical signal for a gluino at a hadron collider would be four jets plus missing energy. Its symbol is $\tilde{g}$.

GLUME: One of the two chaffy basal bracts of a grass spikelet that are found at the base of spikelets of the grasses.

GLUON: A subatomic particle of the type believed to hold quarks (the smallest elementary particle of matter) together. They are indirectly involved with the binding of protons and neutrons together in atomic nuclei through the strong force.

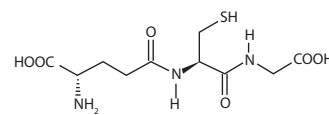
GLUTAMATE: The anion of amino acid (glutamic acid), especially its sodium salt monosodium glutamate, which functions as a neurotransmitter at excitatory synapses in the central nervous systems of vertebrates and at excitatory neuro-muscular junctions in insects and crustaceans.

GLUTAMATE RECEPTOR (GLUR): Any receptor protein that binds neurotransmitter glutamate as a ligand. They fall under two main types: ionotropic and metabotropic.

GLUTAMIC ACID: A colourless amino acid, with the chemical formula $COOH(CH_2)_2CH(NH_2)COOH$. One of its functions is to increase the solubility of its associated proteins by providing them with one negative charge. It also helps in the removal of toxic ammonia from the body. Its sodium salt (sodium glutamate) is used as a food flavouring.

GLUTAMINE (L-GLUTAMINE): A colourless, soluble non-essential amino acid, with the chemical formula $COOHCH(NH_2)(CH_2)_2CONH_2$. It is formed when one of the acidic carboxyl groups of glutamic acid couples with ammonia. It is important in removing excess ammonia, which is carried safely to the liver where it is converted to urea and freely expelled as urine. Glutamine is the most abundant amino acid (the building block of protein) in the body. Dietary sources of glutamine include plant and animal proteins such as beef, pork and poultry, milk, yogurt, ricotta cheese, cottage cheese, raw spinach, raw parsley and cabbage.

GLUTATHIONE: A peptide comprising the amino acid glutamic acid, cysteine and glycine. This is an antioxidant which occurs widely in plants, animals and microorganisms and helps protect cells from reactive oxygen species such as free radicals and peroxides.



Chemical structure of Glutathione

GLUTELIN: Any of a group of plant proteins that is soluble in dilute acids and alkalis but insoluble in neutral salt solutions and in water and alcohol. Examples are glutenin from wheat and oryzenin from rice. Glutenin is the binding agent in flour pastes and dough.

GLUTEN: A mixture of proteins found in wheat and most other cereal grains, which is not readily soluble in water. It is the mixture of gliadin and glutenin that remains when the starch of wheat grains has been removed. Gluten is an elastic material which stretches when bread rises.

GLUTEUS MAXIMUS: The large, powerful muscle in each of the buttocks forming the bulk of the buttocks and which helps give them their rounded shape. It acts to extend the upper leg, spread it and turn it outward. There are two other gluteal muscles called the gluteus medius and gluteus minimus. The **gluteus medius** is an intermediate muscle of the buttocks that originates in the ilium, with insertion to the surface of the greater trochanter, with nerve supply from the superior gluteal nerve. It acts to abduct and rotate the thigh. The **gluteus minimus** is the innermost muscle of the buttocks, involved in the abduction and rotation of the thigh.

GLUTINOUS: 1. Being able to adhere to substances. 2. Having a gummy exudation.

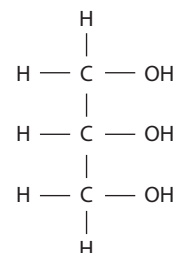
GLYCAEMIC INDEX (GI): An indication of how eating carbohydrate in a particular food will affect the concentration of glucose in the blood or the increase in blood glucose over 2 hours after ingesting 50g of available carbohydrate, expressed as a percentage. The GI of a food is governed by various factors such as the type of carbohydrate, the concentration and nature of the fibre, the proportions of protein and fat and the methods of cooking or processing.

GLYCERALDEHYDE: A sweet colourless crystalline solid, which is the simplest of the monosaccharides containing an aldehyde group, with the molecular formula $C_3H_6O_3$. It is important in many metabolic pathways. In glycolysis glyceraldehyde 3-phosphate is formed, together with dihydroxyacetone phosphate, by the cleavage of fructose 1,6-bisphosphate. Glyceraldehyde 3-phosphate is also one of the compounds formed in the Calvin cycle.

GLYCERATE 3-PHOSPHATE (GLYCERALDEHYDE 3-PHOSPHATE (GA- LP): A phosphorylated three-carbon monosaccharide that is an intermediate in the Calvin cycle of photosynthesis. It was formerly known as 3-phosphoglycerate or phosphoglycerate acid (PGA).

GLYCERIDE (ACYLGLYCEROL): An ester of glycerol (propane-1,2,3-triol) in which a hydroxyl group is replaced by a fatty acid. They are subdivided into triglycerides, 1,2- or 1,3-diglycerides and 1- or 2-monoglycerides, according to the number and position of acyl. Glycerides are one of the four main types of complex lipids. Soaps are formed from the reaction of glycerides with sodium hydroxide.

GLYCERINE (GLYCEROL; PROPANE-1,2,3-TRIOL): A colourless, sweet and sticky liquid that is soluble in water, which is a basic component of nearly all complex lipids. Its chemical formula is $C_3H_8O_3$, density 1.261 g/cm³, melting point 17.8 °C (291.0 K) and boiling point 290 °C (563 K). It is obtained from fats and oils as a by-product of soap manufacture and used as a solvent, antifreeze, a plasticizer and a sweetener and in the manufacture of dynamite, cosmetics, liquid soaps, inks and lubricants.



Chemical structure of Propane-1,2,3-triol

GLYCEROPHOSPHOLIPID: Any derivative of glycerophosphoric acid that contains at least one *O*-acyl or *O*-alkyl or *O*-(1-alkenyl) group attached to the glycerol residue.

GLYCERYL NITRATE: In medicine, the term refers to a vasodilator mainly used in treating the symptoms of angina pectoris. It reduces the amount of blood returning to the heart, thereby reducing the amount of work the heart has to do. When used as an explosive, it is known as **nitroglycerin**.

GLYCINE (AMINOETHANOIC ACID; AMINOACETIC ACID): The simplest amino acid and one of the main components of protein, with the molecular formula $C_2H_5NO_2$. When purified, it is a sweet colourless crystalline compound. It is a nonessential amino acid occurring widely as a component of animal and plant proteins. Synthetically produced glycine is used in solutions for irrigation, in the treatment of various muscle diseases and as an antacid and dietary supplement.

GLYCO-AMINO-ACID: A compound consisting of saccharide linked through a glycosyl linkage (*O*-, *N*- or *S*-) to an amino acid.

GLYCOBIOLOGY: The study of carbohydrates and their complexes particularly glycoproteins and their effects on health and disease.

GLYCOCALYX 1 (CELL COAT): A layer of carbohydrate on the surface of the plasma membrane of most eukaryotic cells, for example, the slime on the outside of a fish. It is made up of the oligosaccharide (carbohydrate with small number of linked monosaccharides) side-chains of the glycolipid and glycoprotein components of the membrane and may include oligosaccharide secreted by the cells.

GLYCOSIDIC BOND: A covalent bond that joins the monomers in a carbohydrate to one another functional group or molecule.

GLYCOGEN (ANIMAL STARCH): A type of carbohydrate similar to starch, used by animals as the secondary long-term energy storage in the muscle cells and liver. It is a polymer of glucose residues linked by α -(1,4)- and α -(1,6)-glycosidic bonds. It is made primarily by the liver and the muscles, but can also be made by glycogenesis within the brain and stomach. The uterus also stores glycogen during pregnancy to nourish the embryo.

GLYCOGENESIS: The conversion of glucose to glycogen, which is stimulated by insulin from the pancreas, in which glucose molecules are added to chains of glycogen for storage. Glycogenesis occurs in skeletal muscle and to a lesser extent in liver. This process is activated during rest periods following the Cori cycle, in the liver and also activated by insulin in response to high glucose levels, for example, after a carbohydrate containing meal.

GLYCOGENOLYSIS: The conversion of polymers (glycogens) to glucose monomers, which occurs in liver and is stimulated by glucagon from the pancreas and adrenaline from the adrenaline medulla.

GLYCOL: A material consisting of ethane-1,2-diol or any of a class of organic compounds, with the general formula $(CH_2)_n(OH)_2$ of the alcohol family in which two hydroxyl groups are attached to different carbon atoms. It is used as antifreeze material in engines. Examples of glycol are propylene glycol (1,2-propanediol), used in foods, cosmetics, as a solvent, preservative and moisture-retaining agent; 1,3-butanediol and 1,4-butanediol are used as raw materials for plastics and other chemicals; 2-ethyl-1,3-hexanediol is used as an insect repellent; and 2-methyl-2-propyl-1,3-propanediol is used as a raw material of the tranquiliser meprobamate.

GLYCOLIC ACID (HYDROXYETHANOIC ACID; 2-HYDROXYETHANOIC ACID; DICARBOUS ACID; GLYCOLIC ACID; HYDROXYACETIC ACID): A colourless, odourless and hygroscopic crystalline compound, with the molecular formula $C_2H_4O_3$, density 1.27 g/cm^3 , melting point 75°C (348 K) and decomposes at boiling point. It is highly soluble in water. It is a

substrate that is oxidised during photorespiration in C_3 plants. It is formed by the oxidation of ribulose bisphosphate to phosphoglyceric acid and phosphoglycolic acid. It occurs in sugar cane and sugar beet. It can be made by the electrolytic reduction of oxalic acid or by boiling a solution of sodium monochloroethanoate. It is used in various skin-care products, in making textiles and leather and for cleaning metals.

GLYCOLIPID: Any one of a group of sugar-containing lipids, present in cell membranes, in which the lipid portion of the molecule is usually based on glycerol or sphingosine, the sugar is typically galactose, glucose or inositol. Examples are glycosphingolipids, which is the most abundant and structurally diverse type of glycolipids in animals; galactosyl ceramide, which is enriched in brain tissue and is a major component of the myelin sheaths around nerves; and glucosyl ceramide, which is present in the cell membranes of many cell types and abundant in serum.

GLYCOLS: Any of various alcohols containing two hydroxyl groups on different carbon atoms, usually but not necessarily adjacent. Examples are ethylene glycol (ethane-1,2-diol) and butane-1,4-diol.

GLYCOLYSIS: The series of 10 biochemical reactions taking place in most cells, in which a molecule glucose is broken down to pyruvate with the release of usable energy in the form of adenosine triphosphate (ATP) in the absence of oxygen. It occurs in the cytoplasm. Glycolysis produces both ATP for cellular energy requirements and building blocks for synthesis of other cellular products.

GLYCOMICS: The study of the entire set of glycomes, consisting of all types of sugars, which are synthesised by an organism.

GLYCOPHYTE: Any plant such as beans and rice that thrives in soils with a low concentration of salt. In contrast, halophytes are plants that tolerate a high concentration of salt in soils.

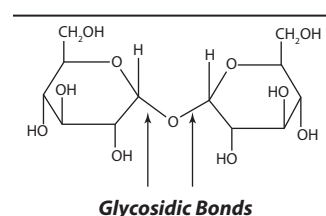
GLYCOPROTEIN (GLYCOPEPTIDES): Any of a class of proteins which have carbohydrate groups attached to the polypeptide chain or a protein linked to an oligosaccharide or a polysaccharide. Glycoproteins are small molecules found on the outside surface of all body cells and includes most secreted proteins (except serum albumin) and proteins exposed at the outer surface of the plasma membrane. Different types of cells have specific, unique types of glycoproteins attached to them. Glycoproteins are vital to a number of important biological functions. They allow certain types of cell-to-cell communication, help coordinate complicated cellular responses to stimuli and activate the action of other types of cells.

GLYCOSAMINOGLYCAN (GAGs; MUCOPOLYSACCHARIDES): Polysaccharide side chain of proteoglycans made up of repeating disaccharide units consisting of a hexose, a six-carbon sugar or a hexuronic acid, linked to a hexosamine, a six-carbon sugar containing nitrogen.

GLYCOSIDASE: The enzyme that is involved in hydrolising glycoside bonds between two sugar molecules.

GLYCOSIDE: Any one of a group of naturally occurring organic compounds in plants, broken down by plant enzymes into sugars. It consists of a pyranose sugar residue such as glucose, linked to a non-carbohydrate residue (R) by a glycosidic bond: the hydroxyl group (-OH) on carbon-1 of the sugar is replaced by -OR.

GLYCOSIDIC BOND (GLYCOSIDIC LINK): The type of functional group that links the monosaccharide units of disaccharide, oligosaccharides and polysaccharides and the hydroxyl group of an organic compound such as an alcohol.



GLYCOSYL GROUP: The structure obtained by removing the hydroxyl group from the hemiacetal function of a monosaccharide and, by extension, of a lower oligosaccharide.

GLYCOSYLAMINES: Compounds that have a glycosyl group attached to an amino group, NR_2 .

GLYCOSYLATION: The process in which a saccharide is joined to another molecule, such as a protein to form a glycoprotein or to a lipid to form a glycolipid. It occurs in the rough endoplasmic reticulum and the Golgi apparatus of cells.

GLYOXAL (ETHANE-1,2-DIONE; ETHANEDAL): A yellow crystalline solid, with the chemical formula $(\text{CHO})_2$, density 1.27 g/cm^3 , melting point 15°C (288 K) and boiling point 51°C (324 K). It is a dialdehyde and gives off green vapour when heated, burning with a violet-coloured flame. It is used to harden the gelatin employed for photographic emulsions, in embalming fluids, for leather tanning and for rayon shrink proofing.

GLYOXYLATE CYCLE (GLYOXALATE CYCLE): A cyclic series of metabolic reactions involved in photorespiration of plants, including oxidation of glycolate to glyoxyate by the enzyme glycolate oxidase within microbodies or peroxisomes. The glyoxylate cycle avoids the carbon dioxide forming steps of the TCA cycle and thus allows net synthesis of carbohydrates from fatty acids via succinate. The enzymes of the glyoxylate cycle are active in germinating oil-bearing seeds and in other plant tissues that metabolise fat. The cycle also takes place in microorganisms but not in higher animals.

GLYOXYSOME: A specialised cell part used in plant metabolism containing the five enzymes of the glyoxylate cycle and the β -oxidation cycle in addition to catalase and oxidase. They are, therefore, found in those cells of higher plants where fats are utilised in this way, for example, in the endosperm or cotyledons of fat-storing seeds. They also contain an enzyme that catalyses the decomposition of hydrogen peroxide to water and oxygen.

GLYPHOSATE N- (PHOSPHONOMETHYL) GLYCINE: A herbicide that kills a wide range of weeds, especially perennials but shows little persistence in soil and has low toxicity on animals.

GNATHOSTOMATA: A subphylum or superclass of chordates consisting of all vertebrates including amphibians, reptiles, birds and mammals that possess upper and lower jaws, teeth, paired appendages and girdles (secondarily lost in some), as well as other advanced features. This is in contrast to the more primitive Agnatha such as the lampreys and hagfishes, which lack jaws and paired appendages. The group has three groupings: **Chondrichthyes** or the cartilaginous fish; **Placodermi**, an extinct clade of armoured fish; and **Teleostomi**, which includes the familiar classes of bony fish, birds, mammals, reptiles and amphibians.

GNEISS: A coarse-grained metamorphic rock formed under conditions of increasing temperature and pressure, in which the minerals have been segregated into parallel layers, creating a banded or laminated structure. It often occurs in association with schists, diorite and granites and also minerals such as biotite and hornblende. It is named after the medium or type of rock in which it occurs, for example, granite gneiss, diorite gneiss and biotite gneiss.

GNETALES: A group of seed plants of the class Gnetales in the division Gnetales. Presently, there are three living genera, which are Gnetum, Ephedra and Welwitschia. They are practically unknown as fossils but considered close to the ancestral line of angiosperms.

GNOMON: In geometry, the figure remaining after a parallelogram has been removed from one corner of a larger similar parallelogram.

GNOME: Acronym for **GNU Network Object Model Environment**, it is a desktop environment made up of free and open source software. It is intended to provide a user-friendly suite of applications and an easy-to-use desktop.

GNOMON MAGIC SQUARE: A 4×4 array in which the elements in each 2×2 corner have the same sum. An example is the Dürer's magic square (shown in the diagram), in which the sums in four quadrants and also the sum of the middle four numbers are all 34.

16	3	2	13
5	10	11	8
9	6	7	12
4	15	14	1

*Dürer's magic square
(an example of Gnomon
magic square)*

GNOMONIC PROJECTION:

The projection of a sphere from its centre onto any tangent plane; for example, the gnomonic projection of a circle on the surface of a sphere onto an ellipse on a tangent plane.

GNOTOBIOTIC: Referring to organisms, born in aseptic conditions and removed from the mother by Caesarean section. They are reared in the laboratory, exposed to or inoculated with known strains of microorganisms that the researchers wish to be present in the animal and live either in a germ-free or a controlled environment.

GNU: 1. A Unix-like OS free for use. It was started in 1983 by Richard Stallman and is part of the Free Software Foundation. 2. Also known as **wildebeest**, it is a type of large antelope, genus *Connochaetes* with a long tail and horns that curve to the sides. It stands higher at the shoulder than at the rump. They live in often large, constantly moving herds and graze on the grasses and scrub of open plains.

GNU ZIP (GZip): In computing, free and open source compression software, commonly used to compress web pages on the server end for decompression in the browser. Files compressed using GZip can be recognised by the file extension 'GZ'. The software is published by the Free Software Foundation and was originally developed for Unix and is also available for other operating systems, including Windows and Mac OS.

GNUTELLA: A file sharing system, which allows networked users to share files on each other's computers. It is an open source software.

GOBLET CELL: Specialised cells in certain epithelia, in intestinal and respiratory tracts of mammals and also in the epidermis of fish. Its function is to secrete mucin, which dissolves in water to form mucus.

GÖDEL NUMBER: A function used by Kurt Gödel for the proof of his incompleteness theorem and uniquely associated with every formula of a formal theory. When used to study statements in arithmetic, it is a unique number for a given statement that can be formed as the product of successive primes raised to the power of the number corresponding to the individual symbols that comprise the sentence. For example, the statement $(\exists x)(x = s y)$ that reads "there exists an x such that x is the immediate successor of y " can be coded $2^8.3^4.5^{13}.7^9.11^8.13^{13}.17^5.19^7.23^{16}.29^9$, where the numbers in the set (8, 4, 13, 9, 8, 13, 5, 7, 16, 9) correspond to the symbols that make up $(\exists x)(x = s y)$. Prime-power coding using Gödel numbers can be used to encode Turing machine rules.

GÖDEL, KURT (1906 - 1978): An Austrian born U.S mathematician who developed Gödel's incompleteness and completeness theorems. He proved the consistency of the continuum hypothesis with the axioms of Zermelo-Fraenkel set theory; and developed the theory of recursive functions.

GÖDEL'S COMPLETENESS THEOREM (COMPLETENESS THEOREM): A theorem in mathematical logic that indicates a correspondence between semantic truth and syntactic provability in first-order logic. It is the result that a theory is consistent if and only if it has a model. It was first proved by Kurt Gödel in 1929.

GÖDEL'S INCOMPLETENESS THEOREMS (GÖDEL'S THEOREM): Two theorems of mathematical logic that establish inherent limitations of all but the most trivial axiomatic systems for mathematics. The propositions are undecidable, using only the axioms of that

mathematical structure. The first incompleteness theorem states that no consistent system of axioms whose theorems can be listed by an "effective procedure" essentially, a computer program is capable of proving all facts about the natural numbers. For any such system, there will always be statements about the natural numbers that are true, but that cannot be proved within the system. The second incompleteness theorem shows that if such a system is also capable of proving certain basic facts about the natural numbers, then one particular arithmetic truth the system cannot prove is the consistency of the system itself. The theorems, proven by Kurt Gödel in 1931, are important both in mathematical logic and in the philosophy of mathematics.

GÖDEL STATEMENT: A statement that asserts its own unprovability, especially that used in Gödel's theorems and given in terms of its Gödel number.

GOETHITE: A yellow-brown mineral, with the chemical formula $\text{FeO} \cdot \text{OH}$ or $\text{Fe}^{3+}\text{O}(\text{OH})$. It was named after the German, Johann Wolfgang von Goethe. Crystallising in the orthorhombic system, it is formed as a result of oxidation and hydration of iron minerals. It is also precipitated by groundwater. Goethite is considered as the most important ore of iron after haematite. It is used in eliminating the chemical element, cadmium, from polluted waters found in industrial liquid trash. It has also got uses in medicine and also used as a pigment.

GOGGLES: A close-fitting pair of special lenses fixed in a flexible frame, made of materials such as rubber or plastic for protecting injury to the eyes from glare, dust, water, chemicals, strong wind, flying objects, blinding light, chemical flash, etc. They are often worn in swimming pools, laboratories, workshops, etc. Some specially made ones are also used for night-vision or against stereoscopic effects.

GOITRE: A swelling of the thyroid gland at the front of the neck caused by a deficiency of iodine, which causes a reduction in thyroid hormone synthesis. Another cause is defects in thyroid hormone synthesis and inflammation of the thyroid.

GOLAY CELL: A transparent cell containing gas, used to detect infrared radiation, in which radiation absorbed in a gas chamber heats the gas, causing a rise in the gas temperature and pressure. The amount of incident radiation can be measured from the pressure rise in the tube.

GOLD: A soft, yellow, corrosion-resistant, unreactive element transition metallic element, with the symbol Au, atomic number 79, atomic weight 196.967, melting point 1063 °C (1336 K), boiling point 2966 °C (3239 K) and specific gravity 19.32. It is the most malleable and ductile metal and a good thermal and electrical conductor. It does not tarnish and used for jewellery, dentistry and to coat parts of some electronic devices and also as an international monetary standard. Gold ore contains particles of the free metal.

GOLD, THOMAS (1920-2004): An Austrian-American astrophysicist and a professor of astronomy at Cornell University, a member of the U.S. National Academy of Sciences and a Fellow of the Royal Society (London). Gold was one of three young Cambridge scientists, who in the 1950s proposed the now mostly abandoned 'steady state' hypothesis of the universe. In the late 1960s he correctly interpreted the newly discovered pulsars in terms of rotating neutron stars. Gold's work crossed academic and scientific boundaries into biophysics, astronomy, aerospace engineering and geophysics.

GOLDBACH'S CONJECTURE: Named after German mathematician Christian Goldbach (1690- 1764), it is the conjecture, which says that every even integer greater than 2 is the sum of two prime numbers. It remains one of the unsolved problems in number theory.

GOLDEN RECTANGLE: A triangular geometric figure that can be cut up into a square and a rectangle similar to the original one. The longer side is constructed with compass and straightedge from the shorter

side. The golden triangle has aesthetic appeals and is frequently found in art and architecture.

GOLDEN RATIO: A proportion of the golden triangle in which a line is divided so that the ratio of the shorter part to the longer part equals the ratio of the longer part to the entire line. Its symbol is ϕ . For example, if 1 represents the length of the shorter part, the solution to the equation $\frac{\phi+1}{\phi} = \frac{\phi}{1}$ is given as $\phi = \frac{1+\sqrt{5}}{2} = 1.61803\dots$

GOLDMAN EQUATION (GOLDMAN-HODGKIN-KATZ VOLTAGE EQUATION): An equation used in cell membrane physiology, which expresses the quantitative relationship between the concentrations of charged species on either side of a membrane and the resting trans-membrane potential.

GOLDSCHMIDT PROCESS: A method of extracting metals, discovered by the German chemist Hans Goldschmidt (1861–1923) by reducing the oxide with aluminium powder, for example, $\text{Cr}_2\text{O}_3 + 2\text{Al} \rightarrow 2\text{Cr} + \text{Al}_2\text{O}_3$. The reaction can also be used to produce molten iron.

GOLDSTONE'S THEOREM: A theorem in relativistic quantum field theory which states that if there is an exact continuous symmetry of the Hamiltonian or Lagrangian defining the system and this is not an asymmetry of the vacuum state (there is broken symmetry), then there must be at least one spin-zero massless particle called a Goldstone boson (Nambu-Goldstone bosons).

GOLGI APPARATUS (GOLGI BODY; GOLGI COMPLEX): A network of vesicles and membranous sacs within the cytoplasm of all eukaryotic cells, but absent from prokaryotes such as bacteria. It distributes, packages and stores the lipids and proteins made in the endoplasmic reticulum for delivery to the plasma membrane for secretion or to destinations within the cell. It was named after the Italian histologist, Camillo Golgi. The Golgi apparatus sorts newly synthesised proteins for delivery to various destinations and modifies the oligosaccharide chains found on glycoproteins and glycolipids.

GOLGI TENDON ORGAN: A small sensory receptor organ located at the junction between a skeletal muscle and its tendon. It consists of small bundles of tendon fibres enclosed in a layered capsule with dendrites coiling between and around the fibres. Its function is to detect contraction of the muscle. The organ is activated by muscular contractions that stretch the tendons.

GOMPERTZ CURVE: A curve used to describe the size of a population (y) as a function of time (t), where the relative growth rate declines at a constant rate. This is given by $y = ae^{-bt}$.

GONAD: A sex gland such as ovaries and testes that produces gametes (sperms, ova) and hormones in animals. The ovary produces the ovum and the testis produces the spermatozoon.

GONADOTROPHIN (GONADOTROPHIC HORMONE): A hormone such as follicle-stimulating hormone (FSH) and luteinizing hormone (LH) secreted by the pituitary gland. The FSH initiates sperm production in men and the development of the ovarian follicles in women and the LH initiates the production of ova by these follicles. In men, LH causes the testes to produce the hormone testosterone. Another gonadotrophin, called **chorionic gonadotrophin**, found in the urine of pregnant women is secreted in women by the placenta and functions to maintain suitable conditions within the uterus for the foetus as it develops.

GONIOMETER: An instrument used to measure angles, particularly faces between crystals that allows an object to be rotated to a precise angular position.

GONORRHOEA: A sexually transmitted disease caused by a spherical bacterium *Neisseria gonorrhoeae* (*gonococcus*), which causes pain in the sexual organs and which can be passed on from one person to another during sexual intercourse. Symptoms usually develop within about 10

days of being infected, although they sometimes may not appear until many months later. Symptoms in men include burning sensations on urination, discharge of pus, inflammation of the foreskin and rarely pain or tenderness in the testicles. In women, the symptoms include vaginal discharge, pain or a burning sensation when passing urine, less commonly pain or tenderness in the lower abdominal area, etc.

GOOCH CRUCIBLE: A porcelain dish with a perforated base over which a layer of asbestos is placed and used for filtration in gravimetric analysis. It was named after Frank Austen Gooch.

GOOD AGRICULTURAL AND ENVIRONMENTAL CONDITION

(GAEC): Certain standards which farmers are obliged to follow, which include maintaining their land in good agricultural and environmental condition. This concept includes the following: the protection of soil against erosion, the maintenance of soil organic matter and soil structure, and the safe-guarding of landscape features.

GOODNESS OF FIT: A procedure in statistics referred to as chi square (χ^2) which is used to test the goodness of fit of a set of observed frequencies to a distribution which is hypothesised. The distribution can be normal, Poisson or binomial. To use the chi square the frequency must be greater than 50, also the individual frequencies

must also be greater than 5. This is given by $\chi^2 = \sum_{i=1}^n \frac{(O_i - E_i)^2}{E_i}$, where

O_i is the observed frequencies and E_i is the expected frequencies.

GOODPUT: In computer networks, it refers to the application level throughput, which is the number of useful bits per unit of time forwarded by the network from a certain source address to a certain destination, excluding protocol overhead and excluding retransmitted data packets. For example, a 5 megabyte file may require 300 kilobytes of header information and acknowledgements to be sent during the data transfer process. Therefore, the throughput would be roughly 5.3 megabytes divided by the transfer time. The goodput would be the original 5 megabytes divided by the transfer time. Therefore, goodput is always less than or equal to the throughput measurement.

GOOGLE: The World-Wide Web search engine that indexes the greatest number of web pages and provides a free service that searches this index in less than a second. The search engine was started by Larry Page and Sergey Brin, two Ph.D. students at Stanford University in 1996. They developed a search engine algorithm that ranked Web pages not just by content and keywords, but by how many other Web pages linked to each page. Every day Google answers more than one billion questions from people around the globe in 181 countries and 146 languages. To google is to search for information on the Internet using this search engine. **Google Drive** is a service offered by Google that allows users to store and share files online. The service was launched on April 24, 2012 and provides 5 GB of free storage. Additional storage can be purchased for a monthly fee. **Google Chrome** is a Web browser from Google that was introduced for Windows in 2008 and the Mac and Linux in 2009. Google Chrome fully integrates into its online search system as well as other applications. Based on the open source Chromium browser, Chrome offers innovative features such as integrating the domain name address bar and search box into one input field. Also included is a display of "most visited pages" when a new tab is opened and a tab can be converted to a new window by dragging it off the tab bar. The memory usage for each tab can be viewed and misbehaving tabs can be closed with Chrome's equivalent to a task manager function. **Google Chrome OS** is an open source lightweight operating system (OS). It uses one-sixtieth as much hard drive space as Windows 7 and is intended for netbooks or tablet PCs that access Web-based applications and stored data from remote servers.

GOOGLE SCHOLAR: A freely accessible search engine that provides access to scholarly works from a variety of sources, such as books, abstracts, online repositories, peer-reviewed articles, professional

societies, publishers, technical reports, theses, universities and other web sites.

GOOGLE+: A popular social media site that is owned and operated by Google that was launched on December 15, 2011. As of August 2018, it has 111,000,000 monthly active users.

GOOGOL: The number 1 followed by a hundred zeros (10^{100}), approximately 70! (factorial of 70). Googol is a large number that exceeds the number of atoms in the observable universe, which is between 10^{78} to 10^{82} atoms. The term was introduced by Milton Sirota in 1920 at the age of nine, when asked by his uncle, Edward Kasner (1878-1955), an American mathematician to suggest a name to such a large number. **Googolplex** is a term for 10 to the power of googol, 1 followed by 10 to the power of 100 zeros numeral.

GOPHER: A computer program designed for distributing, searching and retrieving documents over the internet.

GORGE (CANYON): A deep narrow valley or chasm running through hills or mountains often carved from the landscape by a river or stream. Examples are the Akosombo and Bui gorges in Ghana, where the Volta River runs through and has been used to run plants for hydroelectric power generation; and the Grand Canyon of the Colorado River in Arizona. A gorge may be formed as a waterfall retreats upstream, as a result of the collapse of caves or when the down cutting power of the river is greater than the processes of valley-wall erosion.

GOURD: The hard-shelled fruit of a trailing or climbing plant of the family Cucurbitaceae and order Violales. They are annual herbaceous plants native to temperate and tropical areas. Gourd vines are capable of climbing over 3.5 metres and as annual plants, survive only one growing season. Many varieties are cultivated either as ornamental plants or to provide dried bottle-like containers which can be used as utensils such as water carriers. The two most commonly grown types of gourd are Lagenaria and Cucurbita species. The Cucurbita species produces a variety of gourds, melons, squashes (vegetable marrows) and pumpkins. The five domesticated species include *Cucurbita argyrosperma*, *Cucurbita ficifolia*, *Cucurbita maxima*, *Cucurbita moschata* and *Cucurbita pepo*. *Lagenaria siceraria*, bottle gourd is a vine grown for its fruit, which can either be harvested young and used as a vegetable or harvested mature, dried and used as a bottle or utensil in many countries.

GOUT: A form of acute arthritis that causes severe pain and swelling in the joints. It most commonly affects the big toe, but may also affect the heel, ankle, hand, wrist or elbow. It affects the spine often enough to be a factor in back pain. Gout is different from other forms of arthritis because it occurs when there are high levels of uric acid circulating in the blood, which can cause urate crystals to settle in the tissues of the joints.

GOUT FLY (CHLOROPID GOUT FLY; BARLEY GOUT FLY): A small fly, *Chlorops pumilionis*, of the family Chloropidae with a straw yellow thorax and black ocular triangle head. It is a pest of many cereal crops such as wheat, barley, rye and oats.

GOUY BALANCE: A method of measuring diamagnetic and susceptibility of solid, liquid and solution samples. The sample is suspended from a balance with the bottom part of the sample between the poles of an electromagnet. When the magnetic field is switched on, the sample experiences a field gradient causing an apparent change in weight.

GOUY-CHAPMAN MODEL: A model of the electrical double layer in which thermal motion causing disordering of the layer is taken into account.

GOVERNOR: A device that controls and maintains the speed of an engine at constant revolutions per minute despite variations in the load, using the principle of negative feedback. When an alternator

supplies electrical power at a given frequency, a governor must be used to maintain the prime mover at a constant speed to prevent fluctuations in voltage and frequency by controlling the fuel flow so that the speed of the prime mover remains constant regardless of load and other disturbances.

GPRS (GENERAL PACKET RADIO SOURCE): A packet-based wireless communication service that provides increased bit rates (56-115 Kbps), always on capabilities that allow users to remain connected to the wireless network and improved user experience. It is a technology between 2G and 3G networks; hence it is referred to as a 2.5G network.

GRAAFIAN FOLLICLE (OVARIAN FOLLICLE; SECONDARY FOLLICLE; VESICULAR OVARIAN FOLLICLE): The female reproductive unit composed of an oocyte, granulosa cells and theca layers found inside the ovary. The oocyte inside the follicle would mature into an ovum or egg cell, which is then released from the ovary into the fallopian tube during ovulation.

GRABEN: An elongated block of rock that sinks between a pair of parallel near-vertical faults when they move slightly outwards.

GRACEFUL GRAPH: A graph whose vertical vertices can be numbered with the integers from 1 to n and each edge labelled with the difference between the numbers at the vertices, in such a way that each edge receives a different label.

GRAD (GRADIENT; DEL): A simple generalisation of the one dimensional gradient. It behaves in a similar manner but also obeys both vector laws (adding, subtracting and multiplying vectors) and operator laws such as differentiating a product, quotient and so on.

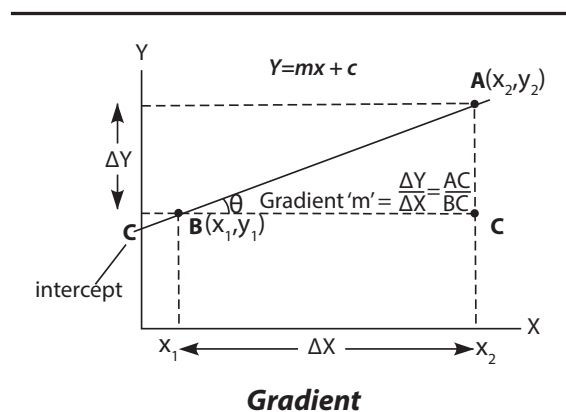
GRADE: 1. Animal that is not purebred or of unknown ancestry or a group of organisms that share certain morphological characteristics but are not necessarily related in evolutionary terms. An example is organisms that possess a codon and can be regarded as a grade. 2. A unit used in measuring angles.

GRADED POTENTIAL: A local voltage change in a neuron membrane induced by stimulation of a neuron. The magnitude of the potential depends on the strength and frequency of the stimulus and having a duration of about a millisecond. These potentials can carry for only a short distance down the plasma membrane and get weaker the farther they travel.

GRADIENT: 1. In mathematics, often denoted **grad**, it is the slope of a straight or curved line at any given point, represented by the slope of the tangent at that point. In Cartesian coordinates, a straight line, $y = mx + c$, has a gradient m , which is given by

$$m = \frac{y_1 - y_2}{x_1 - x_2} = \frac{\Delta y}{\Delta x} = \tan \theta, \text{ or } m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{\Delta y}{\Delta x} = \tan \theta, \text{ where } \theta \text{ is the}$$

angle the line makes with the x axis (horizontal axis). For a curve, $y = f(x)$, the gradient at a point is the derivative dy/dx at that point, i.e. the slope of the tangent to the curve at that point. For a curve, $y = f(x)$, the gradient at a point is the derivative dy/dx at that point, i.e. the slope of the tangent to the curve at that point. 2. The vector whose components parallel to each coordinate axis are the partial derivatives of a given function with respect to the variable mapped on that axis and whose direction is that in which the derivative of the function has its maximum value. In other words, the gradient of a function f on the set of real numbers R^n is defined as the vector field $\sum_{i=1}^n \frac{\partial f}{\partial x_i}(x) \frac{\partial}{\partial x_i}$. 2. In physics, it is the rate at which a physical quantity, such as temperature or pressure, increases or decreases relative to change in a given variable, especially distance. 3. In biology, it is a series of progressively increasing or decreasing differences in the growth rate, metabolism or physiological activity of a cell organ or organism.



GRADIENT ELUTION: A continuous change of solvent composition to increase eluent strength in liquid chromatography (high performance liquid chromatography, HPLC), either by gradually mixing a second solvent of greater eluting power with the first, less powerful solvent or by a gradual change in pH or other property.

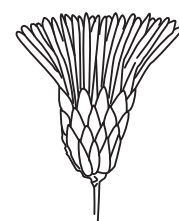
GRADIENT METHOD (STEEPEST DESCENT): In mathematics, an iterative method, due to Fermat, of finding the minimum of a real valued differentiable function of n variables by moving to the minimum of the function (an exact line search method) along the line in the direction of the steepest negative gradient (a steepest descent direction) at successive points on the curve or surface. This is the prototype of non-linear optimisation methods. The corresponding maximisation technique is called **steepest ascent**.

GRADIENT OPERATOR: The operator differential applied to a three-dimensional vector-valued function to yield a vector whose three components are the partial derivatives of the function with respect to its three variables. The symbol for gradient is a dot ($\cdot$). Thus, the gradient of a function f , written $\text{grad } f$ or $f_{\cdot}$ is $f_{\cdot} = i f_x + j f_y + k f_z$ where f_x , f_y and f_z are the first partial derivatives of f and the vectors i , j and k are the unit vectors of the vector space.

GRADING UP: Crossing a purebred male with a grade female.

GRADUALISM: A theory stating that evolutionary change of new species happens slowly by gradual accumulation of small genetic changes continuously over long periods of geological time.

GRADUATE: In botany, it is to divide into regular marked intervals as seen in the involucre bracts (minute scales or bristles) of some *Asteraceae* such as marigold, dandelion and sunflower.



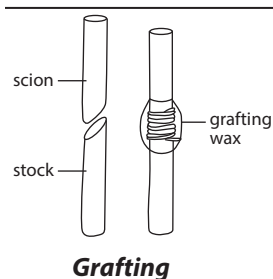
Graduate Involucre

GRADUATED CYLINDER: A narrow, cylindrical piece of laboratory equipment used to accurately measure the volume of a liquid. It is usually marked with horizontal lines to represent units of measurement.

GRAECO-LATIN SQUARES (EULER SQUARE; GRAECO-ROMAN SQUARES; LATIN-GRAECO SQUARES): A square array made by combining n objects of two types such that the first and second elements form a Latin square. For many years, Euler squares were known to exist for $n = 3, 4$ and for every odd n except $n = 3k$. Euler's **Graeco-Roman squares conjecture** maintained that there are no Euler squares of order $n = 4k + 2$ for $k = 1, 2, \dots$. However, such squares were found to exist in 1959 by Bose and Shrikande, refuting the conjecture.

GRAFT: 1. An isolated portion of living tissue that is removed and joined to another tissue, either in the same or a different organism, with the consequent growth resulting in the fusion of the tissues. In horticulture, it is a method of plant propagation in which a segment

(the scion) of the plant to be propagated is inserted onto another plant (the stock) in such a way that their vascular tissues combine, allowing growth of the grafted segment. A composition of rosin, beeswax and tallow, called grafting wax is used to glue and seal the wounds and scions of newly grafted trees to protect them from infection. The technique, which is mainly used on woody species, relies on the natural regenerative capacities of plants following wounding. Grafting is only successful between closely related species. 2. In polymer chemistry, it is a reaction in which one or more species of block are connected to the main chain of a macromolecule as side-chains having constitutional or configurational features that differ from those in the main chain.



GRAFT HYBRID: A type of plant chimaera that may be produced when a part of one plant (the scion) is grafted onto another plant of a different genetic constitution (the stock).

GRAFT MACROMOLECULE: A macromolecule with one or more species of block connected to the main chain as side chains. These side chains have constitutional or configurational features that differ from those in the main chain.

GRAFT POLYMER: A polymer composed of graft macromolecules.

GRAFT-VERSUS-HOST DISEASE (GVHD): A condition that may occur after bone marrow transplants in which the donated bone marrow immune cells regard the host's body cells as foreign and attack them.

GRAHAM, THOMAS (1805-69): Scottish chemist, best known for his research in diffusion of gases and liquids and in colloid chemistry. He became professor of chemistry at Glasgow University in 1830 and moved to the University College, London, in 1837. His 1829 paper on gaseous diffusion introduced Graham's law. He went on to study diffusion in liquids, leading in 1861 to the definition of colloids.

GRAHAM'S LAW OF DIFFUSION (GRAHAM'S LAW): A law in chemistry, formulated by Scottish physical chemist Thomas Graham, relating the diffusion rate of a gas to the inverse square root of its density. It states that at a constant temperature and pressure, the rate of diffusion of a gas is inversely proportional to the square root of its density. Thus, if the molecular weight of one gas is four times that of another, it would diffuse through a porous plug or escape through a small pinhole in a vessel at half the rate of the other. The formula is $R \propto \frac{1}{\sqrt{d}}$ or $R \propto \frac{1}{\sqrt{m}}$, where R is rate of diffusion, d is density and m = molar mass. The law is used in the diffusion method of separating isotopes.

GRAIN: 1. Also known as **grain crop** or **caryopsis**, it is a dry one-seeded fruit, usually containing a high percentage of starch, produced by, for example, corn, oats, barley, rye and other cereal crops. Grain crops have a high food value, store well and are a primary food stuff, contributing over half the world's calorie intake. 2. The smallest unit of mass based on a pound containing 16 ounces or 7,000 grains and equal to 453.59 grams. 3. An extruded length into which solid propellants are formed. 4. A tiny, solid particle, such as salt or sand. 5. The arrangement or direction of objects such as fibres, layers, particles of wood, leather, stone and paper.

GRAIN ALCOHOL (ALCOHOL; ETHANOL; FERMENTATION ALCOHOL; ETHYL ALCOHOL; ETHYL HYDROXIDE; METHYLCARBINOL): A colourless, limpid, volatile, flammable, toxic liquid with a pleasant odour and a pungent taste and the molecular formula C_2H_5OH , relative density 0.7851 at 20 °C, melting point at -112.3 °C (160.85) and boiling point at 78.4 °C (351.55 K). It is soluble in water and most organic liquids. It is produced by chemical synthesis and by fermentation or biosynthetic processes. It is the alcohol found in alcoholic drinks such as beer, wine and spirits. Apart from being a key input into alcoholic beverages, it is used as

a solvent, extractant, antifreeze and intermediate in the synthesis of innumerable organic chemicals.

GRAIN APHID: An insect which lives in crops such as wheat, barley, oats and corn. Grain aphids feed by piercing the plant and sucking out the plant juices.

GRAIN RESERVES: The amount of cereal grain held in a store by a country which is estimated to be above the country's requirements for one year.

GRAIN WEEVIL (GRAIN WEEVILS; GRANARY WEEVILS): A reddish-brown weevil, *Sitophilus granarius*, which lays eggs in stored grains. The larvae bore into the grain, where they feed and pupate.

GRAM (GRAMME): The metric system unit for mass or weight, defined as equal to the mass of one cubic centimetre of water at 4°C. Its value is equal to one-thousandth (1/1000) of a kilogram. Its symbol is *g*.

GRAM MATRIX: The self-adjoint matrix G whose i,j th entry is the inner product y_i, y_j of the i th and the j th elements of a given finite sequence of vectors, $\{y_j\}$ in an inner product space. Its determinant is called **Gram determinant**.

GRAM MOLECULAR WEIGHT: The weight in grams that is numerically equal to the molecular weight of a compound on a scale on which the atomic weight of the ^{12}C isotope of carbon is taken as 12 exactly. In SI, gram-molecular weight is replaced by the mole.

GRAM STAIN (GRAM METHOD): A process for staining micro-organisms, which is used to differentiate Gram negative bacteria, which are stained violet or blue; and Gram positive bacteria, which are stained pink. The bacterial sample is smeared on a microscope slide, stained with a violet dye, treated with acetone-alcohol (a decolouriser) and finally counterstained with a red dye.

GRAM-SCHMIDT PROCESS: An iterative method of converting any linearly independent family of vectors in an inner product space into an orthonormal system.

GRAMINEAE (POACEAE): A family containing the grasses, which number about 9000 species in about 620 genera including mainly herbaceous, but some woody plants consisting of cereals, bamboo, reeds and sugar cane. Grasses are the dominant vegetation in the savannahs, prairies and steppes. Economically they are the most important family of plants as they contain all the cereals, which form man's staple diet.

GRANA: A multilayered membrane unit formed by series of stacked thylakoid disks containing chlorophyll. It is found in the inner membrane of chloroplasts.

Grand UNIFIED BOOTLOADER (GNU GRUB; GRUB): A bootloader from the GNU Project with a graphical interface where a user can select, the desired operating system from the menu for booting an operating system. It is used for various Linux and Unix variants to boot and manage the booting of Linux and Unix operating systems on IBM compatible computers.

GRAND UNIFIED THEORY, GUT (THEORY OF EVERYTHING, ToE): Attempts to combine the three fundamental interactions: strong, weak and electromagnetic interactions into a single gauge theory with a single symmetry group. There are a number of different theories, most of which postulate that the three forces, with the exception of gravity, can be unified into one at some very high energy. The energy above which the interactions are the same is around 10^{15} GeV, which is much higher than those obtainable with existing accelerators.

GRANITE: A very hard crystalline rock with a medium to coarse grained texture, usually containing quartz crystals. Most types of granite are igneous rocks, although some are metamorphic.

GRANULAR: Consisting of small, round piece of grains or particles; grainy.

GRANULAR CARBON: A carbon material consisting of separate particles or grains which are monolithic, on the average larger than about 100 μm in diameter, but smaller than about 1 cm.

GRANULAR ENDOPLASMIC RETICULUM (ROUGH ENDOPLASMIC RETICULUM; ROUGH E.R.): A type of the

endoplasmic reticulum studded with ribosomes on its surface giving it a rough appearance under the microscope.

GRANULAR MATERIALS: A collection of distinct macroscopic particles, such as sand in an hourglass or peanuts in a container. The evolution of the particles follows Newton's equations, with repulsive forces between particles that are non-zero only when there is a contact between particles. Although granular materials are very simple to describe, they exhibit a tremendous amount of complex behaviour, much of which has not yet been satisfactorily explained. They behave differently than solids, liquids and gases which has led many to characterise granular materials as a new form of matter.

GRANULATION: The mottled appearance of the surface of the Sun when seen at high spatial resolution. It is caused by hot gases rising from the solar interior.

GRANULATION TISSUE: The bright red, granular tissue that develops during healing. The tissue consists initially of fine blood vessels and therefore bleeds easily. Later, fibrous tissue is laid down and a scar replaces the granulation tissue.

GRANULE: 1. A small grain or pellet; a particle. 2. In geology, it is a rock or mineral fragment larger than a sand grain and smaller than a pebble, between 2 and 4 millimetres in diameter. 3. In astronomy, it is one of the small, transient, brilliant markings in the photosphere of the Sun. 4. In biology, it is a cellular or cytoplasmic particle, especially one that stains readily.

GRANULITE: Granular metamorphic rock that is derived largely from quartz and feldspar. Granulites often have a banded appearance.

GRANULOCYTE: Any white blood cell that contains granular material and lysosomes in its cytoplasm found in the tissues during acute inflammatory processes and in the superficial surface aspects of a lesion during subacute or chronic inflammation.

GRANULOMA: A growth or nodule of connective tissue and capillaries usually associated with an infection such as tuberculosis or syphilis. Granulomas also occur in sarcoidosis.

GRANULOSA CELLS (GRANULAR CELLS; FOLLICULAR CELL): Secretory cells in a Graafian follicle that surround the developing female gamete, the oocyte in the ovary of mammals. The major functions of granulosa cells include the production of sex steroids, as well as myriad growth factors thought to interact with the oocyte during its development.

GRANUM (GRANA): A stacked membranous structure found in plant chloroplast, responsible for the light-dependent reactions of photosynthesis.

GRAPES: A flowering plant with about 10,000 varieties of the genus *Vitis* with flaky bark and palmately lobed alternate leaves. Fruits produced are berries known as grapes which turn dark purple to blackish with a pale wax bloom. Some species of grapes have bisexual flowers while others have male and female flowers on separate plants. Hence, cross pollination is required. Grapes typically occur in rain forests and along streams. The European grapevine species, native to the Mediterranean and Central Asia is *Vitis vinifera*. Some other species include *Vitis labrusca*, the North American table and grape juice grapevines; *Vitis riparia*, a wild vine of North America, used for winemaking and for jam; *Vitis rotundifolia*, used for jams and wine; and *Vitis amurensis*, the most important Asian species. Some species are seedless. The fruits are eaten fresh, processed into wine, jam or dried to produce raisins. The raisins contain concentrated sources of energy, vitamins, electrolytes and minerals. In addition, they are packed with several health benefiting poly phenolics anti-oxidants, dietary fibre and other phyto-nutrients and may be used to treat tuberculosis and constipation. In the past, unripe grapes were used to treat sore throats.

GRAPEFRUIT: A citrus tree, *Citrus paradisi*, of the rue family and its edible fruit, cultivated in the tropics. The fruit is lemon-yellow or

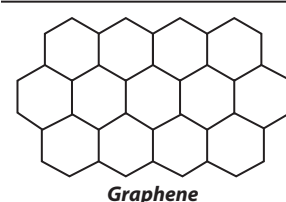
pink when ripe, about twice the size of an orange, with mildly acidic juicy pulp rich in vitamin C. The rind of the grapefruit is made into aromatherapy oil with a highly revitalising and elevating effect on the emotions. It is also used in the treatment of cellulites, dull skin, toxin build-up and water retention.

GRAPH: 1. A method of showing data pictorially or diagrammatically, that illustrates the relationship between two or more variables by means of a series of lines, points, etc., plotted with respect to a set of coordinate axes. 2. A drawing showing a functional relationship between two or more variables by means of a curve, surface, etc. and contains all and only those points whose coordinates satisfy the given relation. 3. The set of points (x, y) where $y = f(x)$, as contrasted with the function f . Also, the set of points (x, y) , where $y \neq f(x)$ as contrasted with the set-valued function. 4. A representation of a set of objects in which some pairs of objects are connected by links called vertices (also called nodes or points) and edges (also called arcs or lines).

GRAPH PAPER: Paper printed with intersecting lines, used as guides for drawing mathematical functions, charts, graphs or diagrams. The lines are usually equally spaced horizontally and vertically, making up a regular grid.

GRAPH THEORY: The mathematical field that deals with graphs and their properties having important applications in topology and in the construction of certain types of algorithms as used in some computer programs.

GRAPHENE: A virtually transparent allotrope of carbon with high electric conductivity, whose structure is one-atom-thick planar sheets of carbon atoms organised into a honeycomb-like crystal lattice arrangement forming a thin sheet that is one atom thick. Graphene is the basic structural element of some carbon allotropes including graphite, charcoal, carbon nanotubes and fullerenes. Graphene can be regarded as a single plane of graphite and is of interest because it has some unusual and potentially useful electrical properties. The material was discovered by Andre Geim, Russian-born British-Dutch physicist and Konstantin Novoselov, also Russian-born British physicist for which they won the 2010 Nobel Prize in Physics. The structure and properties of graphene make it suitable for a broad range of applications and it is being researched for use in such products as smart displays, ultrafast transistors, tough composites and quantum-dot computers.



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GRAPHENE LAYER: A single carbon layer of the graphite structure, describing its nature by analogy to a polycyclic aromatic hydrocarbon of quasi infinite size.

GRAPHIC: 1. In mathematics, using, pertaining to or determined by graphs. 2. In geology, having or denoting a texture formed by intergrowth of the crystals to resemble writing as in graphic granite.

GRAPHIC FILE FORMAT: A format in which graphical images are stored and transmitted. There are two main types: raster graphics, such as TIFF (Tagged Image File Format), JPEG (Joint Photographic Experts Group) and PNG (Portable Network Graphics) in which the image is stored as a bit map (arrangement of dots). The other is vector graphics, such as AI (Adobe Illustrator), PS (PostScript) and WMF (Windows Metafile) in which the image is stored using geometric formulae.

GRAPHIC ORGANISER: A visual learning tool with illustrations of concepts, information, and verbal statements. It can be used to enhance students' thinking skills by encouraging brainstorming, generating new ideas, connecting parts to the whole, drawing sequence, analysing causes and effects. It can take many forms useful in teaching inquiry process skills in science such as descriptive feature charts, T-charts, flow charts, Venn diagrams, tree diagrams, and semantic maps, and others

GRAPHICAL CALCULATOR: An electronic handheld calculator that has an inbuilt facility that is capable of plotting graphs, solving simultaneous equations and performing numerous other tasks with variables. Some can be attached to devices like electronic thermometers, pH gauges, weather instruments, decibel and light meters, accelerometers and other sensors and therefore function as data loggers.

GRAPHICAL REPRESENTATION: A wide variety of techniques that are used to clarify, interpret and analyse data by plotting points and drawing line segments, surfaces and other geometrical forms or symbols. The purpose of a graph is a rapid visualisation of data set. For example, it should clearly illustrate the general behaviour of the phenomenon investigated and highlight any important factors. It can be used, for example, as a means to translate or to complete a frequency table. Therefore, graphical representation is a form of data representation.

GRAPHICAL USER INTERPHASE (GUI): The interactive part of a computer software that allows the user to select commands, call up files, start programs and do other routine tasks by using a mouse to point to pictorial symbols (icons) or lists of menu choices on the screen as opposed to having to type in text commands.

GRAPHICS: Computer graphics are two or three-dimensional images displayed on a computer screen. Graphics are often contrasted with text, which is comprised of characters, such as numbers and letters, rather than images.

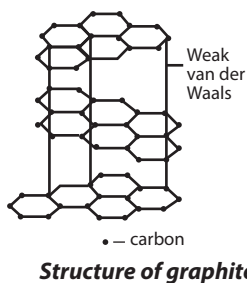
GRAPHICS CARD (GRAPHICS BOARD; VIDEO ADAPTER; VIDEO CARD): A printed circuit board that is plugged into a computer and takes binary data from the CPU and converts it into images to be displayed on a screen.

GRAPHICS INTERCHANGE FORMAT (GIF): A compressed image file format based on indexed colours, which is a palette of at most 256 colours. Together with JPEG (Joint Photographic Experts Group), they are the two most commonly used file formats for pictures on the World Wide Web because pictures saved in these formats take up relatively small amount of space and help greatly reduce their file size. These compressed image files can be quickly transmitted over a network or the Internet, often seen on Web pages. GIF files are useful for small icons and animated images, but they lack the colour range to be used for high-quality photos.

GRAPHICS PROCESSING UNIT (GPU): A single-chip processor used primarily for computing 3D functions such as lighting effects, object transformations and 3D motion.

GRAPHICS TABLET (DRAWING TABLET; DIGITIZER TABLET; PEN TABLET): A highly accurate input device that enables an artist to draw or sketch graphics, similar to the way one draws images with a pencil and paper and commonly used in computer-aided design. It is also used to convert hand-drawn images into a format suitable for computer processing. Images are usually drawn onto a flat surface with a stylus on a computer monitor or screen.

GRAPHITE (PLUMBAGO): A mineral allotrope of carbon and one of the few non-metals that can conduct electricity. It is the third allotrope of carbon. Its chemical formula is C and has a Mohs scale hardness of 1-2. It is soft, dark grey to black, opaque and slippery. In its structure, each carbon atom is sp^2 hybridized and has trigonal planar geometry. A carbon atom is connected to three others by strong sigma bonds using the three hybrid orbitals. The sigma bonded carbon atoms form hexagonal layers of strong covalent networks, arranged into flat, six-member rings as shown in the diagram. These layers are stacked on top of one another to give the three-dimensional structure. There is no actual bonding between the layers, except a loose high electron



system forming van der Waals forces of attraction between the layers. They therefore, can slide over one another, giving it a characteristic slippery feel. The distance between the layers is greater than the distance between the atoms within a layer, therefore graphite has a density of 2.22 grams per cubic centimetre, which is much less than that of diamond. It is used to make pencil leads; electrodes due to the presence of the loose pi electrons; and also high temperature lubricants; crucibles; polishes; arc lamps; batteries; brushes for electric motors; and nuclear reactor cores.

GRAPHITE-MODERATED REACTOR: A nuclear reactor in which graphite is the principal moderating material. They have been used in commercial electricity generation, gas-cooled reactors, water-cooled reactors, etc.

GRAPHITIC COMPOUNDS (GRAPHITE INTERCALATION COMPOUNDS, GICs): Complex materials having the general formula XC_n , in which atoms or molecules are trapped between the layers in graphite. **Graphitic carbon** defines all varieties of substances consisting of the element carbon in the allotropic form of graphite irrespective of the presence of structural defects.

GRAPHITISATION/GRAPHITIZATION: A solid-state transformation of thermodynamically unstable non-graphitic carbon into graphite by means of heat treatment. Graphitisation is also used for the transformation of metastable diamond into graphite by heat treatment, as well as in metallurgy for the formation of graphite from thermodynamically unstable carbides by thermal decomposition at high temperatures.

GRAPTOLITES (GRAPTOLITHINA): A group of extinct marine colonial hemichordates that was common in the Palaeozoic era. Graptolites secreted a protein exoskeleton commonly preserved as a carbon film in shales. An individual lived within a cuplike structure called a **theca** and multiple cups were spaced along one or more branches called **stipes**. The entire colony sometimes was connected, by a threadlike structure known as a **nema**. The preservation, quantity and gradual change over a geologic time scale of graptolites allows the fossils to be used to date strata of rocks throughout the world. They are important index fossils for dating Palaeozoic rocks as they evolved rapidly with time and formed many different species.

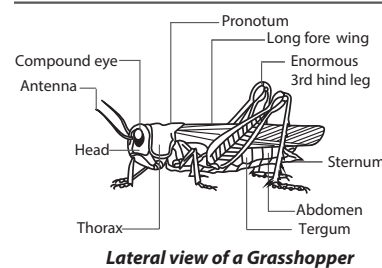
GRASHOF NUMBER (FREE CONVECTION NUMBER): A dimensionless number used in the study of free convection of a fluid caused by a hot body. It is named after the German engineer, Franz Grashof. It is equal to the product of the fluid's coefficient of thermal expansion, the temperature difference between the hot body and the fluid, the cube of a typical dimension of the body and the square of the fluid's density, divided by the square of the fluid's dynamic viscosity.

GRASS: A perennial, nonwoody plant with long narrow leaves, pointed hollow stems, hermaphrodite flowers, which are borne in spikelets and fruits that are grains. They make up the families Poaceae (formerly Gramineae), Cyperaceae (sedges) and Juncaceae (rushes). Examples are maize, bamboo, reed, sugar cane and lemon grasses.

GRASS SICKNESS: A sudden and usually fatal illness affecting sheep, horses and cattle. Symptoms include depression, inflamed membranes and discharge from the nostrils. There is currently no effective treatment.

GRASS WATERWAY: Strip of grass growing in the low area of a field where water can gather and cause erosion.

GRASSHOPPER: A medium to large plant-eating, leaping insect of the family Acrididae (short-horned grasshoppers) or Tettigoniidae (long-horned grasshoppers), both in the order Orthoptera. Short-horned grasshoppers have short antennae; and long-horned grasshoppers have long antennae. There are over 10,000 of known grasshopper species. They



are commonly found in tropical forests, semiarid regions and grasslands. Adult length is between 1 - 7 cm, depending on the species. The body is divided into three main parts: the head, the thorax and the abdomen. A hard shell called an exoskeleton covers the body. Grasshoppers usually have five compound eyes, which they use to see to the front, to the side and to the back. They are usually coloured a combination of brown, grey or green to blend into their environment. Some species sometimes have yellow or red markings to attract females. It has six jointed legs. The two elongated hind legs are used for jumping and the four front legs are primarily used to hold onto prey and for walking. Female grasshoppers are larger than the males and have sharp points at the end of their abdomen that help them to lay eggs underground. It has two pairs of wings. The forewings are narrow and relatively hard. The hind wings are large and membranous. Male grasshoppers sometimes have special structures on their wings that they use to rub their hind legs on or rub together to make sounds to attract females. They are most active during the day, but also feed at night. Some grasshopper species are pests of farm crops. Some of the predators include birds, lizards, mantids, spiders and rodents.

GRASSLAND: A major regional community (biome) in which grasses are the dominant vegetation. Usually the rainfall is insufficient for trees to grow (about 250-500 mm annually in temperate regions and 750-1500 mm in the tropics). Temperate grasslands contain other herbaceous plants, though trees are found only along rivers and streams. They include the North American prairies, the steppes of southwest Russia, the grasslands of Manchuria and Mongolia and the South American pampas. Tropical grasslands known as **savannah** are located near the equator, between the Tropic of Cancer and the Tropic of Capricorn. They cover much of Africa as well as large areas of Australia, South America and India. The soil of the savannah is porous, with rapid drainage of water. It has only a thin layer of humus, which provides vegetation with nutrients. Savannahs have tall coarse tufted grasses and there are often scattered trees. Different savannahs support different grasses due to disparities in rainfall and soil conditions. Also, because the savannah supports such a large number of species competing for living space, usually only one or a few kinds of grass are more successful than the others in a particular area. In Africa, the baobab (*Adansonia digitata*), Acacia and Euphorbia are common species found in the savannah. Some of the animals found in parts of tropical grassland include giraffes, zebras, buffaloes, kangaroos, mice, moles, gophers, ground squirrels, snakes, worms, termites, beetles, lions, leopards, hyenas and elephants.

GRATICULE (RETICLE): Fine lines in the focus of an optical instrument, which have been divided into squares, in order to reproduce it in other dimensions.

GRATING: 1. Partition, covering or frame of parallel or cross bars; a latticework resembling a window grate. 2. In optics, it is also called **diffraction grating** and it is a system of close equidistant and parallel lines or bars, especially lines ruled on a polished surface, used for producing spectra by diffraction.

GRAVEL: 1. A loose deposit of rock fragments or small fragments of rocks sometimes rounded by river erosion originating in the beds of streams or on beaches. Gravel is quarried for use in road-building, railway ballast and for an aggregate in concrete. 2. Hard particles in the kidney or bladder, much smaller than kidney stone, which can pass through the urinary tract.

GRAVIMETER: An instrument for detecting small variations in the gravitational field of the Earth or other body, frequently used in mineral and oil prospecting and in planetary exploration. Variations in the gravitational force on a mass suspended from a spring cause it to stretch or be deflected in a way which is then measured.

GRAVIMETRIC ANALYSIS: Quantitative chemical analysis made ultimately by weighing substances or samples. It is usually separated by precipitation or combustion, to a pure compound or element, of definite known composition.

GRAVIMETRY: The study of the Earth's gravitational field. It is used by geologists to measure the strength of a gravitational field and to map the sub-surface features of the Earth's crust.

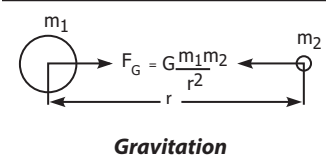
GRAVIPAUSE (NEUTRAL POINT): The point or boundary in space at which the gravitational force of one body is exactly balanced by the gravity pull of another.

GRAVIRECEPTOR: A highly specialised receptor organ and nerve ending in the inner ear, joint, tendon or muscle. They give the brain information about body position, equilibrium, direction of gravitational forces and the sensation of down or up.

GRAVISPHERE: The region within which the force of a given celestial body's gravity predominates over that of other celestial bodies.

GRAVITATE: 1. To obey the law of gravitation; to exert a force or pressure. 2. To tend to move, under the influence of gravitation. 3. To tend in any direction or toward any object.

GRAVITATION: 1. The natural phenomenon of attraction between physical objects with mass or energy. 2. The natural force that causes things such as fruits to fall towards the Earth. 3. The attraction between any two bodies in the universe, which are at a distance r apart from their centres. The greater the mass of each body and the closer they are, the greater the magnitude of the gravitational force and acts along the line joining them and form an action-reaction pair. It is



expressed as: $F_G = G \frac{m_1 m_2}{r^2}$, where m_1 and m_2 are the masses of the particles, r is the distance between their centres and G is a universal constant whose value is $G = 6.67 \times 10^{-11} \text{ N m}^2/\text{kg}^2$.

GRAVITATIONAL ACCELERATION: The acceleration imparted to a body by the attraction of the Earth. It is approximately equal to 980.7 cm/s^2 or 9.81 m/s^2 .

GRAVITATIONAL COLLAPSE: A phenomenon predicted by the general theory of relativity in which matter compressed beyond a critical density collapses as a result of gravitational attraction until it becomes a point singularity. The singularity resulting from gravitational collapse can be interpreted as indicating both the limits of the theory of general relativity and the necessity of constructing quantum gravity.

GRAVITATIONAL CONSTANT (UNIVERSAL GRAVITATIONAL CONSTANT): The numerical factor relating force, mass and distance in Newton's theory of gravitation that yields the force one body exerts on another when multiplied by the product of the masses of the two bodies and divided by the square of the distance between them. Its symbol is G . The attractive force, F , between two bodies of mass m_1 and m_2 and the distance between them, r is expressed

as $G = \frac{Fr^2}{m_1 m_2}$. The numerical value of G is expressed as

$$G = \frac{Fr^2}{m_1 m_2} = 6.67259(85) \times 10^{-11} \text{ m}^3 \text{kg}^{-1} \text{s}^{-2}.$$

GRAVITATIONAL ENERGY: The energy released by an object falling in a gravitational field. Gravitational energy is a form of potential energy.

GRAVITATIONAL EQUILIBRIUM: The state in a celestial body, such as a star, when gravitational forces pulling inward on a particle are balanced by some outward pressure, such as radiation pressure or electron degeneracy pressure, so that no vertical motion results.

GRAVITATIONAL FIELD: A region of space surrounding a body that has the property of mass. In this region, any other body that has mass will experience a force of attraction. The mass of Earth creates a gravitational field around it so anything that is inside the gravitational field of the Earth, would feel the gravitational force the Earth is exerting on it.

GRAVITATIONAL FIELD STRENGTH: The strength of the Earth's gravitational field at a particular point, defined as the gravitational force in Newton that acts on a mass of one kilogram. Its symbol is g . The gravitational field strength of the Earth is equal to the acceleration of free fall at its surface, with a value of 9.81ms^{-2} . It is expressed as gravitational force per unit mass on a point mass: $g = Fg/m$.

GRAVITATIONAL INSTABILITY: A situation in which an object's self-gravity exceeds opposing forces such as internal gas pressure or material rigidity, so that the object collapses. For a gas, gravitational instability sets in when the mass is greater than a certain critical value, known as the **Jean's mass**. In the early universe, instabilities were large enough to produce galaxies and clusters of galaxies.

GRAVITATIONAL LENSING: The bending of light rays by a gravitational field of a massive body producing distorted, magnified or multiple images of more distant objects as predicted by Einstein's general theory of relativity.

GRAVITATIONAL LOADING: The application of force to a bone or part of the body. Under normal circumstances on Earth, gravity "loads" the bones and muscles with the force required to support the body.

GRAVITATIONAL LOCK (CAPTURED ROTATION): The tying of the orbital period of a planet or moon to its axial period through a tidal effect caused by the gravitational pull of the primary. The gravitational lock experienced by the Moon, for example, explains why it always keeps the same face directed toward the Earth.

GRAVITATIONAL MASS: The mass of a body as measured by its gravitational attraction for other bodies. Different gravitational masses have different gravitational forces acting between them.

GRAVITATIONAL POTENTIAL ENERGY (GRAVITATIONAL ENERGY): Energy an object possesses because of its height in a gravitational field above Earth.

GRAVITATIONAL RED SHIFT (GRAVITATIONAL REDSHIFT; EINSTEIN SHIFT): In general theory of relativity, a prediction that the wavelength of electromagnetic radiation emitted by a source in a gravitational field will lengthen as it climbs out of a gravitational field.

GRAVITATIONAL WATER: Soil water of a temporary character that moves downward to the groundwater zone in response to gravity. It represents the amount of water held by the soil between saturation and field capacity. **Field capacity** is the amount of soil moisture or water content held in the soil after excess water has drained away and the rate of downward movement has decreased.

GRAVITATIONAL WAVES: Waves propagated through a gravitational field, which is produced by some change in the distribution of matter. It travels at the speed of light, exerting forces on masses in its path.

GRAVITON: A hypothetical particle with zero electric charge or colour charge and rest mass that is considered to be the quantum particle of the gravitational interaction.

GRAVITROPISM (GEOTROPISM): The growth response to gravity in plants. The upward growth of plant shoots is an instance of negative geotropism; and the downward growth of roots is positive geotropism.

GRAVITY (GRAVITATION): The force of attraction between bodies due to their mass. The gravitational force between two objects depends on their masses and the distance between them. The gravitational force between two bodies, m_1 and m_2 with distance d between them is defined by Newton's law as $Gm_1 m_2 / d^2$, where G is the universal gravitational constant, whose value is $6.673 \times 10^{-11} \text{N m}^2/\text{kg}^2$. Gravity is one of the four fundamental interactions of nature. The others are the strong force, electromagnetism and the weak force. Gravity is the agent that gives weight to objects with mass and causes them to fall to the ground when dropped. It also causes dispersed matter to coalesce, thus accounting for the existence of the Earth, the Sun and most of

the macroscopic objects in the universe. Gravitation is responsible for keeping the Earth and the other planets in their orbits around the Sun; for keeping the Moon in its orbit around the Earth; for the formation of tides; for natural convection, by which fluid flow occurs under the influence of a density gradient and gravity; for heating the interiors of forming stars and planets to very high temperatures; and for various other phenomena observed on Earth.

GRAVITY FEED: The use of the force of gravity to move materials from one location to another.

GRAVITY GRADIENT: The variation in the force of attraction between two bodies expressed as a function of the separation distance.

GRAVITY GRADIENT STABILISATION: A useful passive method to achieve stabilisation of satellites by using the gravity gradient of the primary body. An orbiting spacecraft will tend to align its long axis (more precisely, the axis of minimum moment of inertia) with the local vertical, i.e., in a radial direction.

GRAVURE: One of the three main printing methods, in which printing is done from etched plates or cylinders with a pattern of recessed cells in which the ink is held.

GRAY: The S.I. unit of absorbed dose of ionising radiation named after the British radiobiologist L. H. Gray (1905–65). It is defined as the dose absorbed when one kilogram of matter absorbs one joule of ionisation radiation. Its symbol is Gy and it replaces the rad (1 Gy equals 100 rad).

GRAY CODE: A binary number system, where two successive values differ in only one bit.

GRAZING: The predation method, in which herbivore such as cattle, feeds on growing grass, legumes or other plants.

GRAZING FOOD CHAIN: A cycle, in which vegetation is eaten by animals, digested, then passed into the soil as dung and so taken up again by plants, which are again eaten by animals.

GRAZING PRESSURE: The stress on plant populations due to the grazing of animals. Plants may be lost either directly as a result of animals grazing upon them or indirectly as a result of damage to the vegetation or ecosystem by animals moving through the area.

GRAZING-INCIDENCE TELESCOPE: A type of astronomical telescope designed to form images of celestial x-ray or gamma-ray sources.

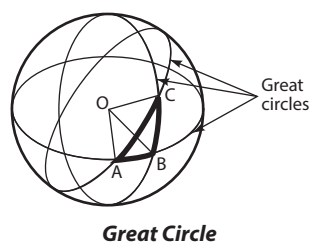
GREASE: 1. A semisolid lubricant possessing a higher initial viscosity than oil. It is used to reduce friction between surfaces from which oils would leak away or cause damage by dripping or where lubrication must be assured for extended periods. Some grease are mixtures of mineral oil and soap. The common ones contain a calcium-base soap that withstands water but not high temperature or a sodium-base soap that withstands higher temperatures and adheres well but dissolves in water. Other soaps used in greases have bases of lithium, aluminium, barium or strontium. Nonsoap thickeners include carbon black, which is unaffected by temperature and is therefore used with extreme low-temperature lubricants; silica gel; and bentonite, a clay developed for universal greases. 2. Rendered animal or vegetable fat that is soft after melting.

GREASEBAND: A strip of paper covered with a sticky substance, wrapped round the trunk of a tree to prevent pests from climbing up the tree.

GREASY PIG DISEASE: A bacterial disease which causes skin abrasions in young pigs caused by *Staphylococcus hyicus*, which can rapidly affect an entire litter.

GREAT ATTRACTOR: A huge concentration of mass, equivalent to about a million galaxies, beyond the Hydra and Centaurus constellations, observable by its effect on the motion of galaxies and their associated clusters over a region hundreds of millions of light years across. Our own galaxy and others near it are heading towards Greater Attractor at the rate of about 600km per second.

GREAT CIRCLE: 1. A circle drawn along the surface of a sphere such that the diameter of the circle is a diameter of the sphere. It is the intersection of a sphere and a plane which passes through the centre point of that sphere. It is the largest circle that can be drawn on a given sphere. All great circles of a given sphere have both the same circumference and the same centre as the sphere. Every circle is a great circle of exactly one sphere. 2. A line on the surface of the earth or its representation on a map. The shortest route between two points on the Earth's surface lies on a great circle.



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GREAT OBSERVATORIES: A series of four major NASA orbiting observatories covering different regions of the electromagnetic spectrum. They are Hubble Space Telescope, Compton Gamma Ray Observatory, Chandra X-ray Observatory and Spitzer Space Telescope.

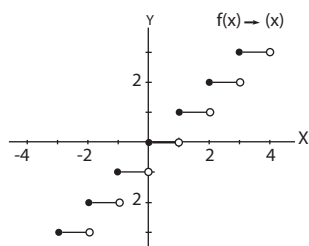
GREAT RIFT: A dark divide in the Milky Way between Cygnus and Sagittarius caused by a succession of large, overlapping clouds of interstellar dust in the equatorial plane of the Galaxy.

GREAT RIFT VALLEY: The largest rift valley on Earth, comprising a long depression in Southwest Asia and East Africa caused by the movement of tectonic plates, largely beginning during the Oligocene epoch. It extends from about 4,285 km from the Jordan valley in Syria into Mozambique. It is marked by a chain of lakes and a series of volcanoes, including Mount Kilimanjaro.

GREATEST: In mathematics, of an element of an ordering, uniquely maximal, that is, greater than every other member of the set. For example, the set of subsets of a given set ordered under inclusion has the given set itself as a greatest element.

GREATEST COMMON DIVISOR, GCD (GREATEST COMMON DENOMINATOR; GREATEST COMMON FACTOR, GCF; HIGHEST COMMON FACTOR, HCF): The largest integer that divides each number without a remainder in a sequence of integers. For example, the GCD of 8 and 12 is 4.

GREATEST INTEGER FUNCTION (FLOOR FUNCTION): The greatest integer function $f(x)$, with the symbol x gives the greatest integer less than or equal to a real number x . Its graph resembles a staircase.



A graph indicating the greatest integer function

GREATEST LOWER BOUND (INFIMUM): The greatest lower bound of a set of real numbers is the largest real number that is smaller than each of the numbers in the set.

GREEDY ALGORITHM: Any of a class of combinatorial algorithms, that looks for simple, easy-to-implement solutions to complex, multi-step problems by deciding which next step will provide the most obvious benefit. It works by recursively constructing a set of objects from the smallest possible constituent parts. It is used in ad hoc mobile networking to efficiently route packets with the fewest number of hops and the shortest delay possible; and also used in machine learning, business intelligence (BI), artificial intelligence (AI) and programming.

GREEN ALGAE: Organisms that belong to the phylum Chlorophyta of the kingdom Protista. It is a large group of marine and freshwater algae, which possess the green pigment chlorophyll. The group includes sea lettuce and spirogyra.

GREEN AUDIT (ENVIRONMENTAL AUDIT): The inspection of an organisation to assess its environmental compliance and management

system in accordance with environmental laws and regulations in order to prevent or minimise harm to the environment.

GREEN BELT: An area of land such as forest, recreational parks, farmland or uncultivated land, which surrounds an urban area and protected to prevent development.

GREEN COKE (RAW COKE): The primary solid carbonisation product from high boiling hydrocarbon fractions obtained at temperatures below 900 K. It contains a fraction of matter that can be released as volatiles during subsequent heat treatment at temperatures up to approximately 1600 K. This mass fraction, the so-called volatile matter, is in the case of green coke between 4 and 15 wt.%, but it depends also on the heating rate.

GREEN ELASTIC (HYPERELASTIC): Of a body, such that there is a scalar function of the deformation gradient whose rate of change is the stress-power per unit volume. Although all elastic bodies in nature have been found to be Green-elastic, in principle a Green-elastic body is elastic but not elastic.

GREEN MANURE: Fast-growing green vegetation such as mustard, grasses, legumes or rape which is ploughed into the soil to rot for its beneficial effects to the soil and subsequent crops. They add nitrogen to the soil, increase the general fertility level, reduce erosion, improve the physical condition of the soil and reduce nutrient loss from leaching.

GREEN MOVEMENT: A group of individuals and organisations such as involved in efforts to protect the environment. The movement includes political parties such as the Green Party and organisations like Friends of the Earth and Greenpeace.

GREEN ROUGHAGE: Plant materials with high moisture content such as pastures and root plants.

GREEN'S FORMULA FOR A PLANE: If C is the boundary of a region S and the functions $P(x, y)$ and $Q(x, y)$ are continuous together with their first order partial derivatives in the closed region $S + C$, then

Green's formula holds that $\oint_C P dx + Q dy = \iint_S \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy$. The

circulation about the contour, C , here is chosen so that the region, S should remain to the left.

GREEN'S FUNCTIONS: Named after George Green (1793-1841), English mathematician, it is a set of integrating kernels used in solving nonhomogeneous differential equations with boundary conditions and involving the fundamental solutions to the corresponding homogeneous differential equations. It finds application in many areas of theoretical physics including mechanics, electromagnetism, acoustics, solid-state physics, thermal physics and the theory of elementary particles.

GREEN'S THEOREM: 1. Named after George Green (1793-1841), English mathematician, it is the relationship between a line integral around a simple closed curve C and a double integral over the plane region D enclosed by C . If P and Q have continuous first order partial

derivatives on D then, $\int_C P dx + Q dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$. 2. The two-dimensional special case of the more general Kelvin-Stokes theorem

that $\int_{\partial\Omega} (f dx + g dy) = \iint_{\Omega} \frac{\partial f}{\partial y} dA$, when f and g are continuously

differentiable on a planar region containing a set n with reasonable boundary. The case where $f = -y, g = x$ gives a useful formula for the area of Ω . There are various corresponding Greens identities involving the Laplacian.

GREENFLY: A type of aphid of the family Aphididae that have mouthparts specially adapted for piercing and feed by sucking sap from plants. It is destructive to various cultivated plants.

GREENHOUSE: A glass or transparent plastic structure, in which plants are grown and provided with warmth, protection and light from the weather. It is heated up by incoming visible solar radiation from the Sun and is absorbed by plants, soil and other things inside the building.

GREENHOUSE EFFECT: The warming effect on the Earth's atmosphere by which radiative energy leaving a planetary surface is absorbed by some atmospheric gases, called greenhouse gases, thereby insulating and warming the planet. Without the thermal blanketing of the natural greenhouse effect, Earth's climate would be about 33 °C (306 K) cooler than at present, which most living organisms cannot survive.

GREENHOUSE GAS: A gas whose molecules contribute to the greenhouse effect and global warming by absorbing energy emitted or reflected from the surface of the Earth; or gases in an atmosphere that absorb and emit radiation within the thermal infrared range. These gases may include water vapour, carbon dioxide, methane, nitrous oxide and ozone. Greenhouse gases greatly affect the temperature of the Earth. Without them, Earth's surface would be on average about 33 °C (306 K) cooler than at present.

GREENING: 1. The process of planting trees and other vegetation in an area. 2. The process of becoming more aware or of increasing others' awareness, of the environment and environmental issues. 3. The process of turning green, which can occur, for example, when potatoes are left too long in the light.

GREENWICH MEAN TIME (GMT): The mean solar time on the zero line of the longitude passing through the Old Royal Observatory in Greenwich, London, from which time in other zones is calculated. It was replaced in 1986 by the Coordinated Universal Time (UTC), which is an internationally accepted standard for calculating time in other zones, based on International Atomic Time.

GREGORY'S SERIES (LIEBNIZ'S SERIES): The Maclaurin series for arctan: $\arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots$ It is the special case of which $x = 1$ sums to $\pi/4$.

GREISS TEST: A chemical analysis test for nitrates or nitrites. It was formerly widely used to detect the possible existence of gunshot residue.

GRENADINE: Having the colour of pomegranate juice.

GREY BODY: A body that emits radiation of all wavelengths at a given temperature, which is a constant fraction of the energy of black-body radiation of the same wavelength at the same temperature. Such an ideal body does not exist in practice but the assumption is a good approximation for many objects used in engineering. Emissivity, with the symbol ϵ is the relative ability of a material's surface to emit energy by radiation and a ratio of energy radiated by a particular material to energy radiated by a black body at the same temperature. A true black body would have an $\epsilon = 1$ while any real object would have $\epsilon < 1$. Emissivity is a dimensionless quantity.

GREY LEAF SPOT: A disease of cereals. In corn, it is caused by the fungus *Cercospora zeae-maydis*, producing yellow to tan lesions in the leaves. After about two weeks the lesions appear tan to brown in colour and rectangular in shape, bordered by the veins of the leaf.

GREY MATTER (GRAY MATTER; *Substantia grisea*): Parts of the nervous tissue that makes up the central nervous system (the brain and spinal cord) of vertebrates that transmits nerve impulses. It is

brown-grey in colour, consisting largely of nerve cell bodies, synapses and dendrites. It is the site of coordination between nerves of the central nervous system. In the brain, it forms the cerebral cortex and the outer layer of the cerebellum. In the spinal cord, grey matter lies centrally and is surrounded by white matter.

GREY SCALES: A method of representing different shades of grey on a screen or printer that is an improvement over monochrome. Each dot in the bit map is represented by a number of bits (from 4 to 8 bits) and requires a larger amount of memory.

GREYOUT: Temporary impairment of vision, due to reduced flow of blood to the brain. It is caused by prolonged high acceleration.

GRID: 1. The system of interconnected overhead wires or underground cables, especially a network of high-tension cable by which electrical power is distributed from power stations. It also refers to a network of interconnected electromagnetic signals. 2. Any grating system, as in a cattle grid, for controlling the movement of livestock across roads. 3. A corrugated or perforated conducting plate in a storage battery. 4. A network or coil of fine wires located between the plate and the filament in an electron tube.

GRID REFERENCE: Cartesian with numbers used to show locations on a map. A four-figure grid reference indicates a specific square, whereas a six-figure grid reference indicates a point within a square. The grid numbers on the horizontal axis are called **eastings** and refers to the eastward-measured distance or the x-coordinate. The grid numbers on the vertical axis are called **northings** and refers to the northward-measured distance or the y-coordinate.

GRIGNARD REAGENT: It is a member of a class of organometallic compounds that have the general formula RMgX, where R is an alkyl group and X is bromine (Br) or iodide (I). The reagents are very reactive, giving rise to highly nucleophilic radical R $\cdot$. They are important in organic synthesis, such as the formation of carbon-phosphorus, carbon-tin, carbon-silicon, carbon-boron and other carbon-heteroatom bonds. Grignard reagents readily react with protic solvents such as water or with functional groups with acidic protons, such as alcohols and amines, factors which may affect the success of a test. It was named after the French chemist, François Auguste Victor Grignard, who discovered the reactions and reagents and was awarded the 1912 Nobel Prize in Chemistry.

GRINDSTONE: A large round sharpening stone usually made from grit or sandstone that is turned by a machine and used to sharpen ferrous tools.

GRIT: 1. Airborne solid particles in the atmosphere which are of natural or manmade origin and which remain in suspension for some time. 2. Small particles of various substances, usually gravels fed to poultry, which is a necessary part of the diet of birds especially layers. It plays a part in the digestive processes that goes on in the gizzard. 3. Sandstone, often used as a grindstone.

GRITSTONE: A medium- to coarse-grained sedimentary rock with angular grains that may rub off easily. Most gritstones are formed in water. The chief mineral components are quartz, feldspar and mica. Those with more than 75% quartz are called **quartz gritstone** with an orange colour. Those with more than 25% feldspar gives rise to **feldspathic gritstone** with brown, sometimes tinged with pink colour.

GROIN: 1. In humans, the hollow area between the abdomen and the thigh on each side of the pubic bone including the external genitals. 2. Also spelt **groynes**, it is a long, low, narrow structure, usually extending roughly perpendicular to the shoreline, designed to protect the shore from erosion by currents, tides, or waves, or to trap sand for the purpose of building up or making a beach.

GROOMING: Rearranging fur or feathers and cleaning the body surface by biting, scratching or licking, which is important for removing parasites, dirt and spreading essential oils over the body surfaces.

GROSS ENERGY (GE): In food, it is the total chemical energy measured from complete combustion of the food in a bomb calorimeter.

GROSS PRIMARY PRODUCTIVITY (GPP): The total primary productivity of an ecosystem.

GROSS PRODUCTIVITY: A measure of the rate at which energy is assimilated by the organisms in a trophic level, a community or an ecosystem.

GROTRIAN DIAGRAM (TERM DIAGRAM): A diagram that shows the energy level and the allowed transitions between the energy levels in an atom, taking into account the specific selection rules related to changes in angular momentum of the electron. The energy is plotted vertically with a horizontal line for each energy level.

GROUND: 1. The solid surface of the earth. 2. In biology, ground tissue comprises all plant tissues other than those of the vascular system and the dermal tissues; it fills the space between the dermal and the vascular tissues. It is composed of three cell types: parenchyma, collenchyma, and sclerenchyma cells. Its functions include support, photosynthesis, and storage. Composed of relatively undifferentiated cells. In electricity, also known as earth, it is an electrical connection between an appliance and the ground so that it shares any charge it has with the earth. As the earth is so large this means in effect that an earthed object is of zero potential and cannot keep a charge hence making it harmless. In the event of a fault in the electrical appliance involving a connection between the live part of the circuit and the outer casing, for example, the current flows to the earth without causing any harm to the user. The earth in an electrical circuit is the reference point from which other voltages are measured.

GROUNDING: In electricity, it is a harmless, low-resistance path between an electric circuit or equipment and the earth or ground for leakage current to flow. It is an important electrical safety procedure.

GROUND COVER: Plants that grow densely close to the ground, either in natural conditions or planted to prevent soil erosion or the spread of weeds.

GROUND FIRE: A fire that consumes all the organic material of the forest floor and also burns into the underlying soil, for example, a peat fire. It differs from a surface fire by being invulnerable to wind. In a surface fire the flames are visible and burning is accelerated by wind, whereas in a ground fire wind is not generally a serious factor.

GROUND LEVEL CONCENTRATION: The concentration of a chemical species, normally a pollutant, in air. It is usually measured at a specific height above the ground.

GROUND LEVEL INVERSION: The inversion of the normal temperature gradient in the atmosphere. The temperature of the air increases with increasing height of the air above the ground; leading to poor mixing of gases released below the inversion.

GROUND MERISTEM: The primary meristem in plant shoots and roots derived from the apical meristem that differentiates into the pith and cortex.

GROUND MORaine: The deposit of clay, sand and varying sizes and shapes of rock that builds up underneath a glacier.

GROUND STATE: The stable lowest energy state of a particle or a system of particles, such as a molecule, atom or nucleus, which may be bound or unbound. When bound, its energy generally is a finite amount less than the energy of the next higher or first excited state.

GROUND SUBSTANCE: The matrix of connective tissue, in which various cells and fibres are embedded. Ground substance includes all the other proteinaceous components, including proteoglycans, matrix proteins and water.

GROUND SYSTEM: One of the four main tissue systems in plants, that occupies the space between the epidermis and the vascular system. It is composed mainly of parenchyma cells with some collenchyma and sclerenchyma cells. It is involved in photosynthesis, water, food storage and support.

GROUND TISSUES: The plant tissues formed by the apical meristems, which consist of parenchyma, collenchymas and sclerenchyma.

GROUND WATER: Water collected underground within the tiny pores of the surrounding aquifer material or porous rock strata and soil pore spaces. Groundwater is replenished with water from the surface through a process called "recharge." This occurs as a part of the hydrologic cycle when water from rainfall percolates into underlying aquifers; naturally emerging at the surface as springs and streams. Most groundwater is free of pathogenic organisms and does not necessarily need purification for domestic or industrial use. Groundwater supplies are not seriously affected by short droughts and are available in many areas that do not have dependable surface water supplies. Some communities and animals all over the world rely on groundwater for their source of drinking water.

GROUND WAVE: A radio wave that travels along the surface of the Earth in approximately a straight line. It supports frequencies of up to 7 MHz. Given enough power, it can be used to establish very long distance communication.

GROUNDNUT (PEANUT; MONKEY NUT; ARACHIS NUT): A branched trailing plant, *Arachis hypogaea* or its oily nutritious seeds. It belongs to the pea and bean family and classified in the division Magnoliophyta, class Magnoliopsida order Rosales, family Fabaceae or Leguminosae (Pea family). The plant is an annual herbaceous plant growing 30 to 50 cm tall, with a small, usually erect, thin stem. The leaves are arranged in alternate pairs and have leaf-like attachments near the stalk. The flowers are yellow orange, cream or white with characteristic floral structures which sink into the ground to grow the pod. They become mature in about 120 to 140 days, when the leaves of the plant turn yellow. Each pod contains between one and three seeds. The seed is a rich source of protein, niacin (a component of the vitamin B complex), vitamins E and B₁, vitamin B₆, copper, zinc, folate and iron. The seeds are pressed for edible oil, ground into peanut butter, eaten as snacks and used in cooking. In some parts of the world, the plant is fed to livestock.

GROUNDWATER RECHARGE: The hydrologic process by which external water from land surface is added to the zone of saturation of an aquifer. It occurs naturally as part of the hydrologic cycle; and through anthropogenic processes (human-induced), either directly through spreading basins or injection wells, or as a consequence of human activities such as irrigation and waste disposal.

GROUP: 1. The vertical column of arrangement in the Periodic Table of the elements consisting of 18 groups. Elements of a group have the same number of valence electrons. 2. In mathematics, a set with a binary associative operation such that the operation admits an identity element and each element of the set has an inverse element for the operation. The group has the following properties: the set is closed

under the law of binary composition; the law of binary composition is associative; there is an identity element; every element has an inverse. 3. Any defined continuously linked collection of atoms within a molecular entity. In organic chemical nomenclature the meaning within this general sense is restricted to any covalently bonded collection of atoms that behaves as a unit in chemical reactions.

GROUP 1 ELEMENTS: The first group of vertically arranged elements from the left hand side of the periodic table, belonging to the group of alkali metals, comprising lithium (Li), sodium (Na), potassium (K), rubidium (Rb), caesium (Cs) and Francium (Fr). They have similar chemical properties because they all have one electron in an outer s-orbital. The alkali metals are all highly reactive, tarnish easily and have low melting points and densities never found in elemental forms in nature. Although hydrogen is placed in this group it rarely exhibits behaviour comparable to the alkali metals.

GROUP 2 ELEMENTS: A group of elements in the Periodic Table consisting of alkaline earth metals comprising beryllium (Be), magnesium (Mg), calcium (Ca), strontium (Sr), barium (Ba) and radium (Ra). They are silver coloured, soft metals, which react readily with halogens to form ionic salts and with water, though not as rapidly as the alkali metals, to form strong alkaline (basic) hydroxides.

GROUP 3-12 ELEMENTS (TRANSITION ELEMENTS; TRANSITION METALS): A set of elements in the Periodic Table, with atomic numbers 21–31, 39–49 and 71–81, which according to IUPAC definition, has atoms with an incomplete d sub-shell or which can give rise to cations with an incomplete d sub-shell. This definition does not include Group 12 elements. It includes elements such as scandium, titanium, vanadium, chromium, manganese, iron, cobalt, nickel and copper, yttrium, zirconium, niobium, molybdenum, technetium, ruthenium, rhodium, palladium and silver.

GROUP 13 ELEMENTS: A group of elements in the Periodic Table consisting of the boron group comprising boron (B), aluminium (Al), gallium (Ga), indium (In) and thallium (Tl), which all have outer electronic configurations ns^2np^1 with no partly filled inner levels. They are the first members of the p-block. The group differs from the alkali metals and alkaline-earth metals in displaying a considerable variation in properties.

GROUP 14 ELEMENTS: A group of elements in the Periodic Table comprising carbon (C), silicon (Si), germanium (Ge), tin (Sn) and lead (Pb). The group, commonly called the carbon group or family consists of one nonmetal (carbon), two metalloids (silicon and germanium) and two metals (tin and lead). They have outer electronic configurations ns^2np^2 with no partly filled inner levels. The main valency of the elements is 4 and the members of the group show a variation from nonmetallic to metallic behaviour in moving down the group. The group tends to be fairly unreactive and the elements form covalent compounds, though tin and lead also form ionic compounds. Thus, carbon is a nonmetal and forms an acidic oxide (CO_2) and a neutral oxide. Group 14 elements exist free in nature and in compounds.

GROUP 15 ELEMENTS: A group of elements in the Periodic Table consisting of the nitrogen group consisting of nitrogen (N), phosphorus (P), arsenic (As), antimony (Sb) and bismuth (Bi), which all have outer electronic configurations ns^2np^3 with no partly filled inner levels.

GROUP 16 ELEMENTS (CHALCOGENS; OXYGEN FAMILY): A group of elements in the Periodic Table consisting of oxygen (O), sulfur (S), selenium (Se), tellurium (Te) and polonium (Po), which all have outer electronic configurations ns^2np^4 with no partly filled inner levels.

GROUP 17 ELEMENTS (HALOGEN ELEMENTS): A group of elements in the Periodic Table consisting of fluorine (F), chlorine (Cl), bromine (Br), iodine (I) and astatine (At). It is the only Periodic Table Group which contains elements in all three familiar states of matter at standard temperature and pressure.

GROUP 18 ELEMENTS (NOBLE GASES): A group of elements in the Periodic Table consisting of helium (He), neon (Ne), argon (Ar), krypton (Kr), xenon (Xe) and radon (Rn). Under standard conditions, they are all odourless, colourless, monatomic gases, with very low chemical reactivity. They are chemically inert or do not readily participate in chemical or biological reactions.

GROUP ADDRESS: 1. In computer science it is a single transport address identifying a collection of users. The collection of users is formed so that they can all receive common multicast data. 2. Software that enables messaging and collaborative work among people.

GROUP OF ELEMENTS: A set of elements which have the same number of electrons in their outermost shell. An example is the halogens.

GROUP PSYCHOLOGY: An area of social psychology dealing with the formation and cohesion of groups, competition and conflict among group members and between groups, the influence of the group on the individual and vice versa, communication within the group and between groups and many other aspects of social interaction.

GROUP REPRESENTATION: A group of mathematical objects that is homomorphic or has the same mathematical structure, as the original group.

GROUP RING: The set of all formal sums $\sum \alpha_x x$, where x ranges over a multiplicative group and the α_x of which all but finitely many are zero, are elements of a field (usually taken to be the complex numbers). The group ring of the group G over a field is usually denoted R_G . Multiplication and addition are defined by $\sum \alpha_x x + \sum \beta_x x = \sum (\alpha_x + \beta_x) x$ and $\left(\sum \alpha_x x \right) \left(\sum \beta_x x \right) = \sum \left(\sum_{st=x} \alpha_s \beta_t \right) x$.

GROUP SELECTION: A mechanism originally proposed by the British ethologist, V. C. Wynne-Edwards in 1962 that alleles can become fixed or spread in a population because of the benefits they bestow on groups, regardless of the alleles' effect on the fitness of individuals within that group. It arose from his observations that individual animals often expose themselves to danger, for example, by warning of predators or forgo reproduction, for example, as with worker bees in a colony for the greater good of the group as a whole. Bees that spend their lives serving their queen, for example and female animals such as elephants that care for the offspring of others in the herd are participating in cooperative acts that may undermine their own chances of reproducing. Hence, groups containing altruistic individuals would have some selective advantage over groups lacking such members. This conflicts with Darwinian orthodoxy, which views natural selection as operating strictly on individuals. Group selection has now been supplanted by the theory of kin selection as an explanation of apparently altruistic acts.

GROUP THEORY: The study of the symmetries that define the properties of a system. It shows how the symmetry of a molecule is related to its physical properties and provides a quick simple method to determine the relevant physical information of the molecule. Invariance under symmetry operation enables much about a system

to be deduced without knowing explicitly the solution to the equations of motion. Newton's law of gravitation, for instance, exhibits spherical symmetry.

GROUPED DATA: Data arranged into various categories using a criterion, each with its frequency attached rather than showing all the original figures.

GROUPING SYMBOLS: In algebra equations that contain multiple groups, symbols such as parentheses - (), brackets - [], braces - { }, and radicals - $\sqrt{\quad}$, that define the order in which operations are to be done. The most frequently used symbols are parentheses - (), and brackets - []. In solving such equations, the innermost grouping, sometimes referred to as nested grouping, receives the first attention. For example, in $(5 + 7) \times [(7 + 3) \div 5]$, the group $(7 + 3)$ is nested within $[(7 + 3) \div 5]$ and is done first, and simplifies the equation to $(5 + 7) \times [10 \div 5]$; so $(5 + 7) \times [10 \div 5] = 12 \times 2 = 24$.

GROUPOID: 1. An algebraic structure on a set with a binary operator under which it is closed. The positive real line with the binary operation $a \cdot b = \sqrt{a^2 + b^2}$ produces a groupoid that is not a semigroup. 2. A category in which every morphism or arrow is invertible. In this case, a groupoid is like a group with multiplication only partially defined.

GROVER'S ALGORITHM: An algorithm, for searching an unsorted database with N entries in $O(N^{1/2})$ time and using $O(\log N)$ storage space invented by Lov Grover in 1996. It is a more efficient method of searching than traditional algorithm.

GROWING PATTERN: In mathematics, the pattern that grows or decreases in value or size from term to term by a constant difference, for example, 1, 2, 4, 8.

GROWTH: 1. The irreversible increase in size and complexity of an organism during development from embryo to maturity, resulting from cell division, cell enlargement and cell differentiation. 2. A mass of cells with no physiological function, for example, a tumour that forms in or on an organ.

GROWTH CONE: The enlarged motile tip of a growing nerve fibre that determines the direction during the development of the nervous system by means of interaction with signalling molecules in its surroundings and its own motile mechanism. It has numerous finger-like projections called **filopodia**, which bear surface receptors.

GROWTH CORRELATION: The relationship that exists between the different growth rates of various parts of a plant body. Growth rates depend on the balance of growth substances in the region and the competition between parts for nutrients.

GROWTH CURVE: 1. A graphic representation of the growth of an organism over time. A single growth curve that results from plotting body mass against age. 2. A curve showing how some quantity associated with a radioactive transformation or induced nuclear reaction increases with time.

GROWTH FACTOR: Any of various naturally occurring chemicals, particularly polypeptides that play a variety of important roles in the stimulation of cell growth, proliferation and cellular differentiation and cell maintenance. Examples are cytokines and hormones that bind to specific receptors on the surface of their target cells.

GROWTH HORMONE (HUMAN GROWTH HORMONE; PITUITARY GROWTH HORMONE; SOMATOTROPHIN; SOMATOTROPIC HORMONE): A hormone secreted by the

Grouped Data Frequency Distribution

Class (Mark)	Frequency
11 - 15	2
16 - 20	3
21 - 25	3
26 - 30	5
31 - 35	6
36 - 40	6
41 - 45	3
46 - 50	2
Total	30

A table of grouped data frequency distribution

mammalian anterior part of pituitary gland that promotes protein synthesis in all cells, increases fat mobilisation and use of fatty acids for energy and decreases use of carbohydrate. Growth effects depend on the presence of thyroid hormone, insulin and carbohydrate. It accelerates the transport of specific amino acids into cells, stimulates the synthesis of messenger ribonucleic acid (RNA) and ribosomal RNA, influences the activity of several enzymes, increases the storage of phosphorus and potassium and promotes a moderate retention of sodium. It is important in controlling linear growth of the bones in the legs and arms. Together with the thyroid hormones and the sex hormones, they are important in determining the final height and development of an individual. It cannot cause elongation of long bones after the epiphyses close, however, so stature does not increase after puberty. Deficiency of the hormone in children results in short stature with the whole body in proportion, whereas deficiency in the adult results in weakness and depression. An excess of growth hormone results in gigantism in children or acromegaly in adults.

GROWTH HORMONE-RELEASING HORMONE (GROWTH HORMONE-RELEASING FACTOR; GRF; GHRF; SOMATOLIBERIN): A 44-amino acid peptide hormone produced in the arcuate nucleus of the hypothalamus that stimulates the release of growth hormone. Growth hormone-releasing hormone (GHRH) is released from neurosecretory nerve terminals of these arcuate neurons and is carried by the hypothalamo-hypophyseal portal circulation to the anterior lobe of the pituitary gland, where it stimulates the production and secretion of growth hormone (GH).

GROWTH INHIBITOR: Any substance that retards the growth of a plant or plant part. Almost any substance will inhibit growth if present at high enough concentrations but two common hormonal inhibitors are abscisic acid and ethylene, which both have effect at very low concentrations, similar to the effective concentrations of auxins, gibberellins and cytokinins. Other inhibitors, which must be present in much higher concentrations to take effect include certain of the phenolics, quinones, terpenes, fatty acids and amino acids.

GROWTH PLATE (EPIPHYSEAL PLATE; PHYSIS): The area of growing tissue near the ends of the long bones in children and adolescents. Each long bone has at least two growth plates, one at each end. The growth plate determines the future length and shape of the mature bone. When growth is complete, sometime during adolescence, the growth plates close and are replaced by solid bone. Because the growth plates are the weakest areas of the growing skeleton, they are vulnerable to injuries called **fractures**.

GROWTH RATE: The rate of increase over a period of time. The percentage change from one period to another is calculated from the formula: $PR = (V_{\text{present}} - V_{\text{past}} / V_{\text{past}}) \times 100$, where: PR = Percent Rate, V_{present} = Present or Future Value, V_{past} = Past or Present Value. The annual percentage growth rate is the percentage growth divided by N, the number of years.

GROWTH RING (ANNUAL RING): The increment of secondary xylem produced in any one growing period in the stems and roots of many plants, especially those in temperate regions.. Although growth rings may not be discernible or are not annual, they have been used in dating ancient wooden structures. In transverse section this is visible as a ring and is due to the difference in radial diameter of xylem elements formed at the beginning and end of the growing season. There may be one or several growth rings in any one year (an annual ring may consist of more than one growth ring). The growth rings are then termed false annual rings.

GROWTH SPIRAL (EQUIANGULAR SPIRAL; LOGISTIC SPIRAL; LOGARITHMIC SPIRAL): A type of spiral that is seen commonly in the natural world. Examples are found in the shells of some molluscs, such as that of the nautilus and the fossil ammonites and also in spider webs. The logarithmic spiral can be constructed from equally spaced rays by starting at a point along one ray and drawing the perpendicular to a neighbouring ray. The logarithmic spiral is an example of a transcendental plane curve.

GROWTH SUBSTANCE (HORMONE; PHYTOHORMONE; GROWTH REGULATOR): Any of a number of organic chemicals that are synthesised by plants or synthetic chemicals that promote or inhibit plant growth when present in very low concentrations. Growth substances may stimulate plant growth by promoting cell division or cell elongation, while others, such as abscisic acid and ethylene, inhibit certain developmental processes. Growth substances include substances produced within the plant and also artificial, often structurally related, chemicals that have similar effects. Endogenous growth substances are often produced in a particular region, for example, auxin is synthesised in shoot apices.

GROWTH YIELD COEFFICIENT: Quantity of carbon formed per unit of substrate carbon consumed.

GROYNE (GROIN): A long, low, narrow structure, usually extending roughly perpendicular to the shoreline, designed to protect the shore from erosion by currents, tides, or waves, or to trap sand for the purpose of building up or making a beach.

GRYKE (GRIKE): A deep fissure that separates slabs in a limestone pavement.

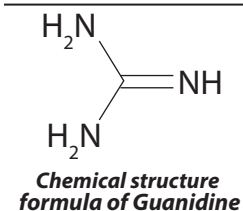
GSM (GLOBAL SYSTEM FOR MOBILE COMMUNICATION): In telecommunications, it is a second generation (2G) standard for mobile networks. It is a circuit-switched technology based on Time Division Multiple Access (TDMA). It operates on three distinct frequencies: 900 MHz, 1800 MHz and 1900 MHz. The 900 and 1800 MHz frequencies are used in Europe and 1900 MHz frequency is used in North America. Data transfer rate of GSM is 9.6 Kbps.

GT DISTRIBUTION (GENERALISED T-DISTRIBUTION): A probability distribution, $f(x)$ which is related to Student's t-distribution

and is given by: $f(x) = \frac{p}{2\sigma q^\beta \beta(p^{-1}, q)} \left[1 + \frac{1 \times 1}{(q\sigma p)} \right]^{q+p-1}$, where β is the

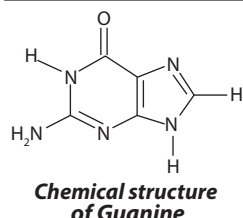
beta function. In $f(x)$, σ is a scale parameter while p and q control the shape of the distribution. Large values of p and q are associated with lighter tails for the distribution. When $p = 2$ and $\sigma = \sqrt{2}a$, this becomes student's t-distribution with 2 degrees of freedom.

GUANIDINE: A crystalline basic compound, with the chemical formula $\text{NHC}(\text{NH}_2)_2$, molar mass of 59.07 g mol⁻¹ and melting point of 50 °C (323 K). It has a strong alkalinity related to urea. It is formed by the oxidation of guanine and also found in urine as a normal product of protein metabolism. It is used in the manufacture of plastics and explosives.



GUANINE (2-AMINO-6-HYDROXYPURINE; 2-AMINOHYPOXANTHINE): A white

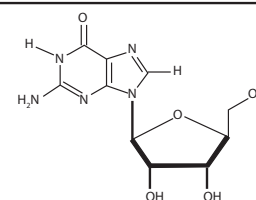
amorphous solid purine derivative often called a base, consisting of two rings, each containing both nitrogen and carbon atoms and an amino group. Its molecular formula is $\text{C}_5\text{H}_5\text{N}_5\text{O}$ with a molar mass of 151.13 g/mol and a melting point of 360 °C (633.15 K). It is relatively insoluble in water, but it is soluble in dilute acids and bases. It occurs freely or combined in various natural sources, including guano, sugar beets, yeast and fish scales. It is one of the major component bases of nucleotides and the nucleic acids DNA and RNA.



GUANO: An accumulation of the droppings (faeces and urine) of birds, bats or seals formed by a colony of animals. They are high in nutrients consisting of ammonia, along with uric, phosphoric, oxalic

and carbonic acids, as well as some earth salts and impurities. It is used as fertiliser and gunpowder ingredient due to its high levels of phosphorus and nitrogen.

GUANOSINE (GUANINE RIBOSIDE): A purine nucleoside consisting of one guanine molecule linked to a D-ribose sugar molecule. Its molecular formula is $\text{C}_{10}\text{H}_{13}\text{N}_5\text{O}_5$ with a molecular weight of 283.24072. It is found in DNA and RNA and its nucleotides play important roles in metabolism.



Chemical structure of Guanosine

GUANOSINE TRIPHOSPHATE (GTP; GUANOSINE 5'-TRIPHOSPHATE): A chemical compound (nucleotide) that is

incorporated into the growing RNA chain during synthesis of RNA. Its molecular formula is $\text{C}_{10}\text{H}_{16}\text{N}_5\text{O}_{14}\text{P}_3$. GTP is similar to adenosine triphosphate (ATP) and plays a special role in microtubule assembly, protein synthesis and cell signalling. It also plays a role in the activation of fatty acids, in gluconeogenesis and in the initiation and elongation of polypeptide chains during protein synthesis.

GUARD BAND: In communication systems, a position of the channel bandwidth between two adjacent channels that is not used. This unused spectrum is called a guard band to help prevent interference between the neighbouring channels.

GUARD BIT: An extra bit that preserves floating-point accuracy. It is used temporarily to handle underflow of an exponent by replacing an extra-bit at the right-hand end of the accumulator.

GUARD CELL: A type of paired epidermal cells surrounding the stoma of plants. Guard cells, unlike epithelial cells, contain chloroplast and control the opening and closing of the stomatal pore.

GUARD COLUMN: A short column in HPLC installed in front of an analytical column, which is intended to protect it from strongly retained impurities, thus prolonging the life of the analytical column.

GUARD DIGITS: Digits that are added to the precision of a calculation so as to allow for rounding error, cancellation error or other numerical error. Guard digits are also used in floating point operations in most computer systems.

GUAVA (*Psidium guajava*): A small tree or shrub in the myrtle family (Myrtaceae) that is native to Caribbean, Central America and South America and can be cultivated in some tropical and subtropical areas worldwide. It grows to between 3 and 10 metres high with a trunk diameter of about 25 centimetres. It has a smooth, thin, copper-coloured bark that flakes off, with a greenish layer underneath. Its evergreen leaves are oppositely arranged, simple, leathery and elliptic to ovate in shape and about 7-15 centimetres long and 3-5 centimetres wide. When crushed, the leaves are aromatic. It produces white, mildly fragrant flowers borne singly or in small clusters in the leaf axils, with numerous stamens and five petals. The stamens are tipped with pale-yellow anthers. The fruits, called guava are round, oval or pear-shaped, usually with a green outer skin when unripe but turns yellow or maroon when matured or ripe. Some matured and ripe fruits even maintain the green outer skin and become soft. Guavas can be eaten raw. It can be made into candies, jams, marmalades and juices. The fruit contains vitamin C, folic acids and roughage. Its yellow to reddish, fine-grained wood is quite strong and used in carpentry and turnery. Its parts such as roots, bark, leaves are used traditionally to treat several ailments such as gastroenteritis, diarrhoea and dysentery.

GUERNSEY: A breed of dairy cattle that originated on the Channel Island of Guernsey. It has a fawn coat with distinct patches of white, with pronounced yellowish milk.

GUEST: 1. In computer science, a **guest account** refers to a user of a system or service, who is not registered and is granted certain guest privileges. Such a user gains very limited access to the service. In Windows and Windows Vista, it is an account for users who do not have a permanent account on a computer or domain. It allows people to use your computer without having access to your personal files. People using the guest account cannot install software or hardware, change settings or create a password. Although some computer operating systems have a default guest account setup after their installation, many times this account has to be setup manually by the administrator before it can be accessed. 2. A **guest computer** is a computer that operates under the control of another computer called the host.

GUI (GRAPHICAL USER INTERPHASE): The interactive part of a computer software that allows the user to select commands, call up files, start programs and do other routine tasks by using a mouse to point to pictorial symbols (icons) or lists of menu choices on the screen as opposed to having to type in text commands.

GUIANA SPACE CENTRE: Europe's main launch facility, owned by ESA (European Space Agency) and operated by CNES (Centre National d'Etudes Spatiales, the French Space Agency). It is located near Kourou in French Guiana, on the northern coast of South America at 5.2° N, 52.8° W and is ideal for launching satellites into geosynchronous orbits and also supports missions into polar orbit. It benefits from Earth's rotation near the equator, which gives a free boost of 460 m/s to eastward-launched rockets.

GUID (GLOBALLY UNIQUE IDENTIFIER): A 128-bit (16 byte) number used by software programs to uniquely identify the location of a data object. Some examples of data that include GUIDs are streaming media files, Windows registry entries, database keys and various file types. GUIDs are typically written in hexadecimal notation, containing 32 digits.

GUIDANCE: In aeronautics, it is the process of directing the movements of an aeronautical or space vehicle, with particular reference to the selection of a flight path or trajectory.

GUIDED MISSILE: An unmanned weapon that is controlled by some system, either internal or external, from its launch until it hits its target. There are four types: (i) surface-to-surface, (ii) surface-to-air, (iii) air-to-air and (iv) air-to-surface. The first guided missiles were built in Germany during the Second World War.

GUIDEPOST CELL: A differentiated neuron that influences the development of the axons of other neurons so that they connect ultimately with their target cells. Guidepost cells are specific cellular cues in the embryonic environment utilised by axonal growth cones in pathfinding decisions.

GUILD: In ecology, it is a group of species within a community that exploits the same resources in a similar way, therefore sharing a similar ecological niche. For example, different species of reptiles may constitute a guild by occupying the same types of underground shelters and different species of seed eating birds may constitute a guild. Some other examples of guilds are forbs, geophytes, graminoids, shrubs, trees, vines and arthropods. Guilds are classified according to how they acquire their nutrients, their state of mobility and their mode of feeding.

GUILTWARE: In computing, it is any shareware that makes the user fulfil some obligations such as registering and paying for it or donating to a charity by eliciting feelings of guilt.

GUINEA CORN (GRAIN SORGHUM; *Sorghum vulgare*): A monocotyledonous cereal grain crop in the family, Gramineae that is native to Africa and cultivated in several countries such as India, Pakistan, northern China, Argentina, Australia, southern Europe and

the United States. It thrives in hot areas with little rainfall, where most crops such as corn and rice would not survive. The plant resembles a cane and grows to heights of 0.5-2.5 metres but has been known to reach 4.5 metres. It is the fifth most important cereal in the world after wheat, rice, maize and barley; and in Africa, it is the second most produced after maize. The leaves are about 5 centimetres broad and 0.75 metres long, with the leaves growing off the stalk. The panicles range from loose to dense, bearing 800–3,000 kernels. The kernels vary widely among different varieties in shape, colour and size, with the colour varying from white to dark brown. Its feed quality is lower than that of corn and contains vitamin B, iron, nicotinic acid, proteins and carbohydrates. It is an important staple food in the semi-arid tropical regions of Asia and Africa. The grain is ground into powder and used to prepare porridge, flatbreads, cakes and alcoholic beverages. It is also used for feeding livestock and poultry. The stalks are used as fodders, building materials and for making brooms and brushes.

GUINEA GRASS (GREEN PANIC GRASS): A large perennial bunch grass *Megathyrsus maximus* that is native to Africa and widely distributed in some parts of the world. It grows naturally in open grasslands, usually under or near trees and shrubs and along riverbanks. It can withstand wildfire and drought. The species has broad morphological and agronomic variability, ranging in height from 0.5-3.5 metres with 5-10 centimetres stems. The panicles are open, with as many as 9000 seeds per plant. It is used as fodder and silage.

GUINEA WORM (DRACUNCULIASIS): A disease caused by the large female nematode, *Dracunculus medinensis*. A native to warm tropical regions, it is a long thin worm that lives as a parasite under the skin of people and animals. The adult female is larger than the adult male and can grow to several centimetres in length. It attacks several parts of the body including the legs and feet, the head, torso, upper extremities, buttocks and genitalia, forming painful ulcerating blisters.

GULAR: Of, or pertaining to the throat, for example, gular pouches are found on the throats of some birds.

GULAR FLUTTERING: A cooling mechanism adopted by birds to counteract overheating, by rapidly vibrating the flap membranes in the throat to increase evaporation; as seen in birds such as cormorants, pelicans, etc.

GULAR POUCH: A distinguishing feature in some birds such as pelicans, consisting of a bare sac that can be expanded to scoop up a large prey; it is also inflated and used to attract females during courtship.

GULF STREAM: A narrow (100 km) and swift (up to 5 knots or 9.26 km/h) warm ocean current that flows north from the warm waters of the Gulf of Mexico. It originates north of Grand Bahama Island where the Florida Current and the Antilles Current meet and which eventually becomes the eastward-flowing North Atlantic Current.

GULF WAR SYNDROME: A group of symptoms occurring in some Gulf War veterans who fought in the 1991 Gulf War characterised by headaches, memory loss, muscle weakness, depression, respiratory problems, lethargy, nausea and pain. Although its cause is uncertain, it has been attributed to exposure to chemical weapons, vaccines, infectious diseases or other factors.

GULLET: 1. Also known as **oesophagus**, it is a tube which is part of the alimentary canal with an inner lining of folded mucous membrane, in which food travels down from the mouth to the stomach. In humans it is a tube running the full length of the neck and the thorax, held in its position ventral to the vertebral centra by a tunica adventitia of loose connective tissue. 2. An invagination into the cytoplasm of certain ciliates, used for food intake. 3. A gully or ravine.

GULLY: A narrow, stony valley or channel with steep sides, created by running water eroding sharply into soil, typically on a hillside.

Gullies resemble large ditches or small valleys, but are metres to tens of metres in depth and width.

GULLY EROSION: Removal of soil to form relatively narrow and deep trenches known as gullies.

GUM: 1. The soft tissues surrounding the necks of the teeth and the alveolar bone of the jaw of mammals. 2. Any substance that swells in water to form gels or sticky solutions. Structurally they are mostly complex, highly branched polysaccharides, although a few gums with relatively simple structures are known. Three main classes of gum are recognised. Acidic polysaccharides of glucuronic or galacturonic acids form glassy, hard gums. These are often produced by plants as a result of injury. Examples include gum arabic and tragacanth. In algae, another class of acidic gums are found in which the acidity is due either to sulfate acid ester groups or to uronic acids. Examples are agar, alginic acid and carrageenin. The third class contains gums that are obtained from seeds; examples are the seeds of the carob tree (*Ceratonia siliqua*). Natural gums are increasingly being replaced by synthetic substitutes in the manufacture of adhesives.

GUM ARABIC (ACACIA GUM): A polysaccharide, found in the cell walls of certain plants, especially the acacia tree. It is made up of D-galactose and D-glucuronic acid and arabinose and rhamnose. It is a water-soluble, gummy exudate. It is obtained commercially from Acacia and is used in glues and pastes and as a mounting medium in microscopy, in the food industry as a stabiliser, in printing, paint production, glue, cosmetics, pharmaceuticals and various industrial applications.

GUMMOSIS: A symptom of certain plant diseases in which there is an abundant formation of gum. Gummosis often occurs on trees but may occur on herbaceous plants, such as infection of young cucumber fruits with the cucumber scab fungus (*Cladosporium cucumerinum*) resulting in secretion of gum at the edge of the lesions.

GUN-METAL: A type of bronze or an alloy which is high in copper (88%) and also containing tin and zinc. It has a tensile strength of 221 to 310 MPa, specific gravity 8.7, a density of 8,719 kg/m³ and a Brinell hardness of 65 to 74. It is resistant to corrosion from salt. It is used to make steam and hydraulic castings, valves and gears and also statues and various small objects such as buttons. It was once used to cast canons, giving it the name gun metal.

GUNPOWDER (BLACK POWDER): A mixture of potassium nitrate, charcoal and sulfur. It is widely used for quarry blasting, fuses and fireworks and it is the oldest known explosive. It burns rapidly, producing a volume of hot gas made up of carbon dioxide, water and nitrogen and a solid residue of potassium sulfide.

GUNTER'S CHAIN (THE SURVEYOR'S CHAIN): A standard unit of length named after the English mathematician Edmund Gunter or a measuring band equivalent to 20.1 metres subdivided into 100 links of 20.1 cm each. In engineering, a chain equals 30.5 m or 100 links of 30.5 cm each. 80 chains is 1.6 km long and 4046.856 m² has 10 square chains.

GUSTINESS: Intensity of turbulence; the ratio of the root mean square of wind velocity fluctuations to the meanwind velocity.

GUT (ALIMENTARY CANAL, DIGESTIVE TRACT): The mucous membrane-lined tube in the digestive system of an animal extending from the mouth to the anus, responsible for processing food and preparing it for entry into the blood and from which wastes are eliminated. It includes the pharynx, oesophagus, stomach and intestines.

GUT-ASSOCIATED LYMPHOID TISSUE (GALT): The digestive tract's immune system, which is any of various lymphoid tissues and structures that protect the digestive tract against infectious organisms and other ingested antigens. GALT includes the palatine and other

adenoids (pharyngeal tonsils), appendix, Peyer's patches and other clustered or diffuse lymphoid cells formed throughout the gut wall.

GUTTA-PERCHA (PALAQUIUM): A rubbery substance derived from the sap of any of several tropical trees of the genera *Palaquium* and *Payena*, found in Southeast Asia. Chemically, gutta-percha is a polyterpene (trans 1.4-polyisoprene). The latex is bio-inert, very resilient and a good electrical insulator. It is also used as rubber, but also as insulation material, a waterproofing compound and in golf balls.

GUTTATION: A secretion of xylem sap on the tips or edges of leaves of some vascular plants, such as grasses through specialised pores called hydathodes. The process occurs most frequently during conditions of high humidity, when the rate of transpiration is low, especially at night.

GUYOT (TABLEMOUNT): A flat-topped circular underwater volcanic mountain that occurs mostly in deepwater over 200 metres below the surface of the sea and are thought of to be conical volcanic peaks levelled off by movement of water current. The diameters of these flat summits can exceed 10 km.

GWS MODEL (GLASHOW-WEINBERG-SALAM MODEL): The first step toward finding a universal force in the 1960s by American physicists, Sheldon Glashow and Steven Weinberg and Pakistani physicist Abdus Salam with their standard model theory of particle physics. They used a branch of mathematics called group theory, to show how the weak force and the electromagnetic force could be combined mathematically into a single electroweak force. For contributions to the unification of the weak and electromagnetic interaction between elementary particles, Abdus Salam, Sheldon Glashow and Steven Weinberg were awarded the Nobel Prize in Physics in 1979.

GYMNOSPERM: The woody vascular plants in which the ovules are carried naked on the scales of a cone: their seeds are not enclosed in an ovule. They usually have needles that stay green throughout the year. Examples are pines, cedars, spruces and firs.

GYMNOSPERMAE: A class of the Tracheophyta or Spermatophyta containing about 700 species of chiefly arborescent seed-bearing plants. These are divided into some 67 genera and 12 families. Gymnosperms differ from the angiosperms in having naked seeds not enclosed in an ovary. Double fertilisation does not occur and thus no true endosperm is developed.

GYNAECANDROUS: An inflorescence having the carpel borne above the stamen.

GYNAECOLOGY: The branch of medicine and surgery often linked with obstetrics specialising in diseases of women, specifically disorders of the female reproductive tract (uterus, vagina and ovaries). Some of the disorders include the following: menstrual disorders, abnormal uterine bleeding, urinary incontinence, urinary tract infections, uterine fibroids, cancer of the cervix, cancer of the ovary, cancer of the uterus, disorders of the vulva, pelvic pain and dyspareunia (feeling pain while during sexual intercourse). A person, who practises gynaecology is called a **gynaecologist**.

GYNANDROMORPHY: A phenomenon in an organism that contains both male and female characteristics because it is genetically a mosaic, due to the loss of X chromosomes in a stem cell of a female. It occurs in organisms such as butterflies, some birds and crustaceans, especially lobsters and crabs.

GYNANDROUS (GYNOSTEMIAL): 1. Having stamens and styles united in a column as found, for example, in orchids. 2. Characterised by hermaphroditism.

GYNOBASE: An enlarged or elongated receptacle that supports a multilocular ovary.

GYNOBASIC STYLE: A style which arises near the base of the carpel. It is common in flowers of *Boraginaceae*.

GYNOECIUM (GYNAECIUM): A collective term for the female reproductive organs (innermost whorl) of a flower, consisting of one or more carpels either free or fused together. In mosses, liverworts and hornworts, it refers to a cluster of archegonia and any associated modified leaves or stems present on a gametophyte shoot.

GYNODIOECIOUS: Having pistillate flower on one plant and a bisexual or hermaphrodite flower on another plant of the same species. Examples of such plants are *Lobelia siphilitica* and *Lobelia spicata*.

GYNOPHORE: An elongated floral stalk that holds the gynoecium above other floral parts.

GYNOSTEGIUM: A complex structure which is formed by the fusion of the stamens and the gynaecium. This forms a caplike mass of tissue and covers the gynaecium. It is characteristic of members of the milkweed family (Asclepiadaceae).

G **GYPSUM (PLASTER OF PARIS, POP; DRYWALL):** A common white or colourless mineral composed of hydrous calcium sulfate, with the chemical formula $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$, Mohs scale hardness 1.5-2 and specific gravity of between 2.31-2.33. It is a widely distributed form of sedimentary rock, formed by the precipitation of calcium sulfate from seawater and is frequently associated with other saline deposits such as halite and anhydrite, as well as with limestone and shale. Fine-grained gypsum called **alabaster** is used for ornamental works.

GYRE: A circular surface rotation of ocean water in each major sea. Gyres are large and permanent and occupy the northern and southern halves of the three major oceans.

GYROCOMPASS: A north-seeking form of gyroscope that is driven continually so that it can be used as a non-magnetic compass. It combines the action of two devices, a pendulum and a gyroscope to produce alignment with the Earth's spin axis. It is used as the prime navigational instrument on nearly every ship and on major aircraft and missiles.

GYROFREQUENCY (CYCLOTRON FREQUENCY; LARMOR FREQUENCY): The frequency of rotation of a charged particle or ion as it spirals a magnetic field.

GYROMAGNETIC RATIO: The ratio of the angular momentum of an atomic system to its magnetic moment. Its symbol is δ . The inverse of the gyro-magnetic ratio is called the **magneto-mechanical ratio**.

GYROPILOT: An automatic device for keeping a ship or airplane on a given course using signals from a gyroscopic reference. The

marine version operates a ship's rudder by displacement signals from the gyrocompass. In an airplane, the device is usually known as the **automatic pilot** and consists of sensors to detect deviations in direction, pitch and roll, which pass signals via a computer to alter the controls as necessary.

GYROSCOPE: A mechanical or optical device used to maintain orientation during motion. It is a disc with a heavy rim mounted in a double gimbal so that its axis can adopt any orientation in space. When the disc is set spinning the whole contrivance has two useful properties. These are the gyroscopic inertia and the precession. The gyroscope turns about an axis at right angles both to the axis about which the torque was applied and to its main axis of spin. An optical gyroscope, laser or fibre, measures the interference pattern generated by two light beams, travelling in opposite directions within a mirrored ring or fibre loop, in order to detect very small changes in motion. Gyroscopes are used in compasses, in automatic pilots on ships and aircraft, in the steering mechanisms of torpedoes, in antiroll equipment on large ships and in inertial guidance systems.

GYROSTABILISER: A gyroscopic device for stabilising a ship, airplane and instrument mounting. Ship stabilisers, for example, consist of fins protruding from the ship's hull, moved hydraulically to oppose roll under the control of signals from small gyroscopes that sense roll and velocity.

GYRUS (CONVOLUTION): Any of the irregular convex folds or ridges of the surface of the brain especially of the cerebrum.

GYTTJA: A Swedish word, which defines a nutrient-rich sedimentary peat that is found at the bottom of a body of water consisting mainly of plankton, other plants, animal residues, and mud.

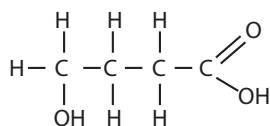
GZip (GNU ZIP): In computing, free and open source compression software, commonly used to compress web pages on the server end for decompression in the browser. Files compressed using GZip can be recognised by the file extension 'GZ'. The software is published by the Free Software Foundation and was originally developed for Unix and is also available for Windows and Mac.

GZK LIMIT (GREISEN-ZATSEPIN-KUZMIN LIMIT): A theoretical upper limit on the energy of cosmic rays imposed by the interaction between the cosmic rays and the cosmic microwave background radiation. It was independently computed in 1966 by Kenneth Greisen and Vadim Kuz'min and Georgiy Zatsepin.



4-HYDROXYBUTANOIC ACID (γ -HYDROXYBUTYRIC ACID; SODIUM OXYBATE; GAMMA HYDROXYBUTYRIC ACID, GHB):

A naturally occurring carboxylic acid with the chemical formula $C_4H_8O_3$. It is found in small amounts in the central nervous system of most living things, wine, beef and small citrus fruits. It is used as a therapeutic drug to treat insomnia, depression and alcoholism, narcolepsy and to enhance athletic performance. It is also used as a recreational drug with effects similar to alcohol.



*Chemical structure Of
4-hydroxybutanoic Acid*

4-HYDROXYBUTANOIC ACID LACTONE (DIHYDROFURAN-2(3H)-ONE; GAMMA-BUTYROLACTONE; 1,4-BUTYROLACTONE; 1,4-LACTONE): A colourless oily liquid lactone with the chemical formula $C_4H_6O_2$. It can be synthesised from *gamma*-hydroxybutyric acid (GHB) by removal of water or by distillation from such a mixture. It may also be obtained via oxidation of tetrahydrofuran (THF). It is an intermediate for the manufacture of butyric acid compounds and pyrrolidones. It is used as a plasticiser, solvent for polymers, nylon precursor and in insecticides. It is also used in the petroleum industry for natural gas, acetylene, paraffins and naphthene separations.

H: Symbol for henry, the SI unit of inductance, equal to the inductance of a closed circuit with a magnetic flux of 1 weber per ampere of current.

H ZONE: The middle of the dark band at the centre of a sarcomere that contains only thick filaments.

HAAR CONDITION: Named after the Hungarian mathematician Alfred Haar (1885- 1988), it is the condition for continuous functions $g_1, \dots, g_n$ on a set that the determinant $\det[g(x_j)]$ does not vanish for any subset of distinct points $x_1, \dots, x_n$. This holds for $1, x, \dots, x_n$ on any interval.

HAAR MEASURE: Introduced by Hungarian mathematician Alfred Haar (1885- 1988) in 1933, it is non-zero measure, μ on a sigma-ring, S , generated by compact subsets of a topological group that is locally compact, such that the measure is either left-invariant ($\mu(xA) = \mu(A)$ for all x in G and A in S) or right-invariant ($\mu(Ax) = \mu(A)$ for all x in G and A in S). In the commutative case these coincide and on a compact group any left-invariant measure is also right-invariant and vice versa. Such a measure is unique up to a multiplicative constant. An example is Lebesgue measure.

HABER PROCESS (HABER-BOSCH PROCESS): An industrial process developed by German chemist and Nobel laureate Fritz Haber, for manufacturing ammonia economically from nitrogen and hydrogen by direct combination of its elements, nitrogen and hydrogen. The hydrogen is obtained from seawater by passing an electric current through the water. The reaction process can be represented by $N_{2(g)} + 3H_{2(g)} \rightarrow 2NH_{3(g)}$.

HABIT: In biology, it is the instinctive actions of animals and the mode of growth of a plant.

HABITAT: A place where an organism, an animal or a plant lives successfully, including all living and nonliving factors or conditions of the surrounding environment resulting in adaptation to the local conditions there. Examples of habitats are seashore, woodland, pond, etc. Microhabitats are much smaller areas, for example, a bird's nest, a cow pat or a log.

HABITAT ACTION PLAN (HAP): A detailed description of a specific habitat together with the detailed actions and targets proposed for conserving it.

HABITAT COMPRESSION: The restriction of a local population within a set boundary, due to increase in the population.

HABITAT EXPANSION: Increase in the habitat (overall area) distribution of the species.

HABITAT PATCH: A location that encompasses a distinct habitat type used by a species for breeding or obtaining other resources.

HABITAT RESTORATION: Activities carried out to return a degraded or former habitat to a healthy, self-sustaining condition that resembles as closely as possible its pre-disturbed state. Examples include removing material from a filled wetland, increasing tidal flow to a restricted wetland, re-establishing natural river flow, enhancing degraded seafloor habitats, treating runoff to improve water quality, cleaning up contaminated habitats and managing invasive species.

HABITAT SELECTION: Locations chosen by an organism over other ones to suit climatic and environmental conditions, as well as to enhance the opportunities for survival and fitness of individuals.

HABITUATION: 1. A condition in which an individual involuntarily tends to be physiologically but not physically dependent on a drug with a desire to continue its use but not to increase the dosage. 2. A simple type of learning in which behaviours not needed are eliminated, consisting of a gradual waning in the response of an animal to continuous or repeated stimuli that is not associated with reinforcement.

HACKER: A programmer or computer user that works in a skillful but undisciplined manner. Such a user may alter codes in other users' file or may insert virus. The main interest of the hacker is to defeat security systems.

HACKING: An unauthorised access to a computer using weaknesses in a computer system or computer network, without a specific, constructive purpose or for fraudulent purposes.

HACKLES: 1. Any of the long, slender, often glossy feathers on the neck of a bird, especially a male domestic fowl. 2. The erectile hairs along the back of the neck of an animal, especially of a dog.

HADAL ZONE (HADOPELAGIC ZONE; TRENCH ZONE): The deepest level of the ocean, below the abyssal zone. It begins at depths of 6000 metres to the bottom of the ocean. Few species are adapted to these conditions because of lack of light and intense pressure creates hostile living conditions. Organisms such as jellyfish, viperfish, tube worms, sea cucumbers and the deep sea angler fish adapted to this region will die in the lower-pressure areas above.

HADAMARD DESIGN: Named after the French mathematician Jacques Hadamard, it is a block design with $4\lambda + 3$ points for some integral λ and with $2\lambda + 1$ blocks, any two distinct points of this design being contained in exactly λ blocks; that is, a $(4\lambda + 3, 2\lambda + 1, \lambda)$ design. A Hadamard design is a symmetric design.

HADAMARD INEQUALITY: 1. The inequality that the determinant of a positive definite matrix is not greater than the product of the diagonal entries of the matrix. 2. The inequality that the determinant of a square matrix is no greater than the product of the Euclidean norms of each row in the matrix.

HADAMARD MATRIX: An $n \times n$ matrix whose entries are equal to +1 or -1 and with an inverse that is its transpose divided by n ,